TravelMate Capstone Manuscript

TRAVELMATE

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College of Computer Studies

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Document control and reading guide

Prepared for the TravelMate team, adviser and panel. Version 2.0 is the source-reconciled documentation baseline dated 15 October 2026. Approval and academic administrative details remain unconfirmed.

This companion PDF is independently typeset from the same structured content as the editable Word files. It is not a verified Word export. The packaged DOCX renderer could not run because soffice.exe was unavailable. Word fields and pagination require a compatible renderer and final review.

This revision documents the inspected working files on 15 October 2026. “Complete” in the inventory means a code path was located across its applicable layers; it does not mean end-to-end acceptance passed. Static inspection, historical reports, current isolated tests and unexecuted acceptance procedures have separate evidence records. [1][2][6][7]

Evidence key: [1] frontend; [2] backend; [3] old SRS; [4] old SPMP; [5] format reference; [6] current checks and captures; [7] historical reports. The complete references and traceability registers are included.

Acknowledgement

The TravelMate team acknowledges the academic context of Cordova Public College and its College of Computer Studies. Specific advisers, panel members, contributors and forms of assistance have not been confirmed in the supplied material. [TO BE CONFIRMED: names, roles and supported contributions for the final acknowledgement.] This statement is prepared for completion and review; it does not attribute unverified assistance.

Approval Sheet

This manuscript for TravelMate is submitted in partial fulfillment of the requirements for the Bachelor of Science in Information Technology at Cordova Public College, College of Computer Studies. Final academic year, submission date and final proponents are [TO BE CONFIRMED]. This page records no approval until the authorized reviewers complete it.

Table 1 Approval fields requiring authorized completion

Capacity	Name	Signature and date
Capstone adviser	[TO BE CONFIRMED]	[NOT SIGNED]
Panel chair	[TO BE CONFIRMED]	[NOT SIGNED]
Panel member	[TO BE CONFIRMED]	[NOT SIGNED]
Panel member	[TO BE CONFIRMED]	[NOT SIGNED]
Academic approver if required	[TO BE CONFIRMED]	[NOT SIGNED]

Abstract

TravelMate addresses the coordination of destinations, schedules and group expense assumptions in a travel plan. The project combines a public interactive Earth with authenticated traveler and administrator workspaces. Its current implementation uses Next.js/React, Express and Prisma/PostgreSQL. Source-backed geographic and attraction records establish place identities before a configured AI provider arranges only permitted venue IDs and times. Deterministic validation and scheduling reconcile day windows, transfers, costs and honest incomplete results. Travelers can retain and edit saved versions, manage day activities, navigate saved trips, record actual visited places and submit feedback; administrators review user state, reports, appeals, feedback and operational records. The documentation was reconciled against current working files rather than old feature descriptions. It identifies 93 feature entries, 50 backend routes and 20 quality requirements, with 133 reproducible acceptance procedures. Isolated regression checks in this documentation task passed 152 frontend and 521 backend tests using copied sources and blocked external services. Anonymous current screens were captured. Those results do not establish live provider delivery, database persistence acceptance, user satisfaction, production performance or deployment recovery. Static concerns, one unresolved non-national geographic photo and missing administrative evidence remain explicit review items. The contribution is a coherent, source-grounded planning workflow and an auditable documentation baseline; effectiveness and release acceptance require the pending authorized system, operational and user evaluations. [1][2][6]

Keywords: travel planning; grounded itinerary; group budget; saved trip; source provenance; verification.

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Software Project Management Plan

Version 2.0 documentation reconciliation. Prepared for the TravelMate project team, adviser and capstone panel. This plan records supported project context and provides an explicitly proposed management baseline for remaining verification and maintenance. It does not reconstruct undocumented meetings, costs or completed milestones.

SPMP 1 Overview

TravelMate is a travel planning system that helps a traveler identify a destination, describe a trip and budget, obtain a source-grounded itinerary, and manage saved activities. The current implementation combines an interactive public discovery globe, authenticated traveler workspace and administrator workspace. It provides planning guidance rather than commercial reservations, payment processing or live crowd surveillance. [1][2]

The institutional context retained from the supplied TravelMate documents is Cordova Public College, College of Computer Studies, Cordova, Cebu, Philippines, and the Bachelor of Science in Information Technology. The final roster, approved academic year, adviser and panel membership, submission date, and signatures remain [TO BE CONFIRMED]. The old title’s Cebu qualifier is treated as historical context because current destination search is worldwide. [3][4]

This revision documents the inspected working files on 15 October 2026. “Complete” in the inventory means a code path was located across its applicable layers; it does not mean end-to-end acceptance passed. Static inspection, historical reports, current isolated tests and unexecuted acceptance procedures have separate evidence records. [1][2][6][7]

Table 2 Management baseline and deliverables

Item	Supported position	Required action
Scope	Guest discovery; traveler planning, saved trips, maps, visits and feedback; administrator monitoring/moderation.	Approve the reconciled SRS; explicitly exclude marketplace/payments.

Item	Supported position	Required action
Delivery set	Manuscript; SPMP; SRS; SDD; STD; manual; inventory; RTM; diagram sources; evidence/limitations.	Review as one versioned package.
Acceptance	Source inspection and isolated regression available; manual system, UAT and deployment acceptance pending.	Execute STD with disposable fixtures, then authorized staging checks.
Constraints	Existing data, source code and original documents remained read-only in this task.	Keep corrections as separately approved development changes.

SPMP 2 References

The consolidated References section identifies the current frontend and backend snapshots [1][2], old TravelMate SRS [3], old SPMP [4], supplied formatting reference [5], task test evidence [6], and dated engineering reports [7]. These sources have different authority: current code defines implementation; approved requirements would define intended acceptance; archived proposals do not prove historical execution.

SPMP 3 Definitions

Table 3 Management and engineering terms

Term	Meaning
Requirement	An identified intended behavior or quality condition; an implementation defect does not become a requirement.
Grounding	Binding a venue to source-supported identity, geographic context and provenance before scheduling.
Estimate	A supplied assumption or reference amount, not an offer, booking or guaranteed current price.
Version	A persisted itinerary snapshot and its restorable request envelope.
Verification	Evidence tied to a particular procedure, source state and environment.
Acceptance	A reviewer-approved decision after required functional, operational and user evidence is complete.

Term	Meaning
Legacy	Retained code/schema or old intention without a current supported end-user flow.

SPMP 4 Project organization

The old SPMP identifies project-manager, frontend and backend responsibilities, but does not supply a reliable confirmed mapping for the final team. The old SRS contains one clear name entry and another combined name string. This revision does not split, join or assign those names. The team must confirm the roster and role allocation in the Development Team Profile. [3][4]

Table 4 Proposed responsibility allocation requiring confirmation

Responsibility	Accountable person	Work and evidence
Project coordination	[TO BE CONFIRMED]	Scope, risk review, schedule/budget approvals and consolidated review records.
Frontend engineering	[TO BE CONFIRMED]	Current forms/navigation, globe, maps, responsive behavior, accessible feedback and generated contract consumption.
Backend and data engineering	[TO BE CONFIRMED]	Validation, authorization, grounding, scheduling, persistence, schema and provider controls.
Verification and documentation	[TO BE CONFIRMED]	Reproducible tests, RTM maintenance, defect retest evidence and manuscript consistency.
Adviser and panel	[TO BE CONFIRMED]	Academic guidance, review and formal acceptance signatures.
Deployment/data operator	[TO BE CONFIRMED]	Secrets, backups, migrations, provider agreements and incident handling.

SPMP 5 Managerial process

The current repositories show incremental source, tests and dated audit artifacts. They do not establish a particular Agile ceremony, sprint velocity, meeting attendance or percent contribution. The proposed process is a small change batch: agree on a requirement/defect, implement on an isolated branch, run the affected tests, inspect the user flow, update the RTM, and obtain review before release. Historical completion claims require dated evidence. [1][2][7]

Table 5 Work breakdown and remaining acceptance work

Work package	Deliverable or exit condition	Status
WBS 1 Scope and provenance	Reconcile old requirements, source snapshot, features, routes and processes.	Documented in this package.
WBS 2 Requirements and design	SRS, SDD, business rules, schema register and diagrams.	Documented; team review pending.
WBS 3 Isolated verification	Recorded unit/service regression and disposable system tests.	152 frontend and 521 backend tests passed; manual system cases pending.
WBS 4 Corrections	Resolve D-01 through D-09 where applicable and rerun relevant checks.	Proposed development work; not performed here.
WBS 5 Operational acceptance	Provider permissions, production-like auth/mail, persistence, recovery and performance.	Not executed.
WBS 6 Academic acceptance	Confirmed front matter, UAT, final document pagination review and signatures.	Pending team/adviser/panel input.

Historical schedule context: the old SPMP proposed initiation on 30 April 2026; requirements on 1 June; risk review on 15 June; QA on 1 July; prototype on 15 July; database/API work on 15 August; AI integration on 30 August; testing on 1 September; and documentation/defense on 1 October. It also described roughly sixteen weeks and 15-20 hours per person per week. These are archived planning statements, not verified actual dates or effort. The old SRS has separate

March-May change records and defense claims that have not been independently validated. [3][4]

Table 6 Proposed schedule control without invented dates

Gate	Dependency	Approved date and owner
G1 Scope review	Review current inventory and requirements.	[TO BE CONFIRMED]
G2 Correctness review	Resolve acceptance-blocking defects and run boundary/ownership cases.	[TO BE CONFIRMED]
G3 Operations review	Authorize staging services and execute recovery/performance checks.	[TO BE CONFIRMED]
G4 User review	Approve tasks/participants; execute UAT and record results.	[TO BE CONFIRMED]
G5 Submission	Confirm people/year/signatures and review Word pagination.	[TO BE CONFIRMED]

The old SPMP proposed a PHP 2,000 total with domain/API ranges of PHP 500-1,000 and a table allocating PHP 1,000 to each. This is an archived proposal, not an approved current budget or expense record. Actual cloud, email, AI, map/geocoding, media rights, domain and equipment costs depend on providers, agreements and usage. No paid service was called for this documentation task. [4]

Table 7 Budget and resource confirmation form

Cost/resource	Current evidence	Approved amount or record
Domain/hosting/database	Configuration exists; invoices and deployed ownership not provided.	[TO BE CONFIRMED]
AI/mail/data providers	Optional credentials and quotas; no usage billing inspected.	[TO BE CONFIRMED]
Equipment/connectivity	Browser, Node and PostgreSQL required; no asset register supplied.	[TO BE CONFIRMED]
Team effort	No verified timesheets or contribution percentages.	[TO BE CONFIRMED]

Cost/resource	Current evidence	Approved amount or record
Contingency and maintenance	Recommended operational allocation; no approved figure.	[TO BE CONFIRMED]

Table 8 Risk register and proposed responses

Risk	Effect	Response and review evidence
R1 Wrong identity/media	Wrong-country photo, attraction or route misleads a traveler.	Use exact IDs/provenance; fail honestly; run TM-XT-014 and grounding tests.
R2 Missing/stale prices	A low apparent total is mistaken for a complete payable price.	Separate known/estimated/unpriced costs; disclose references and verify externally.
R3 Provider outage/quota	Mail, AI, maps or search becomes unavailable.	Bounded timeouts/limits; labeled fallback; rehearse with stubs before staging.
R4 Concurrent edits	A stale preview/save/visit batch overwrites current state.	Revision hashes, row/advisory locks, unique keys and disposable contention tests.
R5 Scope drift	Retained schema is mistaken for a supported marketplace.	Inventory status, current UI reconciliation and explicit excluded scope.
R6 Privacy/access control	User records, tokens or GPS exposed.	Owner/role checks, secret handling, minimized evidence and privacy review.
R7 Recovery failure	Image rollback follows an incompatible schema migration.	Backup/restore rehearsal and compatibility gate; image rollback is not schema rollback.
R8 Academic evidence gap	Unverified dates, UAT or performance presented as fact.	Placeholders, evidence registers and explicit reviewer sign-off.

SPMP 6 Technical process

The frontend uses Next.js App Router and React; Express services validate and process requests; Prisma and PostgreSQL hold application data. Local GeoNames/PSGC records support discovery, and external providers enrich grounded planning. The SDD describes the actual scheduling and cost rules. Development and release procedures shall use the dependency locks and compatible paired repository revisions. Generated contract checks, unit tests, integration fixtures, production HTTP tests and browser tests have distinct purposes. Their presence in CI does not prove this task executed the pipeline. [1][2]

SPMP 7 Supporting process

Configuration management: retain an immutable release identifier for each repository, its dirty-file diff when relevant, migration set, provider configuration names and evidence package. Do not commit .env files, credentials or test session cookies. Changes to scope, provider rights, currency rules or persisted JSON need a requirements/design review. Keep corrective changes out of this documentation-only output.

Quality assurance: map each requirement to its feature, implementation, test and manual entry; record actual results and defects separately. Review source-backed financial examples, ownership negatives, asynchronous country identity, provider failures and version concurrency. Review Word fields and every rendered page with a Word-compatible renderer before final academic submission. The available environment could not render DOCX; the supplied companion PDF is independently typeset from the same content, not a verified Word export. [6]

Communication and issue control are proposed: use a dated issue entry containing the affected requirement, evidence, owner, severity, decision and retest. No meetings or approvals are asserted. Critical data/security failures should block release; a reviewer must approve the disposition of lesser defects. Record decisions in the unresolved-details register rather than silently marking them complete.

SPMP 8 Additional plans

Maintenance planning should cover dependency/security review, provider quota and license review, curated geographic/media refresh, database capacity, incident contact, session revocation and recovery exercises. No automatic daily backup worker was located. Retention periods, service-level objectives, recovery point/time objectives and actual maintenance ownership remain [TO BE CONFIRMED]. This plan proposes those controls without claiming they have been operated.

Software Requirements Specification

Version 2.0. This specification defines the current TravelMate scope and intended acceptance rules, including the places where implementation evidence is partial. The separate 113-row RTM links 93 functional and 20 quality requirements to source, design, tests and manual sections.

SRS 1 Introduction

Trip planning requires coordinating destination identity, time, transport, activities, accommodation assumptions and group money. The project addresses that coordination problem by combining a structured trip brief with source-grounded places, a constrained schedule and explicit cost evidence. This problem statement is a design rationale derived from the implemented workflow and supplied project objectives; no survey, quantified market need or effectiveness study is claimed. [1][2][3]

The general objective is to provide a coherent traveler planning and saved-trip workspace. Specific objectives are to support reliable destination selection; construct bounded multi-day schedules from permitted venues; show budget assumptions and unpriced gaps; retain editable versions and actual visit history; support saved-trip maps and managed day actions; and provide administrator visibility and moderation. These objectives are evaluated through requirements and reproducible cases rather than invented satisfaction percentages.

Table 9 Stakeholders and permission boundaries

Actor	Authorized scope	Excluded authority
Guest	Public discovery, legal notices and account entry/recovery.	No saved-trip or admin data.
Traveler	Own trips, versions, visited places, reports, feedback, notifications and profile; permitted generation/maps.	No other traveler’s records, administrative moderation or public role assignment.
Suspended traveler	Sign-in and dedicated notice/appeal workflow.	Ordinary protected workspace operations remain blocked.

Actor	Authorized scope	Excluded authority
Administrator	User-state moderation, appeals/reports, feedback, metrics, notices and system/audit views; own profile.	No public admin creation; no implemented owner marketplace, payment settlement or prompt editor.
Adviser/panel/ operator	Academic or operational review as separately authorized.	No additional application role inferred from these stakeholder labels.

The application role enum contains traveler and admin. Guest is an unauthenticated state, and suspension is an account status. Legacy Listing.ownerId does not establish a current owner role. Current global geography extends beyond the old Cebu-only/fixed-seven-day descriptions. The current supported trip range is one to thirty-one inclusive days. [1][2]

SRS 2 References and interpretation

This revision documents the inspected working files on 15 October 2026. “Complete” in the inventory means a code path was located across its applicable layers; it does not mean end-to-end acceptance passed. Static inspection, historical reports, current isolated tests and unexecuted acceptance procedures have separate evidence records. [1][2][6][7]

Each TM-FR ID is a requirement identity, not a test result. The 86 complete, 4 partial and 3 unavailable inventory entries are code-path classifications. Partial/absent features are explicitly constrained below. “Shall” expresses intended acceptance; D-03, D-04, D-06 and D-07 record possible deviations requiring disposable reproduction. The quality requirements distinguish implemented controls from proposed measurable targets. [1][2][6]

SRS 3 Specific requirements

SRS 3.1 External interfaces

User interface: public landing/discovery and account dialogs; /dashboard traveler workspace; /admin/dashboard administrator workspace; /account/appeal for a suspended account; public /privacy and /terms. Dashboard navigation uses internal workspace state, so the Saved trips label may be backed by the legacy bookings state key. /gps and /dashboard/gps now redirect into saved-trip navigation rather than supporting an independent planning workspace. [1]

Software interface: the browser sends same-origin HTTP requests through Next.js rewrites/proxy handlers to Express, normally at localhost:5000 in development. Server-only BACKEND_URL selects the API origin. JSON schemas and explicit role/ownership checks govern protected operations. Google OAuth has dedicated start/status/callback forwarding. Administrator updates use text/event-stream SSE; no Socket.IO requirement is inferred. The API route reconciliation includes technical readiness and session endpoints with no distinct screen. [1][2]

Hardware/browser interface: an internet-connected modern browser is needed for remote services; WebGL supports the 3D Earth, and Leaflet renders saved-trip maps. A country selector and visible failure states provide alternatives when globe rendering/data fails. GPS needs explicit browser permission and an appropriate secure context. No minimum RAM, GPU model, physical-device matrix or server capacity was validated. Production requires Node >=22.13, PostgreSQL and configured provider credentials where used. [1][2]

SRS 3.2 Functional requirements

SRS 3.2.1 Public discovery

TM-FR-001 Landing and navigation [complete; Guest / all]. The public site shall present planning guidance, anchored information sections, FAQ, account entry and legal links.

Acceptance and scope: Landing samples are illustrative. Some marketing copy still mentions removed travel-style/comparison controls; see discrepancy D-01.

TM-FR-002 Interactive Earth controls [complete; Guest / all]. The globe shall support rotation, drag, zoom, pause and reduced-motion handling, with a country selector fallback.

Acceptance and scope: WebGL or dataset failure shows fallback feedback. Selection focuses Earth; it does not create a trip or open a destination-details panel.

TM-FR-003 Country hit detection and border identity [complete; Guest / all]. Hover detection shall use actual country geometry and stable geographic codes, including small-area hit targets, and display that country’s border.

Acceptance and scope: Natural Earth geometry has finite resolution. Disputed/non-national areas are data records, not claims of sovereignty. Decorative arcs, dots and rings are absent.

TM-FR-004 Country destination photo and identity card [partial; Guest / all]. A hover card shall use the same stable code for the destination photograph, English name, available flag and border, and hide on pointer exit.

Acceptance and scope: 264/265 geographic areas have verified photos; Bir Tawil is unresolved and receives no unrelated fallback. All 249 ISO entries plus Kosovo are covered. Antarctica and some non-national areas have no national flag.

TM-FR-005 Image and map credits [complete; Guest / all]. The public footer shall expose image/map source, author, licensing and adaptation credits without a permanent caption under Earth.

Acceptance and scope: Credits describe asset rights, not provider availability or travel recommendations.

TM-FR-006 Privacy and terms pages [complete; Guest / all]. Public privacy and terms pages shall describe the planning scope and applicable content handling.

Acceptance and scope: Current privacy text emphasizes Google photos although ordinary production enrichment uses reusable providers; notice reconciliation is recommended (D-02).

SRS 3.2.2 Accounts and access

TM-FR-007 Traveler registration [complete; Guest]. Registration shall validate names, email, optional Philippine mobile number, strong password and confirmation, and create only a traveler account.

Acceptance and scope: Names 2-80 characters combined; password 8-64 with upper/lower/digit/special and no whitespace. Mobile is optional. Email configuration is required; accepted mail is not confirmed inbox delivery.

TM-FR-008 Email verification [complete; Guest with pending account]. Verification shall consume a valid account-bound, short-lived code once and permit subsequent password sign-in.

Acceptance and scope: Eight hexadecimal characters, 5-minute lifetime and 5 attempts. Verification is separate from legacy profileStatus.

TM-FR-009 Resend verification [complete; Guest with pending account]. Resending shall replace only an eligible account’s verification code, enforce cooldown/limits and report mail failures honestly.

Acceptance and scope: Three eligible resends per hour; failed send invalidates that code while retaining cooldown. No automatic switch from Resend to Gmail.

TM-FR-010 Password sign-in and role redirect [complete; Guest]. Password sign-in shall validate credentials and email verification, issue a signed session and route traveler/admin/suspended accounts appropriately.

Acceptance and scope: Wrong credentials return 401; unverified email returns 403. Suspended accounts can sign in for appeals, but ordinary protected APIs reject them.

TM-FR-011 Google OAuth [complete; Guest / existing account]. Google sign-in shall validate state, PKCE and verified provider identity before resolving the application account and role.

Acceptance and scope: Requires provider configuration and authorized redirect origin. Existing verified local accounts require the documented linking-safe behavior; no admin role is granted from public input.

TM-FR-012 Authoritative sessions and access control [complete; Traveler / administrator]. Protected API requests shall validate the signed session, expiry, current role, account state, session version and per-session revocation.

Acceptance and scope: Custom HMAC-SHA256 session format, not a standard JWT refresh-token system. Eight-hour cookie; proxy checks presence only.

TM-FR-013 Logout and local cleanup [complete; Traveler / administrator / suspended traveler]. Logout shall clear application/OAuth cookies, attempt individual session revocation and disclose unconfirmed revocation.

Acceptance and scope: If database revocation fails, local cookies clear but server revocation is explicitly unconfirmed. Other logins are not intentionally revoked.

TM-FR-014 Forgot/reset password [complete; Guest with verified active account]. Recovery shall issue a short-lived reset code, validate the replacement password, consume the token once and invalidate old sessions.

Acceptance and scope: Reset lifetime 15 minutes. Actual email delivery/provider authorization was not tested. Reset is not an in-profile password-change screen.

TM-FR-015 Suspension appeal access [complete; Suspended traveler]. A suspended traveler shall read account notices/history and submit one pending appeal with a bounded message.

Acceptance and scope: Three attempts/hour; a pending appeal blocks duplicates. Approval/restoration is an administrator decision.

SRS 3.2.3 Destination selection

TM-FR-016 Worldwide direct search [complete; Traveler]. Destination search shall return source-backed place identity, country, hierarchy and coordinates without inventing missing source records.

Acceptance and scope: Full imported catalogue records 5,721,309 records/250 source country entries; fallback compressed index is bounded. Coverage is not proof of complete geography.

TM-FR-017 Country and administrative browsing [complete; Traveler]. The destination browser shall follow recorded parent IDs/types with pagination and preserve an existing confirmation on cancellation.

Acceptance and scope: Available levels/terminology vary. Philippine independent cities retain official parents; coordinates absent from official data require separate resolution.

TM-FR-018 Manual resolution and clear [complete; Traveler]. Manual destination input shall require verified disambiguation; clearing shall remove confirmation and dependent plan/stay state.

Acceptance and scope: Country/region/island selections need narrowing before generation. Wrong-country or unresolved coordinates cannot be promoted to confirmed destinations.

TM-FR-019 Area suggestions and comparison [complete; Traveler]. Area browsing shall expose verified child localities and a limited comparison without implying researched prices.

Acceptance and scope: One to six city identifiers; comparison uses entered spending allowance and mapped interest samples, not live tourism popularity or confirmed costs.

TM-FR-020 Starting location and optional geolocation [complete; Traveler]. An origin shall be entered or explicitly obtained through browser location permission, with manual entry after denial/failure.

Acceptance and scope: Geolocation requires secure context/permission. Origin selection clears when account changes; it is optional for current generation.

TM-FR-021 Currency detection and fallback [complete; Traveler]. Destination country codes shall map to supported planning currency or require an explicit supported fallback.

Acceptance and scope: 20 supported codes; JPY/KRW/VND use whole units. Fallback is a planning choice, not currency conversion.

TM-FR-022 Reference-currency equivalents [complete; Traveler]. Reference conversion shall preserve the trip currency and label source/date/freshness, retry limits and unavailable rates.

Acceptance and scope: Frankfurter mid-market reference, not card/cash conversion. Same-code rate is 1; wrong pair/invalid rate is rejected; Retry-After is honored.

TM-FR-023 Destination transport guidance [complete; Traveler]. The planner shall label curated destination transport modes as guidance and allow progress when no guidance is available.

Acceptance and scope: Guidance exists for selected country/city patterns. It is not a live timetable, capacity or fare source.

TM-FR-024 Destination inspiration carousel [complete; Traveler]. An inspiration selection shall populate its source destination/preferred activity for review before generation.

Acceptance and scope: Inspiration is not a booking or a guarantee the attraction fits the final schedule.

SRS 3.2.4 Trip planning and enrichment

TM-FR-025 Required trip brief and bounds [complete; Traveler]. Generation shall require a confirmed locality, positive supported-currency group budget, valid party and 1-31 inclusive future/current dates.

Acceptance and scope: Maximum 20 travelers; solo=1, couple=2, family/friends=2-20. Budget ceiling 1,000,000,000. Server start-day check uses UTC while browser uses local date (D-03).

TM-FR-026 Preferences with Balanced generation policy [complete; Traveler]. New/regenerated plans shall use Balanced style/pace while accepting interests, preferred activities, stay preference and transport.

Acceptance and scope: No current travel-style or pace selector; old style values remain readable. Arrays max 20; bounded item lengths.

TM-FR-027 Optional accommodation suggestions [complete; Traveler]. The current planner shall browse relevant property suggestions without requesting room prices or requiring a stay selection.

Acceptance and scope: Current UI passes suggestionsOnly. LiteAPI sandbox / Rapid / OSM sources vary; no room reservation. Empty/provider-error stays do not block generation.

TM-FR-028 Manual stay and retained property details [complete; Traveler]. A manual/mapped stay shall persist as an unpriced planning preference and remain recognizable after reopen.

Acceptance and scope: Provider identity/allowed photos can be retained. Selection does not establish availability, rate, star quality or a completed booking.

TM-FR-029 Expense assumptions [complete; Traveler]. Explicit room, food, fare, fee and contingency assumptions shall be validated and calculated without invented prices.

Acceptance and scope: Default contingency 10%; permitted 0-50%. Missing costs are excluded from known subtotal, not free. Currency precision applies.

TM-FR-030 Optional pre-trip expenses [complete; Traveler]. Pre-trip user estimates shall be included once as whole-party costs with matching starting location and currency.

Acceptance and scope: No government fee advice is generated. Amounts are entered estimates; verify actual fees separately.

TM-FR-031 Place grounding and geographic validation [complete; Traveler]. The backend shall select physical attractions from supported records and reject invented, wrong-country or unsupported administrative/object entries.

Acceptance and scope: Wikipedia/Wikidata/OSM references can be incomplete. Max 30 km straight-line is only one filter; membership evidence remains required. Coordinates alone do not prove a venue.

TM-FR-032 AI structured schedule and repair [complete; Traveler]. AI shall arrange only supplied place IDs/times in batches of at most seven days and pass independent schema, identity, date and transfer checks.

Acceptance and scope: OpenAI default gpt-4o-mini / Gemini default gemini-3.1-flash-lite are code defaults and can be configured. One invalid-output repair per batch, sequential provider fallback; 45s per adapter call, 540s outer generation signal.

TM-FR-033 Deterministic full-day scheduling [complete; Traveler]. The scheduler shall reconcile catalog-backed visits, travel buffers, meal/free-time blocks and honest incomplete days within usable local-day windows.

Acceptance and scope: Normal current Balanced cap is three catalog visits/day; no place repetition to fill a long trip. Arrival/departure/rest constraints exist in API but current planner has no input controls (see the API-only constraint inventory entry).

TM-FR-034 Catalog fallback and incomplete results [complete; Traveler]. When AI fails or references are sparse, the system shall retain catalog-backed suggestions/allowances with explicit incomplete or unavailable disclosures.

Acceptance and scope: Catalog fallback is not a guest chatbot. Mock label is possible only with explicit non-production flag; production does not invent replacement venues.

TM-FR-035 Generation idempotency and fencing [complete; Traveler]. Repeated identical generation keys shall replay completed results, reject conflicting input and fence stale concurrent attempts.

Acceptance and scope: 8-100-character keys. A pending record under 10 minutes conflicts; failed/stale attempts use compare-and-set fencing. This does not guarantee paid remote computation cancellation.

TM-FR-036 Automatic save after generation [complete; Traveler]. Successful generation shall attempt an owned atomic initial save or regeneration update and preserve a recoverable draft if saving fails.

Acceptance and scope: Generation success and save success are separate. Save key is generated by saveIdentity; ordinary API calls without one may have different deduplication behavior.

TM-FR-037 Activity photographs and provenance [complete; Traveler]. Activity photos shall retain exact place/source identity, reusable rights and credits, with separately labeled illustrative/destination fallbacks.

Acceptance and scope: No photo upload endpoint. Stored selection -> matched Wikimedia/Wikipedia -> Unsplash illustration -> trip destination -> neutral placeholder. Ordinary production flow bypasses Google provider.

TM-FR-038 Date-matched weather [complete; Traveler]. Weather shall distinguish current reference conditions from the dates actually covered by a forecast and disclose missing or stale coverage.

Acceptance and scope: OpenWeatherMap when configured, then Open-Meteo (14-day request). Thirty-one trip days do not imply 31 forecast days.

TM-FR-039 Crowd estimates [complete; Traveler]. Crowd context shall remain an explicitly low-confidence calendar heuristic rather than live congestion data.

Acceptance and scope: Weekend and broad Philippines/December heuristics only; no foot-traffic feed or automatic live rerouting.

TM-FR-040 Advisory refresh without overwriting edits [complete; Traveler]. Refreshing conditions shall update advisory metadata without regenerating or replacing manual activity content.

Acceptance and scope: A refresh is not confirmation of unchanged future conditions or automatic rescheduling.

SRS 3.2.5 Saved trips and traveler records

TM-FR-041 Saved-trip list and pagination [complete; Traveler]. Saved trips shall show owned summary cards with status/source/cost context and load further pages without exposing other accounts.

Acceptance and scope: Summary responses omit full itinerary/weather. Internal tab key bookings means Saved trips, not an active booking feature.

TM-FR-042 Reopen, day navigation and return [complete; Traveler]. Reopening and navigating days shall preserve the selected trip/draft and never silently regenerate missing saved content.

Acceptance and scope: Opening a sparse day does not repair/populate it. Legacy URL/history normalization is separate from AI generation.

TM-FR-043 Activity editing [complete; Traveler]. The editor shall permit bounded title, time, description, location, category and estimated-cost edits, recalculating totals as user estimates.

Acceptance and scope: Venue identity changes clear stale place/photo evidence. Unsupported changed price claims become estimates; server-owned management history cannot be forged.

TM-FR-044 Add, remove and reorder activities [complete; Traveler]. Draft activity changes shall preserve chronological day identity and existing evidence where appropriate, with explicit removal confirmation.

Acceptance and scope: At most eight saved activities/day in relevant validation. Completed/skipped management records are protected against silent removal.

TM-FR-045 Save manual changes and recover drafts [complete; Traveler]. Manual edits shall remain a browser draft until saved, with owner-scoped recovery and guards against unintended navigation loss.

Acceptance and scope: Browser storage may be blocked/full. A browser draft is not a server backup; invalid/wrong-owner drafts are rejected.

TM-FR-046 Regenerate existing trip [complete; Traveler]. Regeneration shall require deliberate input review and produce a version of the same active trip only after valid generation/save.

Acceptance and scope: Current Balanced defaults apply to regenerated plans. Failed generation/save must leave the saved plan available; no automatic regeneration on open.

TM-FR-047 Version history and whole-version restoration [complete; Traveler]. Version restoration shall atomically recover the selected owned snapshot and its matching original trip context as a new version.

Acceptance and scope: 50 metadata records/page. Legacy versions can lack party/context and fail safely; unrecorded context is not guessed.

TM-FR-048 Archive and restore trip [complete; Traveler]. Archiving shall retain the trip and its prior completion state; restoring shall return it to active/completed accordingly.

Acceptance and scope: Archived trips cannot be directly completed/reopened; restore first. Archiving is distinct from permanent deletion.

TM-FR-049 Complete and reopen trip [complete; Traveler]. Trip completion/reopening shall update lifecycle state only and shall not automatically create/remove visit history.

Acceptance and scope: Completion does not prove attendance. Completed itineraries remain readable but editing/management requires reopen.

TM-FR-050 Delete owned trip [complete; Traveler]. Deletion shall require deliberate confirmation, delete the owned trip/version records and preserve detached visit/feedback history according to foreign keys.

Acceptance and scope: No normal UI undelete is implemented. Documentation task did not execute deletion.

TM-FR-051 Saved cover photos and stay details [complete; Traveler]. Saved cards shall preserve relevant destination cover/photo attribution and any selected unpriced stay without altering itinerary content.

Acceptance and scope: Cover selection is not attraction verification. Unsplash selection tracking is separate and must not run on every render.

TM-FR-052 Manual visited-place recording [complete; Traveler]. A traveler shall explicitly record a named place and past/present actual visit date under their own account.

Acceptance and scope: Future visit dates are rejected using the supplied valid timezone. Manual records are not automatically geographically verified.

TM-FR-053 Record places from a saved trip [complete; Traveler]. The batch visit modal shall list actual saved place candidates across days, prevent stale/duplicate selections and save selected actual visits atomically.

Acceptance and scope: Allowances/rest/placeholders excluded. Archived trip requires restore. Batch duplicate prevention is stronger than the date-specific database unique key.

TM-FR-054 Visited history search and removal [complete; Traveler]. Visit history shall be owner-scoped, searchable, date-ordered and paginated, with explicit individual removal.

Acceptance and scope: Deleting a trip leaves its visited records detached. Removing a visit does not remove the saved itinerary.

TM-FR-055 Itinerary feedback and updates [complete; Traveler]. Feedback shall bind one 1-5 rating, supported category and 10-2000-character comment to an owned saved trip, updating the same record on resubmission.

Acceptance and scope: Category budget_accuracy/weather_info/ai_cohesion/route_efficiency. Sentiment remains not_analyzed; no sentiment AI is implemented.

TM-FR-056 Traveler problem reports [complete; Traveler]. Travelers shall submit bounded problem reports for administrator review and receive resolution notices.

Acceptance and scope: Reports are not an itinerary chat command or a guarantee of support response time.

TM-FR-057 Traveler notifications [complete; Traveler]. The traveler notification view shall display owned notices and permit marking one/all read without changing another recipient's data.

Acceptance and scope: Traveler updates use request/polling behavior; administrator live SSE is a distinct feature. Notices are not push/email delivery guarantees.

TM-FR-058 Traveler profile and contact settings [complete; Traveler]. Profile editing shall persist only allowed name/contact/avatar/bio fields and preserve account role/email/security state.

Acceptance and scope: Name 2-80, phone <= 30, bio <= 500, avatar HTTPS <= 2048. Email is read-only; unknown protected keys do not become writable.

SRS 3.2.6 Saved trip maps

TM-FR-059 Owned saved-trip map and timeline [complete; Traveler]. The saved-trip map shall derive days/stops from the selected owned itinerary and separate daily allowances from visit stops.

Acceptance and scope: Opening a map does not complete an activity or record a visit. Skipped activities remain visible but are excluded from relevant routing.

TM-FR-060 Exact map-place resolution and photos [complete; Traveler]. An uncertain activity shall require an exact source-matched candidate before navigation, without silently promoting a vague name.

Acceptance and scope: Vague allowances/activities cannot become precise stops by guess. Location-provider failure leaves unresolved feedback.

TM-FR-061 Selected-day ordered route [complete; Traveler]. Day routing shall follow recorded verified-stop order for the selected walking/cycling/driving mode and disclose omitted unresolved stops.

Acceptance and scope: OSRM road/path estimates, no live traffic. Public provider quota, no-route, excessive snap distance and unavailable routes are explicit.

TM-FR-062 Location sharing and route guidance [complete; Traveler]. Live route guidance shall begin only after an explicit location-sharing action and permit stopping, manual fallback and mode changes.

Acceptance and scope: HTTPS/permission required. Denial, inaccurate/stale position and route failure retain recovery paths. No safety-critical navigation guarantee.

TM-FR-063 Activity-specific weather [complete; Traveler]. Activity weather shall keep current and forecast requests distinct and use the exact resolved stop/date/time within provider coverage.

Acceptance and scope: Exact location needed; scheduled weather outside actual coverage remains unavailable. Global provider request budgets apply.

TM-FR-064 Retired GPS URL compatibility [complete; Traveler / guest]. Legacy GPS URLs shall redirect into Saved trips navigation while retaining only valid trip/day identifiers.

Acceptance and scope: The standalone GPS workspace is removed; compatibility redirect does not restore removed navigation tabs.

SRS 3.2.7 Managed day activities

TM-FR-065 Confirm completion, skip or restore activity [complete; Traveler]. Managed activity status changes shall require owner, active trip, current revision, saved activity identity and explicit confirmation.

Acceptance and scope: Completed details are protected until undo. Skip retains cost/history and excludes remaining-day routing/replanning. Status change does not record a VisitedPlace.

TM-FR-066 Confirmed activity edit/reschedule [complete; Traveler]. Managed edit/reschedule shall reject stale revisions, new duplicate places, time overlaps and changes to completed details until undone.

Acceptance and scope: Duration 15-720 minutes. Moving to another day requires Reschedule; target max 8 activities. Changed venue/photo evidence is revalidated.

TM-FR-067 AI remaining-day preview [complete; Traveler]. Replan My Day shall propose times for every remaining saved activity exactly once without writing the itinerary or changing costs/completed history.

Acceptance and scope: Three previews per owner/5 minutes. Existing Gemini/OpenAI required; no mock fallback. Missing saved durations must be entered first. Preview can consume provider quota.

TM-FR-068 Accept or cancel revised day [complete; Traveler]. Acceptance shall validate a signed owner/trip/revision-bound preview and persist one reviewed change; cancellation shall leave itinerary/version data unchanged.

Acceptance and scope: Preview token expires after 15 minutes. Browser abort does not prove remote model billing stopped. Completed/skipped records and saved costs remain.

TM-FR-069 Activity history and legacy compatibility [complete; Traveler]. Management shall preserve stable activity identity/history and default legacy records to Planned without writing on open.

Acceptance and scope: Legacy IDs are derived read-only until confirmed persistence. Management metadata is server-owned, separate from a free-form manual edit.

SRS 3.2.8 Administration

TM-FR-070 Administrator overview [complete; Administrator]. The overview shall show stored aggregate account/generation/feedback counts and seven calendar days without invented activity.

Acceptance and scope: Calendar uses UTC+8/Asia/Shanghai. Counts describe stored records, not active concurrent users or measured satisfaction.

TM-FR-071 User search and access state [complete; Administrator]. User management shall search/filter permitted directory fields and suspend/restore non-admin accounts with confirmation and notifications.

Acceptance and scope: Admin accounts cannot be changed through this action. No public admin creation/role reassignment is exposed. Unknown status currently normalizes to active (D-04).

TM-FR-072 Send traveler warning [complete; Administrator]. A warning shall target an existing traveler with a 10-2000-character message and persist a recipient-scoped notice/audit event.

Acceptance and scope: In-app notice only; not automatic email/SMS. The documentation task sent no warning.

TM-FR-073 Review/resolve problem reports [complete; Administrator]. Administrators shall inspect full traveler reports and resolve only pending problem reports with an audited recipient notification.

Acceptance and scope: Resolution is a recorded decision, not technical proof the underlying defect is fixed. Already resolved/wrong-kind requests are rejected.

TM-FR-074 Review suspension appeals [complete; Administrator]. Appeal review shall validate a pending appeal and current suspended traveler state before approved/rejected decision and bounded explanation.

Acceptance and scope: Approve restores account access; reject retains suspension. Stale/double review and already-restored account are refused.

TM-FR-075 Live reports and reconnect snapshots [complete; Administrator]. The live report stream shall authenticate repeatedly, send scoped snapshots, reconcile cross-instance changes and clean up closed/stalled connections.

Acceptance and scope: SSE (not Socket.IO): 5-second reconciliation, 5-minute connection rotation, event-triggered local refresh and bounded write buffer.

TM-FR-076 System Health and selected audit activity [complete; Administrator]. System Health shall distinguish service configuration from verified availability and show recent generation failures and selected account/audit activity.

Acceptance and scope: No live uptime/latency dashboard, AI prompt editor or automated model curation. Configured means key presence, not a successful provider call.

TM-FR-077 Feedback filters and exact details [complete; Administrator]. Feedback review shall paginate/search/filter stored feedback and open exact linked feedback/related itinerary without inferring sentiment.

Acceptance and scope: Deleted trip/user links can be null. Sentiment defaults not_analyzed; filters do not prove automated analysis.

TM-FR-078 Administrator notifications [complete; Administrator]. Admin notifications shall retain recipient isolation, meaningful category/deep link and accurate unread count while allowing one/all read.

Acceptance and scope: Category is encoded in link metadata for current notices; compatibility handles older notices. Only that admin's read state changes.

TM-FR-079 Administrator own profile [complete; Administrator]. Account Settings shall update the administrator's own permitted contact/profile fields without exposing password/role mutation.

Acceptance and scope: Same allowed-field and length constraints as the traveler profile; no email/role/security-field editing.

SRS 3.2.9 Technical services

TM-FR-080 API error contracts and safe recovery [complete; All callers]. The API shall return stable safe error/code/retryability metadata, sanitized malformed-JSON feedback and distinct dependency failures.

Acceptance and scope: Database outages can return retryable 503 rather than false logout. No exhaustive penetration/security certification is asserted.

TM-FR-081 Database retries and pooled access [complete; API process]. The API shall use PostgreSQL/Prisma pooled TCP access with bounded retries only for recognized connection failures.

Acceptance and scope: An ambiguous limiter retry may count twice (fail-closed). Not a blanket retry of every business write or a proven backup/restore service.

TM-FR-082 Shared rate limits and request-driven cleanup [complete; API process]. Rate limits shall use shared fixed-window PostgreSQL buckets and opportunistically prune old buckets/revocations.

Acceptance and scope: Cleanup is a 1% request-triggered chance, not a scheduled worker. Authentication itself depends on database availability for rate limiting.

TM-FR-083 Provider cache and unavailable disclosures [complete; API process / traveler]. Provider caching shall coalesce identical in-flight requests, bound entries, honor freshness/stale cutoffs and expose unavailable results honestly.

Acceptance and scope: General cache max 300 entries. Flights/hotels 5m fresh/15m cutoff; activities 30m/2h; weather 15m/1h; FX 6h/24h.

TM-FR-084 Security headers, HTTPS and same-origin writes [complete; All callers / operator]. Production shall enforce HTTPS/security headers and reject untrusted cross-site browser writes with server-side authorization.

Acceptance and scope: No full encryption-at-rest/legal compliance or exhaustive XSS/CSRF proof. Frontend CSP includes unsafe-inline and is an explicit review item.

TM-FR-085 Operational health and graceful shutdown [complete; Operator]. Operational endpoints shall expose basic API identity/database readiness and the process shall close HTTP/database resources on termination.

Acceptance and scope: Health is a point observation, not measured uptime or a restore drill. The healthcheck script forwards HTTPS scheme for container checks.

TM-FR-086 Contracts and paired-revision validation [complete; Maintainer]. API contracts shall remain aligned with generated frontend snapshots and be checked against a pinned compatible backend revision.

Acceptance and scope: contracts:sync writes files; documentation task uses copies/check mode only. CI definitions are configuration evidence, not proof a run passed.

TM-FR-087 Container deployment and rollback configuration [complete; Maintainer]. A proposed release shall build a non-root image, apply reviewed migrations explicitly and require health before retaining the new app image.

Acceptance and scope: Configured pipeline, not executed here. Image rollback does not undo applied database migrations. Domain/hosting/backup acceptance needs actual operational evidence.

SRS 3.2.10 Compatibility and excluded scope

TM-FR-088 Flight and activity comparison API [partial; Traveler API consumer; UI removed]. Existing authenticated comparison endpoints shall normalize source-backed flight/activity results and unavailable states, without claiming a current traveler comparison screen.

Acceptance and scope: Unreferenced comparison component retained in source. Amadeus 1-9 travelers; configured production/test distinction. Mark partial because API exists but UI is retired.

TM-FR-089 Hotel quote/revalidation API [partial; Traveler API consumer; current UI suggestions only]. Existing quote APIs shall bind owner/context/occupancy/expiry and reject changed/sold-out/invalid quotes without asserting current UI booking.

Acceptance and scope: API occupancy 1-8 rooms, adults/child ages 0-17 and total <= 20. LiteAPI remains sandbox; Rapid rights gated. No current price/occupancy controls in main planner.

TM-FR-090 Generation arrival/departure/rest API constraints [partial; Traveler API consumer; no current form controls]. The generation API shall validate optional destination-local arrival/departure clocks and in-range rest dates without implying an exposed UI.

Acceptance and scope: HH:MM clocks; <= 31 in-range rest dates. Arrival adds 90 minutes; departure reserves 120. API-only capability is partial for normal users.

TM-FR-091 Removed owner/marketplace/payment workflows [unavailable; No active user role]. Removed owner registration, booking, payment, promotion and trust/profile-moderation actions shall remain unavailable rather than be documented as working product features.

Acceptance and scope: Schema presence is not feature availability. PayMongo is not integrated; owner role no longer exists. Legacy data is not deleted by this documentation work.

TM-FR-092 Retired preferences/comparison tabs and duplication [unavailable; Traveler]. Retired Preferences/Travel options tab URLs shall normalize safely, and the removed duplicate-trip action shall reject requests.

Acceptance and scope: Unreferenced TripPlannerForm still contains a travel-style prop; current dashboard does not render it. Do not infer a current style selector from that file.

TM-FR-093 Guest chatbot, live crowd feed and automatic backups [unavailable; Unavailable / proposed]. Unimplemented historical requirements shall be identified explicitly and must not be presented as observed system capabilities.

Acceptance and scope: Future work requires separate requirements, provider/security design and execution evidence. Old 99.9% uptime, 500-user capacity and automatic daily backup statements remain unverified targets.

SRS 3.3 Input and business rules

Table 10 Principal validation bounds

Input	Current rule	Qualification
Registration	Combined name 2-80; optional Philippine mobile; password 8-64 with upper/lower/digit/special and no whitespace.	Eight-character hexadecimal verification, not the old six-digit OTP.
Trip dates/party	1-31 inclusive days; nonpast start; travelers 1-20; solo 1, couple 2, family/friends 2-20.	Local-browser versus UTC-server today requires D-03 review.
Budget/currency	Positive whole-party ceiling up to 1 billion; 20 supported codes; JPY/KRW/VND zero decimals, others two.	Raw save normalization requires D-07 review; do not silently redefine intended precision.
Preferences	Up to 20 bounded interest/activity entries; up to 8 stay preferences; new generation Balanced style/pace.	No current travel-style/pace chooser.
Expenses	Explicit positive/zero estimates; contingency 0-50 percent, default 10; shared and per-person bases explicit.	Unknown costs are unpriced, not free; no government fee or current tariff guarantee.
Managed activity	Duration 15-720 minutes; cross-day limit eight activities with conflict checks; explicit confirmation and fresh revision.	Completed details protected until undo; skipped cost/history retained.
Feedback/report/appeal	Feedback rating integer 1-5 and comment 10-2000; report subject 5-120/message 10-2000; appeal/decision 10-2000.	Owned itinerary association and duplicate/terminal-state checks apply.
Profile	Name 2-80; phone <=30; HTTPS avatar <=2048; bio <=500.	Email, role, account state and security fields are protected.

SRS 3.4 Financial calculations

The group budget is a ceiling. It is not an instruction to spend the entire amount. Food estimate equals entered per-person daily amount times travelers times days. Accommodation equals validated quoted full stay total, or entered nightly rate times rooms times nights; ordinary nights are days minus one. Local fare is counted once per day when shared, or multiplied by travelers when per person. Activity admissions use supported participant counts. Explicit pre-trip/extra fees are counted once, and an already committed exact-name cost is not added again as an activity charge. [2]

Known subtotal is the sum of counted categories. Contingency is rounded in the currency's minor units from subtotal times the percentage divided by 100. Total equals subtotal plus contingency; remaining equals max(0, budget minus total); shortfall equals max(0, total minus budget). Displayed daily average is total divided by days. Exact per-person shares distribute the minor-unit remainder to the first person. A retained one-day selected overnight offer can carry a legacy one-night case. Reference estimates, live quotes, assumptions and unpriced entries remain distinguishable. [1][2]

Worked deterministic acceptance example: PHP 100 food per person/day for two people over three days = PHP 600; PHP 1,000 room/night for one room over two nights = PHP 2,000; PHP 200 shared fare/day = PHP 600; whole-party pre-trip cost = PHP 1,500. Subtotal is PHP 4,700, 10 percent contingency PHP 470, total PHP 5,170 and remaining from a PHP 10,000 ceiling PHP 4,830. This is a specified example for TM-XT-002, not a user record or a newly executed system result.

SRS 3.5 Quality requirements

TM-NFR-001 Authoritative access. Protected requests shall recheck role, ownership, suspension, expiry, session version and token revocation on the server.

Evidence position: Implemented control; live multi-account acceptance pending.

TM-NFR-002 Credential protection. Passwords shall use per-account random salts and scrypt; browser cookies shall be HttpOnly, SameSite Lax and Secure in production.

Evidence position: Implemented control; production cookie inspection pending.

TM-NFR-003 Request origin and transport. Production shall reject insecure backend transport and unexpected cross-origin writes; browser security headers shall be applied.

Evidence position: Implemented control; deployed reverse-proxy acceptance pending.

TM-NFR-004 Financial precision. Amounts shall respect the selected supported currency and its minor units; explicit costs shall not be charged twice.

Evidence position: Intended acceptance rule; inspect D-07 before sign-off.

TM-NFR-005 Grounded identities. AI shall not introduce unapproved place identities, prices or media; country and venue evidence shall be validated separately from coordinates.

Evidence position: Implemented validation; reference completeness and provider behavior not guaranteed.

TM-NFR-006 Honest degraded service. Failures shall produce a labeled unavailable, retryable or catalog fallback result without fabricating successful delivery, pricing or reservations.

Evidence position: Implemented control; external outage rehearsals pending.

TM-NFR-007 Atomic persistence. Multi-record changes shall use transactions and concurrency checks; stale generation, versions and activity previews shall not overwrite newer state.

Evidence position: Implemented control; live PostgreSQL contention tests pending.

TM-NFR-008 Duplicate prevention. Generation, save, batch visits, feedback and pending appeals shall apply their documented identity keys and duplicate semantics.

Evidence position: Implemented control; isolated service evidence only.

TM-NFR-009 Responsive layout. Public and role workspaces shall remain usable without unintended horizontal overflow at 320, 768, 1024 and 1440 CSS pixels.

Evidence position: Proposed release acceptance set; historical emulated evidence is limited to named reports.

TM-NFR-010 Keyboard and motion access. Core actions shall be keyboard reachable, inputs shall have meaningful labels, images shall have useful alt text, and Earth shall respect reduced motion.

Evidence position: Partly inspected; complete keyboard, contrast and assistive-technology audit not executed.

TM-NFR-011 Timeout and cancellation. Provider adapters and generation shall use bounded deadlines and cleanup; cancellation shall not be represented as guaranteed provider billing cancellation.

Evidence position: Implemented deadlines; 45 s AI adapter, 540 s outer generation and 580 s proxy are ceilings, not performance measurements.

TM-NFR-012 Data minimization. Application outputs and documentation shall not expose credentials, tokens or confidential user records; optional GPS shall require explicit permission and stop on cleanup.

Evidence position: Implemented controls inspected; retention policy and privacy notice reconciliation pending.

TM-NFR-013 Attribution and provenance. Photos, map tiles and geographic assets shall retain required attribution and source metadata; unresolved media shall not borrow another place identity.

Evidence position: Implemented manifest/credits; Bir Tawil geographic-area photo remains unresolved.

TM-NFR-014 Operational recovery. A release shall have an approved backup, restore, migration and rollback procedure tested against disposable data before production use.

Evidence position: Proposed operational acceptance; no automatic backup worker or executed recovery evidence located.

TM-NFR-015 Contract consistency. Generated frontend API contracts shall match the authoritative backend schemas; published OpenAPI shall cover actual routes.

Evidence position: Partial; generated checks exist but 10 actual route operations lack exact OpenAPI entries.

TM-NFR-016 Traceable changes. Version changes, managed activity actions and moderation shall preserve attributable history without claiming cryptographic audit immutability.

Evidence position: Implemented records; role-restricted read access, not a tamper-proof ledger.

TM-NFR-017 Shared rate limits. Rate limits shall persist across instances where configured and provider requests shall honor bounded request pacing and retry advice.

Evidence position: Implemented PostgreSQL buckets and in-memory provider pacing; distributed provider quotas not proven.

TM-NFR-018 Reproducible verification. A release decision shall separate static inspection, historical evidence, newly executed checks and unexecuted acceptance procedures.

Evidence position: Satisfied for this documentation evidence boundary; release approval pending.

TM-NFR-019 Measured performance. Performance acceptance shall record hardware, data volume, network, load, p50/p95/p99 and error rate against an approved target.

Evidence position: Proposed acceptance method; legacy sub-2-second/500-user/99.9-percent targets are unapproved and unmeasured.

TM-NFR-020 Date and time consistency. Trip date eligibility and activity-local clocks shall use an explicit, consistent time-zone policy across client and server.

Evidence position: Intended rule; browser-local versus server-UTC past-date edge remains D-03.

SRS 3.6 Data requirements and constraints

PostgreSQL stores users, session/token support, trips and generation history, itinerary versions, visited places, feedback, notices, moderation and audit records. JSON itinerary documents retain schedule, evidence, photos and cost provenance. Distinct relational constraints protect owner/idempotency keys and version numbers; deleting a trip detaches preserved visits and feedback while cascading itinerary versions. Seven marketplace-oriented models remain in the schema but do not establish current commercial workflows. The complete entity/field and route registers accompany the package. [2]

Dependencies include source coverage, connectivity, provider availability/rights, compatible geographic catalog files and correctly provisioned server secrets. Geographic labels and disputed areas are dataset entries rather than sovereignty claims. No system can guarantee current opening hours, route accessibility, weather, crowd occupancy, room stock or final payable amounts from the inspected sources alone. [1][2][8]

SRS 3.7 Security and privacy

Server checks are authoritative even when navigation is hidden. Password hashes use salted scrypt; sessions are custom HMAC signatures with an eight-hour cookie, not a JWT refresh-token pair. Account-bound verification/reset tokens are short-lived and single-use. Shared limits constrain authentication and replan abuse. Production transport/origin controls and sanitized errors reduce exposure, but a full penetration test, at-rest encryption proof and formal privacy compliance assessment were not performed. [2]

Optional coordinates and profile fields should be minimized and protected. GPS watching is opt-in and cleaned up on stopping/navigation. The supplied material does not establish approved retention periods, controller contact, user-data export/deletion procedure or external provider agreements. These are unresolved operational/policy requirements. The Philippine Data Privacy Act is legal context for a qualified institutional review, not a certification that this implementation complies. [11]

SRS 4 Verification

Each functional requirement has a TM-TC procedure and a manual/design mapping. Twenty TM-QT quality procedures and twenty TM-XT focused negative/boundary procedures supplement them. All 133 system/quality procedures are Not executed in this task; they are not retroactively marked passed from related unit tests. Current EX-01 and EX-02 record 152 frontend and 521 backend isolated passes. EX-03 records anonymous captures only. Historical evidence remains labeled by date and original report. Acceptance requires the unexecuted persistence, provider, permission, operational and user tasks to be run with authorized disposable/staging data. [6][7]

Software Design Description

Version 2.0. This design describes the inspected implementation and the interfaces needed to understand its limits. Source paths and symbols in the inventory locate each feature; retained schema and API-only compatibility are explicitly identified.

SDD 1 Introduction

This revision documents the inspected working files on 15 October 2026. “Complete” in the inventory means a code path was located across its applicable layers; it does not mean end-to-end acceptance passed. Static inspection, historical reports, current isolated tests and unexecuted acceptance procedures have separate evidence records. [1][2][6][7]

SDD 2 System architecture

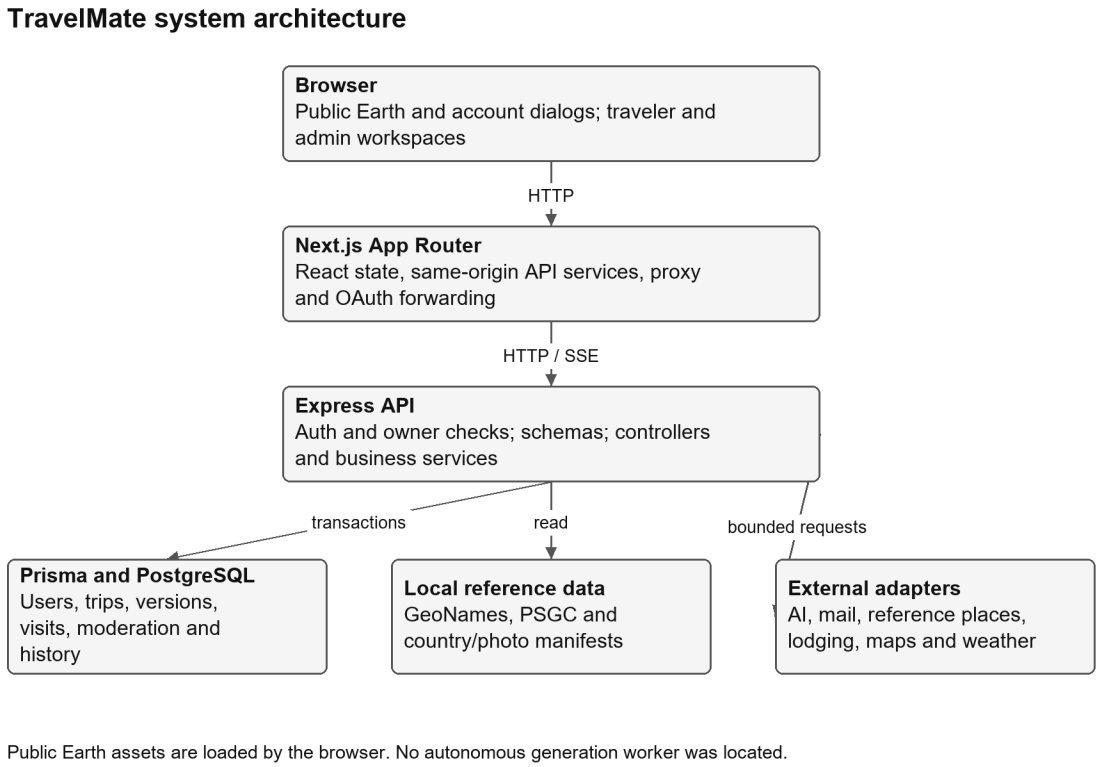


Figure 1 Layered request and persistence architecture from the inspected frontend and backend [1][2]

The browser renders Next.js App Router pages, account dialogs and stateful traveler/admin workspaces. The Earth canvas uses Three.js with geometric country hit testing; Leaflet handles saved-trip maps. Frontend services send same-origin JSON requests through the Next API origin/proxy layer to Express routes. Controllers perform schema, role, owner and state validation, services apply business rules, and repository/Prisma operations transact with PostgreSQL. Read-only local geography and external provider adapters supply reference evidence. No separate autonomous generation worker was located. [1][2]

Table 11 Declared technology baseline

Layer	Inspected dependency declarations	Interpretation
Frontend	Next 16.3.8; React 19.2.4; TypeScript ^5; Tailwind ^4; Three 0.186.1; Leaflet 1.9.4; GSAP ^3.15.0.	package.json declarations; caret ranges are not claimed as installed exact versions.
Backend	Node >=22.13; Express ^5.2.1; Prisma/client/adapter-pg ^7.10.0; pg ^8.23.0; Zod ^4.6.1.	ES modules; generated Prisma client; relational PostgreSQL.
Identity/mail	google-auth-library ^11.1.0; nodemailer ^10.0.15; server crypto.	Google optional; Resend production email, explicit Gmail development alternative.
Verification	Node test runner; Playwright ^1.63.0; ESLint ^9; CI with disposable Postgres.	Definitions do not prove a pipeline ran this task.

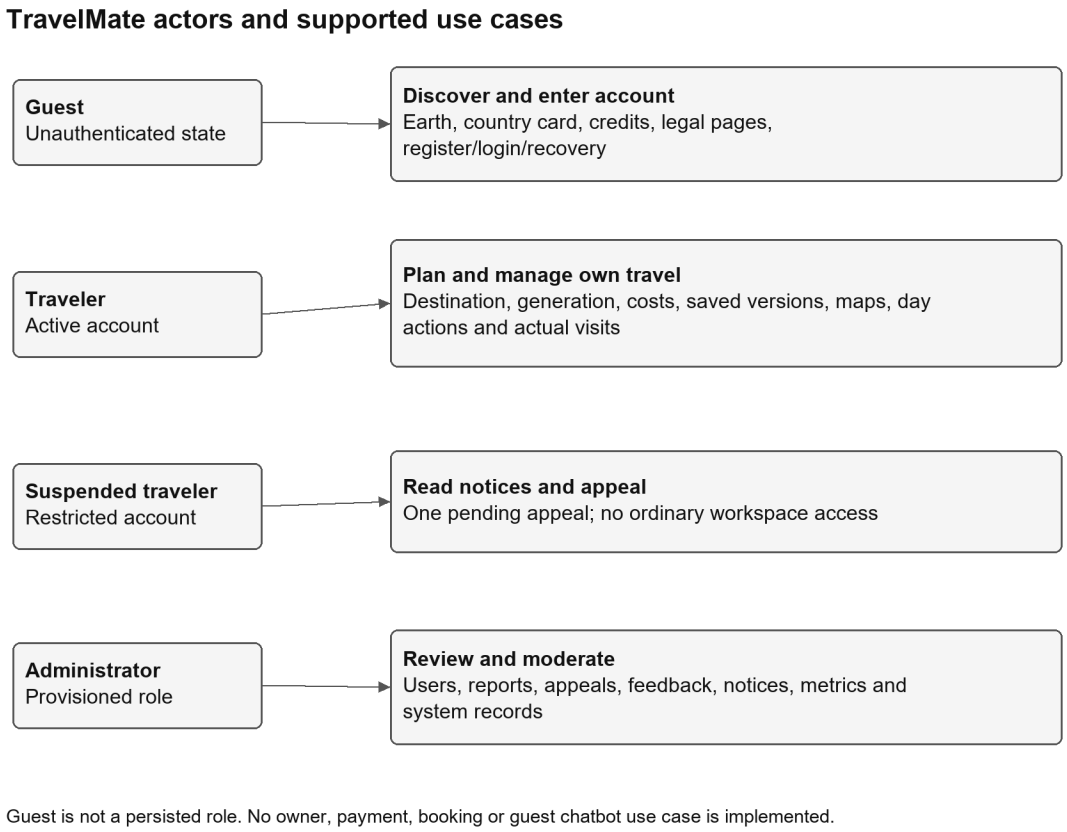


Figure 2 Actor and use case groups for the current supported interfaces [1][2]

SDD 3 Database design

SDD 3.1 Entity relationships

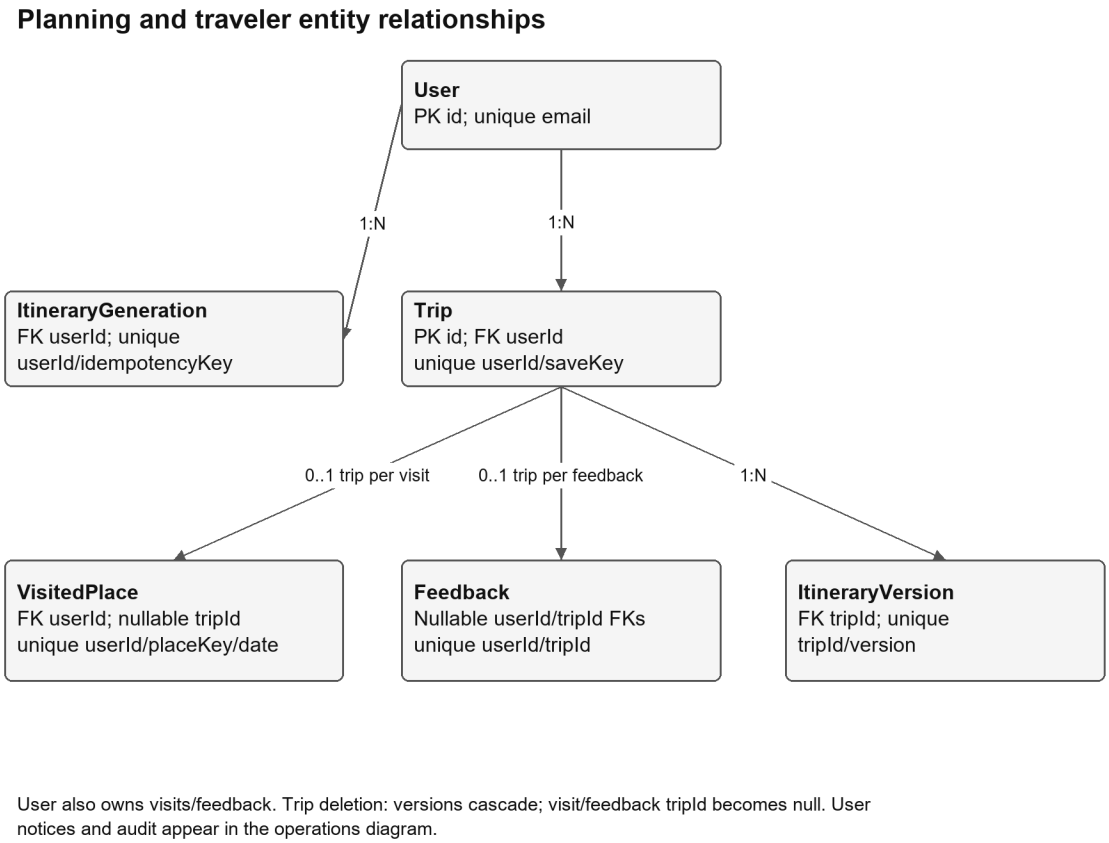


Figure 3 Active planning and traveler data relationships using declared foreign keys [2]

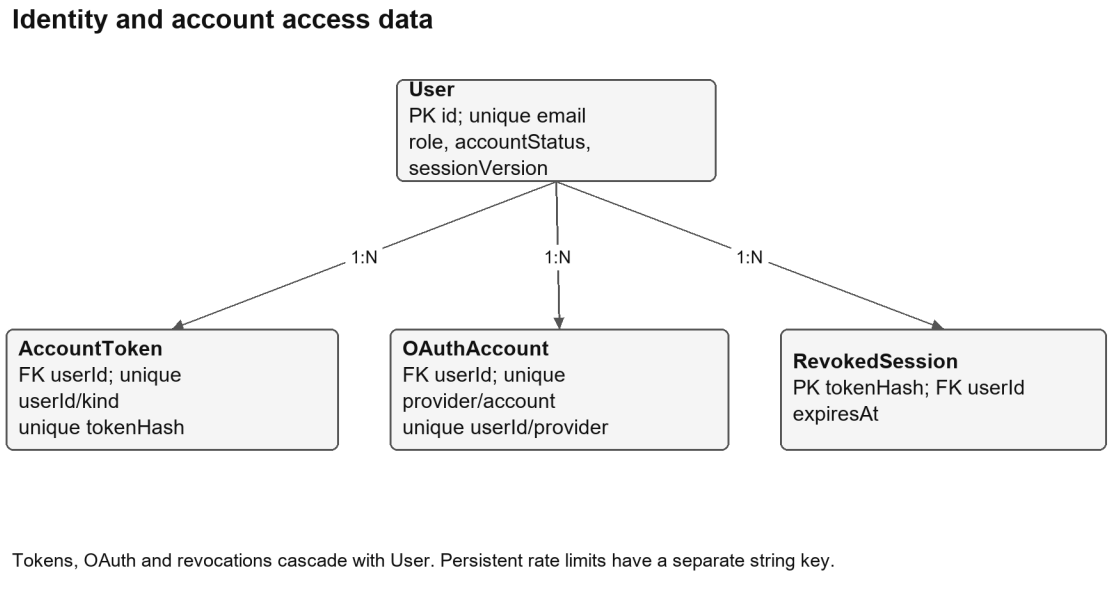


Figure 4 Identity and access support [2]

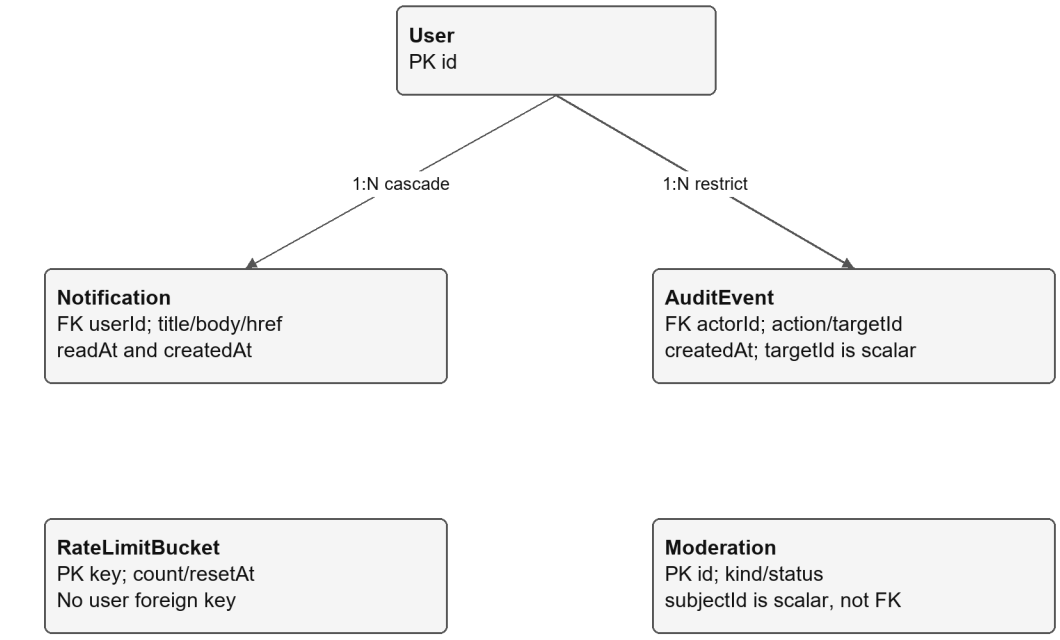


Figure 5 Notification, audit and moderation relations; scalar identifiers are not foreign keys [2]

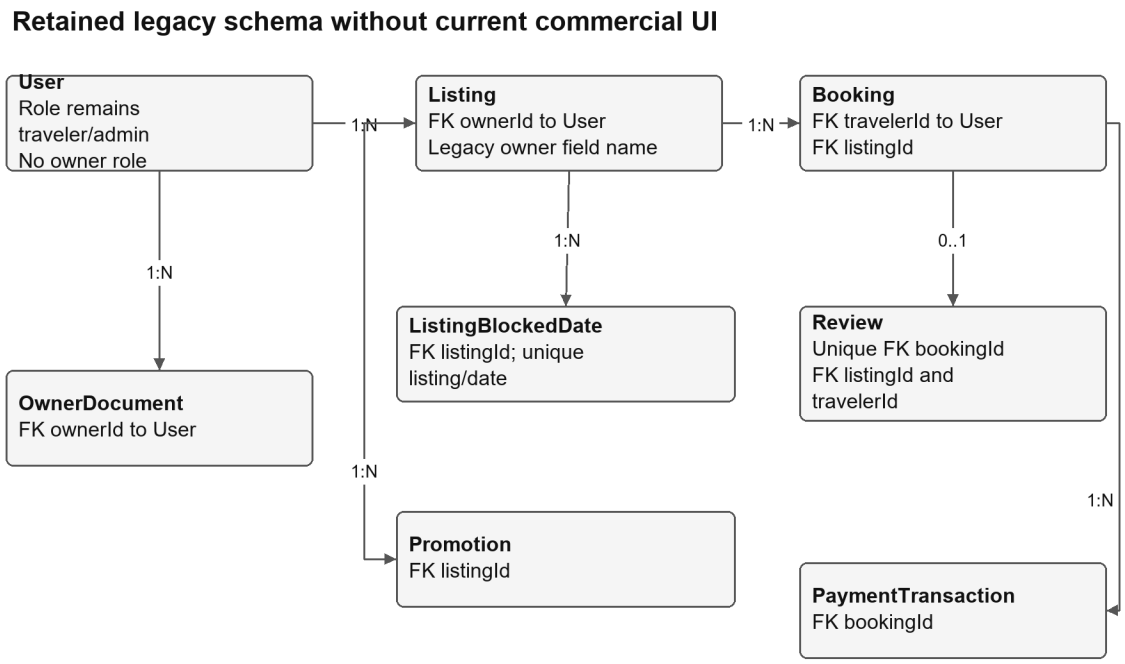


Figure 6 Retained marketplace relations with no current owner or payment workflow [2]

There are twenty Prisma models. Foreign keys, nullable relations and deletion behavior are taken from schema.prisma, not inferred from labels. Moderation.subjectId and AuditEvent.targetId are scalar identifiers, not declared foreign keys. AuditEvent.actorId references User with Restrict semantics. Moderation's pending-appeal uniqueness is strengthened by a partial SQL index in migrations. Itinerary details, activity history, ground-truth proof and pricing evidence remain nested JSON rather than separate relational activity rows. [2]

Table 12 Entity register and current responsibility

Entity	Use and important keys
User	Email unique; traveler/admin role; emailVerified separate from legacy profileStatus; accountStatus and sessionVersion.
AccountToken	Account/kind unique; tokenHash unique; expiry/attempt count and consumed state.
OAuthAccount	Unique provider/account and user/provider pairs; User cascade.

Entity	Use and important keys
RevokedSession	tokenHash primary key; expiry; User cascade.
RateLimitBucket	String key primary key, count/resetAt; no User foreign key.
Trip	Owner, group budget Decimal(12,2), dates, currency(3), coords Decimal(9,6), party, JSON itinerary/weather, status/completedAt/archivedAt; unique user/saveKey.
ItineraryGeneration	Owner/idempotencyKey unique; requestHash, attemptId, status/response/error; destination varchar(120).
ItineraryVersion	Trip/version unique; kind, snapshot JSON, weather, createdAt; Trip cascade.
VisitedPlace	User/placeKey/date unique; optional Trip SetNull; destination/place/media provenance.
Feedback	Nullable User/Trip SetNull; user/trip unique; rating/category; sentiment default not_analyzed.
Notification	Owned title/body/href/readAt and createdAt; User cascade.
AuditEvent	Actor FK Restrict; action and scalar targetId with createdAt; no immutability mechanism claimed.
Moderation	Report/appeal kind; scalar subjectId, details, status; pending appeal partial uniqueness in SQL.
Listing	Legacy owner-named User FK; price/capacity/status and location/media.
Booking	Legacy traveler User and listing FKs; guest/night/amount/status/date fields.
ListingBlockedDate	Legacy listing/date unique; Listing cascade.
Promotion	Legacy listing FK; percent/date/active.
Review	Legacy unique booking FK plus listing/traveler FKs and rating.
PaymentTransaction	Legacy booking FK, kind/status/amount; no accessible payment settlement workflow.
OwnerDocument	Legacy owner-named User FK and file/status; no current owner role.

SDD 3.2 Constraints and migrations

Twenty-four migration directories were inspected as schema history; their presence does not prove they were applied to any live database. SQL checks include bounded trust score, nonnegative listing/booking amounts and capacity rules, trip budget greater than zero, travelers 1-20, ordered dates within a thirty-one-day inclusive span and coordinate ranges. Later role migration converts legacy owner accounts to traveler and increments session version. Unique constraints cover generation/save keys, itinerary version numbers, visited identity/date and feedback user/trip. The schema field register preserves every declared model field and constraint for review. [2]

SDD 3.3 Transactions and concurrency

Generation uses owner/idempotency key plus normalized request hash. A completed request replays its saved response; mismatched input conflicts; pending attempts younger than ten minutes block duplicates; failed/stale attempts obtain a new attemptId and compare-and-set fencing rejects late completion. This limits duplicate records and stale overwrites but cannot guarantee that remote computation stopped or was unbilled. [2]

Trip saves validate signed grounding and use owner/saveKey identity and request hash to avoid duplicate autosaves. Version writes transact with the trip. Restores use complete request envelopes, create a new version, and lock the current row; older snapshots missing required context may be blocked. Managed activities use a revision fingerprint, stable activity IDs, explicit confirmation and row locking. Batch visits verify candidate fingerprints and use a PostgreSQL advisory lock plus de-duplication. Appeals and decisions lock account/state and preserve transactionally related notices/audit. [2]

SDD 4 Module and interface design

SDD 4.1 Public discovery

The discovery canvas loads local country geometry and a code-keyed curated photo manifest. A ray/coordinate hit resolves a geographic record, including small-area proximity targets, and the same record owns its boundary, English name, flag and async photo request. Animation-frame hit refresh maintains the identity beneath a stationary pointer as Earth turns. The overlay is bounded inside the globe stage, uses pointer-events none, and clears on exit/drag conditions without moving layout. The permanent destination panel, flight arcs, markers and bottom caption are removed; attribution is exposed in the footer credits. Coverage is 264/265 geographic areas: all 249 ISO entries plus Kosovo are covered, and Bir Tawil is an explicit non-national missing-photo record. Flags follow the available ISO/code mapping and are not invented for Antarctica/non-national areas. [1][2]

Table 13 Public discovery feature implementation map

Feature and requirement	Source flow and persistence
TM-F-001 / TM-FR-001 Landing and navigation	travelmate-frontend-new/components/features/home/TravelMateLanding.tsx:109 (TravelMateLanding); None; public presentation
TM-F-002 / TM-FR-002 Interactive Earth controls	travelmate-frontend-new/components/features/home/TravelDiscoverySection.tsx:32 (TravelDiscoverySection) -> travelmate-frontend-new/components/features/home/DiscoveryGlobeCanvas.tsx:17 (DiscoveryGlobeCanvas); Static geographic assets
TM-F-003 / TM-FR-003 Country hit detection and border identity	travelmate-frontend-new/lib/discovery-countries.ts:54 (countryAt) -> travelmate-frontend-new/lib/discovery-globe.ts:18 (countryBoundaryPositions) -> travelmate-frontend-new/components/features/home/DiscoveryGlobeCanvas.tsx:81 (setOutline); 265 local geographic records
TM-F-004 / TM-FR-004 Country destination photo and identity card	travelmate-frontend-new/lib/discovery-country-photos.ts:22 (loadCountryPhoto) -> travelmate-frontend-new/lib/discovery-country-flags.ts:4 (discoveryCountryFlag) -> travelmate-frontend-new/components/features/home/DiscoveryGlobeCanvas.tsx:1; Local country-photos.v2.json; 500 WebP assets

Feature and requirement	Source flow and persistence
TM-F-005 / TM-FR-005 Image and map credits	travelmate-frontend-new/components/features/home/DiscoveryCredits.tsx:6 (DiscoveryCredits); Public attribution manifests
TM-F-006 / TM-FR-006 Privacy and terms pages	travelmate-frontend-new/app/privacy/page.tsx:1 -> travelmate-frontend-new/app/terms/page.tsx:1; None

SDD 4.2 Accounts and access

Account action dispatch separates register, login, verify/resend, forgot/reset and logout. Password hashing is salted scryptSync with a 64-byte output; sessions sign a base64url payload with HMAC-SHA256. Current-user middleware consults the database, including revocations and sessionVersion. Google uses state/nonce/PKCE with a ten-minute flow cookie, token audience/nonce and verified-email checks; authoritative Gmail/hosted-domain identity permits guarded linking, while unsafe email association requires local verification. Reset consumes a token and increments sessionVersion. Resend/mail failures never silently switch providers. A suspended account uses the dedicated appeals view. [1][2]

Table 14 Accounts and access feature implementation map

Feature and requirement	Source flow and persistence
TM-F-007 / TM-FR-007 Traveler registration	travelmate-frontend-new/components/features/home/TravelMateLanding.tsx:46 (register) -> travelmate-backend-api/src/controllers/auth-controller.ts:15 (register) -> travelmate-backend-api/src/services/auth/auth-service.ts:60 (registerUser) -> travelmate-backend-api/src/schemas/auth/registration-schema.ts:3 (registrationDetails); User, AccountToken; email unique; transaction
TM-F-008 / TM-FR-008 Email verification	travelmate-frontend-new/components/features/home/TravelMateLanding.tsx:328 (verify-email) -> travelmate-backend-api/src/controllers/auth-controller.ts:44 (verify-email) -> travelmate-backend-api/src/services/auth/auth-service.ts:104 (verifyEmail); AccountToken, User; single-use transaction

Feature and requirement	Source flow and persistence
TM-F-009 / TM-FR-009 Resend verification	travelmate-frontend-new/components/features/home/TravelMateLanding.tsx:367 (resend-verification) -> travelmate-backend-api/src/controllers/auth-controller.ts:44 (resend-verification) -> travelmate-backend-api/src/services/auth/auth-service.ts:176 (reissueVerificationCode); AccountToken, RateLimitBucket
TM-F-010 / TM-FR-010 Password sign-in and role redirect	travelmate-frontend-new/components/features/home/TravelMateLanding.tsx:48 (login) -> travelmate-backend-api/src/controllers/auth-controller.ts:44 (login) -> travelmate-backend-api/src/services/auth/auth-service.ts:54 (authenticateUser); User, AuditEvent, RateLimitBucket
TM-F-011 / TM-FR-011 Google OAuth	travelmate-frontend-new/app/api/auth/oauth/google/route.ts:1 -> travelmate-frontend-new/app/api/auth/oauth/google/callback/route.ts:1 -> travelmate-backend-api/src/controllers/google-oauth-controller.ts:55 (GOOGLE_CALLBACK) -> travelmate-backend-api/src/services/auth/google-oauth-service.ts:160 (findOrCreateGoogleUser); User, OAuthAccount; provider identity unique
TM-F-012 / TM-FR-012 Authoritative sessions and access control	travelmate-frontend-new/proxy.ts:4 (proxy) -> travelmate-frontend-new/services/session.service.ts:1 -> travelmate-backend-api/src/middlewares/auth-middleware.ts:53 (currentUser); User, RevokedSession
TM-F-013 / TM-FR-013 Logout and local cleanup	travelmate-frontend-new/services/session.service.ts:4 (signOut) -> travelmate-backend-api/src/controllers/auth-controller.ts:44 (logout) -> travelmate-backend-api/src/middlewares/auth-middleware.ts:70 (revokeRequestSession); RevokedSession, best-effort AuditEvent
TM-F-014 / TM-FR-014 Forgot/reset password	travelmate-frontend-new/components/features/home/TravelMateLanding.tsx:333 (forgot-password) -> travelmate-backend-api/src/controllers/auth-controller.ts:44 (reset-password) -> travelmate-backend-api/src/services/auth/auth-service.ts:184 (resetPassword); AccountToken, User.sessionVersion
TM-F-015 / TM-FR-015 Suspension appeal access	travelmate-frontend-new/app/account/appeal/page.tsx:1 -> travelmate-backend-api/src/controllers/appeal-controller.ts:1 -> travelmate-backend-api/src/services/platform/appeal-service.ts:12 (submitAppeal); Moderation(kind=appeal), Notification, AuditEvent, User lock

SDD 4.3 Destination selection

Destination search uses local GeoNames SQLite when available, fallback compressed records, and Philippine PSGC hierarchy. The SQLite metadata records 5,721,309 records across 250 source-country entries; the bounded fallback has 132,826 records and 252 country/territory entries. These source counts do not prove exhaustive current administrative coverage. Parent/type/country and source IDs distinguish namesakes. Country/admin-area selections must resolve to a locality for generation. Nominatim is invoked for explicit manual resolution with pacing, not unrestricted keystroke autocomplete. Multi-area comparisons accept one to six choices and remain source-backed. [1][2]

Table 15 Destination selection feature implementation map

Feature and requirement	Source flow and persistence
TM-F-016 / TM-FR-016 Worldwide direct search	travelmate-frontend-new/components/features/traveler/DestinationSelector.tsx:1 -> travelmate-backend-api/src/controllers/locations-controller.ts:12 (GET) -> travelmate-backend-api/src/services/destination/global-location-search.ts:1 -> travelmate-backend-api/src/services/destination/global-catalogue.ts:1; Read-only public GeoNames SQLite + compressed index/PSGC files; no Trip until save
TM-F-017 / TM-FR-017 Country and administrative browsing	travelmate-frontend-new/components/features/traveler/DestinationSelector.tsx:1 -> travelmate-frontend-new/components/features/traveler/DestinationAreaPicker.tsx:1 -> travelmate-backend-api/src/services/destination/global-catalogue.ts:84 (searchCatalogue) -> travelmate-backend-api/src/services/destination/philippine-catalogue.ts:1; Read-only catalogue and PSGC administrative IDs
TM-F-018 / TM-FR-018 Manual resolution and clear	travelmate-frontend-new/components/features/traveler/DestinationSelector.tsx:1 -> travelmate-backend-api/src/controllers/locations-controller.ts:57 (GET_RESOLVE) -> travelmate-backend-api/src/services/destination/location-resolution.ts:1; No personal write; bounded source caches
TM-F-019 / TM-FR-019 Area suggestions and comparison	travelmate-frontend-new/components/features/traveler/DestinationAreaPicker.tsx:1 -> travelmate-frontend-new/hooks/use-destination-areas.ts:1 -> travelmate-backend-api/src/controllers/locations-controller.ts:88 (GET_AREA_COMPARISON) -> travelmate-backend-api/src/services/destination/area-comparison.ts:1; Public geography/cache only

Feature and requirement	Source flow and persistence
TM-F-020 / TM-FR-020 Starting location and optional geolocation	travelmate-frontend-new/hooks/use-starting-location.ts:1 -> travelmate-frontend-new/components/features/traveler/StartingLocationField.tsx:1 -> travelmate-backend-api/src/controllers/locations-controller.ts:35 (GET_CURRENT) -> travelmate-backend-api/src/services/destination/location-search-service.ts:170 (reverseGeocodeLocation); User-scoped browser origin preference; no live-position database storage in this flow
TM-F-021 / TM-FR-021 Currency detection and fallback	travelmate-frontend-new/hooks/use-destination-planning.ts:1 -> travelmate-frontend-new/lib/domain.ts:1 -> travelmate-backend-api/src/controllers/destination-context-controller.ts:9 (GET) -> travelmate-backend-api/src/services/destination/destination-context-service.ts:58 (resolveDestinationContext); Trip.currency VARCHAR(3) after save
TM-F-022 / TM-FR-022 Reference-currency equivalents	travelmate-frontend-new/hooks/use-exchange-rates.ts:1 -> travelmate-frontend-new/services/exchange-rate-client.ts:1 -> travelmate-frontend-new/components/common/ReferenceAmount.tsx:1 -> travelmate-backend-api/src/services/currency/exchange-rate-service.ts:49 (getExchangeRate); In-process bounded cache; browser display preference
TM-F-023 / TM-FR-023 Destination transport guidance	travelmate-frontend-new/hooks/use-destination-planning.ts:1 -> travelmate-backend-api/src/services/destination/destination-context-service.ts:36 (transportFor); Saved itinerary.preferences.transportation
TM-F-024 / TM-FR-024 Destination inspiration carousel	travelmate-frontend-new/components/features/traveler/DestinationSpotCarousel.tsx:1 -> travelmate-frontend-new/lib/destination-spots.ts:1; Static source-backed spot data; trip only on save

SDD 4.4 Trip planning and enrichment

The trip planner confirms location identity, currency, dates, party and budget, with optional origin, interests, stay and explicit expenses. New generation uses Balanced defaults. It gathers physical catalog venues, obtains constrained AI ordering where available, independently validates it, and reconciles complete/incomplete days deterministically. The response carries schedule, grounding, costs and provider/fallback notices. Autosave is a separate persistence step, and an owner-scoped local draft is retained on save failure. Accommodation is recommendation-only in the current UI, with optional manually retained stay and no booking promise. [1][2]

Table 16 Trip planning and enrichment feature implementation map

Feature and requirement	Source flow and persistence
TM-F-025 / TM-FR-025 Required trip brief and bounds	travelmate-frontend-new/components/features/traveler/TravelerDashboard.tsx:886 (generationRequirements) -> travelmate-backend-api/src/controllers/itinerary-controller.ts:450 (POST); Generation record then Trip only after save
TM-F-026 / TM-FR-026 Preferences with Balanced generation policy	travelmate-frontend-new/components/features/traveler/TravelerDashboard.tsx:139 (preferredActivities) -> travelmate-backend-api/src/services/itinerary/travel-preferences.ts:5 (generationDefaults) -> travelmate-backend-api/src/controllers/itinerary-controller.ts:756 (preferenceResult); Trip itinerary.preferences JSON
TM-F-027 / TM-FR-027 Optional accommodation suggestions	travelmate-frontend-new/components/features/traveler/AccommodationPicker.tsx:1 -> travelmate-frontend-new/hooks/use-destination-planning.ts:1 -> travelmate-backend-api/src/controllers/accommodations-controller.ts:37 (suggestionsOnly) -> travelmate-backend-api/src/services/destination/accommodation-places.ts:1; Trip itinerary.plannedAccommodation when saved
TM-F-028 / TM-FR-028 Manual stay and retained property details	travelmate-frontend-new/components/features/traveler/AccommodationPicker.tsx:1 -> travelmate-frontend-new/components/features/traveler/SavedAccommodation.tsx:1 -> travelmate-backend-api/src/services/destination/planned-accommodation.ts:1; Trip.itinerary plannedAccommodation JSON

Feature and requirement	Source flow and persistence
TM-F-029 / TM-FR-029 Expense assumptions	travelmate-frontend-new/components/features/traveler/ExpenseAssumptions.tsx:1 -> travelmate-frontend-new/lib/expense-assumptions.ts:1 -> travelmate-backend-api/src/controllers/itinerary-controller.ts:753 (minimumEnteredCosts) -> travelmate-backend-api/src/services/itinerary/grounded-plan.ts:29 (calculateExpense); Trip.itinerary grounding/priceEvidence JSON
TM-F-030 / TM-FR-030 Optional pre-trip expenses	travelmate-frontend-new/components/features/traveler/TravelerDashboard.tsx:179 (preTripCosts) -> travelmate-backend-api/src/services/budget/pre-trip-cost-service.ts:45 (normalizePreTripCosts); Trip.itinerary selectedTravelCosts.preTrip JSON
TM-F-031 / TM-FR-031 Place grounding and geographic validation	travelmate-frontend-new/components/features/traveler/GroundingSummary.tsx:1 -> travelmate-backend-api/src/services/itinerary/grounded-places.ts:374 (loadGroundedPlaces) -> travelmate-backend-api/src/services/itinerary/place-boundaries.ts:1 -> travelmate-backend-api/src/services/itinerary/place-search-policy.ts:1; Source caches; signed evidence saved in Trip JSON
TM-F-032 / TM-FR-032 AI structured schedule and repair	travelmate-frontend-new/services/api.service.ts:129 (fetchTripData) -> travelmate-backend-api/src/services/ai/provider.ts:1 -> travelmate-backend-api/src/services/ai/grounded-schedule.ts:46 (generateGroundedSchedule); ItineraryGeneration response/status; no AI-authored price fields
TM-F-033 / TM-FR-033 Deterministic full-day scheduling	travelmate-frontend-new/components/features/traveler/DayActivityTimeline.tsx:1 -> travelmate-backend-api/src/services/itinerary/generated-day-scheduler.ts:43 (completeGroundedSchedule); Trip.itinerary day scheduleBlocks/planningStatus
TM-F-034 / TM-FR-034 Catalog fallback and incomplete results	travelmate-frontend-new/components/features/traveler/TravelerDashboard.tsx:133 (providerMessage) -> travelmate-backend-api/src/controllers/itinerary-controller.ts:843 (AI scheduling unavailable); ItineraryGeneration, saved Trip if accepted by save validation
TM-F-035 / TM-FR-035 Generation idempotency and fencing	travelmate-frontend-new/components/features/traveler/TravelerDashboard.tsx:190 (generationRequestRef) -> travelmate-backend-api/src/controllers/itinerary-controller.ts:537 (beginGeneration); ItineraryGeneration unique(userId,idempotencyKey), attemptId

Feature and requirement	Source flow and persistence
TM-F-036 / TM-FR-036 Automatic save after generation	travelmate-frontend-new/components/features/traveler/TravelerDashboard.tsx:521 (It was saved automatically) -> travelmate-backend-api/src/services/platform/platform-service.ts:183 (saveTrip) -> travelmate-backend-api/src/services/platform/save-idempotency.ts:1; Trip, ItineraryVersion(saved/regenerated), AuditEvent
TM-F-037 / TM-FR-037 Activity photographs and provenance	travelmate-frontend-new/components/common/ActivityPhoto.tsx:1 -> travelmate-frontend-new/components/common/ActivityPhotoCredit.tsx:1 -> travelmate-frontend-new/services/activity-photo.service.ts:1 -> travelmate-backend-api/src/services/itinerary/activity-photo-fallback.ts:1 -> travelmate-backend-api/src/services/itinerary/saved-activity-photos.ts:1 -> travelmate-backend-api/src/controllers/activity-photo-controller.ts:1; Trip JSON imageUrl/imageAttribution; optimistic media-only merges
TM-F-038 / TM-FR-038 Date-matched weather	travelmate-frontend-new/lib/conditions.ts:1 -> travelmate-frontend-new/services/provider-response.ts:1 -> travelmate-backend-api/src/controllers/weather-controller.ts:1 -> travelmate-backend-api/src/services/weather/providers.ts:1 -> travelmate-backend-api/src/services/weather/weather-domain.ts:51 (weatherForTripDates); Trip.weather advisory snapshot
TM-F-039 / TM-FR-039 Crowd estimates	travelmate-frontend-new/components/features/traveler/DayActivityTimeline.tsx:1 -> travelmate-backend-api/src/services/crowd/crowd-service.ts:13 (estimateCrowd); Saved day advisory metadata
TM-F-040 / TM-FR-040 Advisory refresh without overwriting edits	travelmate-frontend-new/components/features/traveler/TravelerDashboard.tsx:531 (refreshConditions) -> travelmate-frontend-new/lib/conditions.ts:4 (applyConditionSnapshot) -> travelmate-backend-api/src/controllers/weather-controller.ts:1; Browser draft; persisted snapshot only through supported save path

SDD 4.5 Saved trips and traveler records

Saved trip lists are owned and paginated, and opening a trip renders its stored snapshot without inventing missing visits or regeneration. Manual edits validate and recalculate costs; place identity changes discard stale place media/evidence. Regeneration appends a version; restoring a prior version restores the full available brief. Archive retains the prior completion state. Complete/reopen does not populate visited history. Deletion cascades versions while SetNull preserves linked visits/feedback. Visited recording has a distinct actual-date confirmation, and feedback/report/notice/profile operations are scoped and validated. [1][2]

Table 17 Saved trips and traveler records feature implementation map

Feature and requirement	Source flow and persistence
TM-F-041 / TM-FR-041 Saved-trip list and pagination	travelmate-frontend-new/components/features/traveler/SavedTripLifecycle.tsx:1 -> travelmate-backend-api/src/controllers/platform-controller.ts:24 (GET) -> travelmate-backend-api/src/repositories/trip-workspace-repository.ts:25 (readTripPage); Trip; pages of 20 summary records
TM-F-042 / TM-FR-042 Reopen, day navigation and return	travelmate-frontend-new/hooks/use-itinerary-navigation.ts:1 -> travelmate-frontend-new/components/features/traveler/ItineraryWorkspace.tsx:1 -> travelmate-backend-api/src/repositories/trip-workspace-repository.ts:47 (readTripDetails); Read owned Trip and optional version metadata
TM-F-043 / TM-FR-043 Activity editing	travelmate-frontend-new/components/features/traveler/TravelerItineraryDialogs.tsx:1 -> travelmate-frontend-new/lib/itinerary-editor.ts:1 -> travelmate-backend-api/src/services/itinerary/itinerary-edit-service.ts:92 (applyManualItineraryChanges); Trip JSON + manual_edit version on persisted save
TM-F-044 / TM-FR-044 Add, remove and reorder activities	travelmate-frontend-new/lib/itinerary-editor.ts:1 -> travelmate-frontend-new/components/features/traveler/ItineraryWorkspace.tsx:1 -> travelmate-backend-api/src/services/itinerary/itinerary-edit-service.ts:1; Trip JSON on save; versions preserve earlier snapshot

Feature and requirement	Source flow and persistence
TM-F-045 / TM-FR-045 Save manual changes and recover drafts	travelmate-frontend-new/lib/itinerary-draft.ts:1 -> travelmate-frontend-new/components/features/traveler/RestoreDraftDialog.tsx:1 -> travelmate-frontend-new/components/features/traveler/UnsavedTripDialog.tsx:1 -> travelmate-backend-api/src/services/platform/platform-service.ts:287 (updateTripItinerary); Trip, ItineraryVersion(manual_edit), AuditEvent; user-scoped localStorage
TM-F-046 / TM-FR-046 Regenerate existing trip	travelmate-frontend-new/components/features/traveler/TravelerDashboard.tsx:159 (regenerationTargetId) -> travelmate-backend-api/src/services/platform/platform-service.ts:243 (updateTrip); Trip, ItineraryVersion(regenerated), AuditEvent
TM-F-047 / TM-FR-047 Version history and whole-version restoration	travelmate-frontend-new/components/features/traveler/SavedTripLifecycle.tsx:41 (versionLabels) -> travelmate-backend-api/src/services/platform/platform-service.ts:348 (restoreItineraryVersion) -> travelmate-backend-api/src/services/itinerary/version-snapshot.ts:1; ItineraryVersion unique(tripId,version), row lock, Trip update
TM-F-048 / TM-FR-048 Archive and restore trip	travelmate-frontend-new/components/features/traveler/SavedTripLifecycle.tsx:1 -> travelmate-backend-api/src/services/platform/platform-service.ts:327 (setTripLifecycleStatus); Trip.status, archivedAt, completedAt; AuditEvent
TM-F-049 / TM-FR-049 Complete and reopen trip	travelmate-frontend-new/components/features/traveler/SavedTripLifecycle.tsx:1 -> travelmate-backend-api/src/services/platform/platform-service.ts:327 (setTripLifecycleStatus); Trip.status/completedAt; AuditEvent
TM-F-050 / TM-FR-050 Delete owned trip	travelmate-frontend-new/components/features/traveler/DeleteTripDialog.tsx:1 -> travelmate-backend-api/src/services/platform/platform-service.ts:309 (deleteTrip); Trip delete; ItineraryVersion cascade; VisitedPlace.tripId and Feedback.itineraryId SetNull; AuditEvent retained

Feature and requirement	Source flow and persistence
TM-F-051 / TM-FR-051 Saved cover photos and stay details	travelmate-frontend-new/components/features/traveler/SavedTripCoverPhoto.tsx:1 -> travelmate-frontend-new/lib/saved-trip-photo-selection.ts:1 -> travelmate-frontend-new/components/features/traveler/SavedAccommodation.tsx:1 -> travelmate-backend-api/src/services/destination/destination-cover-photo.ts:1 -> travelmate-backend-api/src/controllers/destination-context-controller.ts:45 (GET_PHOTO); Stored Trip media and destination lookup cache
TM-F-052 / TM-FR-052 Manual visited-place recording	travelmate-frontend-new/components/features/traveler/VisitedPlaces.tsx:1 -> travelmate-backend-api/src/services/platform/visited-place-service.ts:89 (recordVisit); VisitedPlace unique(userId,placeKey,visitDate)
TM-F-053 / TM-FR-053 Record places from a saved trip	travelmate-frontend-new/components/features/traveler/RecordTripVisitsDialog.tsx:1 -> travelmate-backend-api/src/services/platform/trip-visit-candidates.ts:23 (buildTripVisitCandidates) -> travelmate-backend-api/src/services/platform/visited-place-service.ts:66 (recordTripVisits); VisitedPlace, owned Trip; advisory lock and transaction
TM-F-054 / TM-FR-054 Visited history search and removal	travelmate-frontend-new/components/features/traveler/VisitedPlaces.tsx:1 -> travelmate-backend-api/src/services/platform/visited-place-service.ts:141 (readVisitedPlaces) -> travelmate-backend-api/src/services/platform/visited-place-service.ts:135 (removeVisit); VisitedPlace; 20 records/page
TM-F-055 / TM-FR-055 Itinerary feedback and updates	travelmate-frontend-new/components/features/traveler/TripFeedbackForm.tsx:1 -> travelmate-backend-api/src/controllers/feedback-controller.ts:22 (POST) -> travelmate-backend-api/src/services/platform/feedback-service.ts:60 (submitFeedback); Feedback unique(userId,itineraryId), Notification, AuditEvent
TM-F-056 / TM-FR-056 Traveler problem reports	travelmate-frontend-new/components/features/traveler/ReportProblem.tsx:1 -> travelmate-backend-api/src/services/platform/report-service.ts:15 (submitReport); Moderation(kind=report), Notification, AuditEvent

Feature and requirement	Source flow and persistence
TM-F-057 / TM-FR-057 Traveler notifications	travelmate-frontend-new/components/features/traveler/TravelerNotifications.tsx:1 -> travelmate-backend-api/src/controllers/platform-controller.ts:83 (GET_NOTIFICATIONS) -> travelmate-backend-api/src/services/platform/platform-service.ts:386 (readNotification); Notification.userId/readAt; latest 50
TM-F-058 / TM-FR-058 Traveler profile and contact settings	travelmate-frontend-new/components/features/traveler/TravelerAccountCenter.tsx:1 -> travelmate-backend-api/src/controllers/profile-controller.ts:27 (PATCH) -> travelmate-backend-api/src/services/profile/profile-service.ts:17 (validateProfileUpdate); User, AuditEvent transaction

SDD 4.6 Saved trip maps

Saved navigation uses an owned trip manifest with verified/unresolved stops. Exact-stop disambiguation precedes routing. Day routing preserves itinerary order, excludes skipped entries and supports walk/cycle/drive via OSRM; two resolved stops are needed. Provider pacing, 15-second deadline, five-minute cache and payload/stop/snap limits constrain requests. Explicit GPS watch is browser memory state and cleans up on stop/navigation. Per-stop Open-Meteo weather is constrained to provider forecast coverage and quotas. Legacy standalone GPS URLs redirect into Saved trips with valid trip/day context. [1][2]

Table 18 Saved trip maps feature implementation map

Feature and requirement	Source flow and persistence
TM-F-059 / TM-FR-059 Owned saved-trip map and timeline	travelmate-frontend-new/components/features/traveler/SavedTripMapDialog.tsx:1 -> travelmate-frontend-new/components/features/traveler/TripDayMapCanvas.tsx:1 -> travelmate-backend-api/src/controllers/trip-navigation-controller.ts:26 (GET_TRIP_MAP) -> travelmate-backend-api/src/services/navigation/trip-map-service.ts:153 (loadTripMapManifest); Owned Trip read; client per-trip caches

Feature and requirement	Source flow and persistence
TM-F-060 / TM-FR-060 Exact map-place resolution and photos	travelmate-frontend-new/components/features/traveler/SavedTripMapDialog.tsx:1 -> travelmate-frontend-new/services/trip-navigation.service.ts:1 -> travelmate-backend-api/src/controllers/trip-navigation-controller.ts:32 (POST_TRIP_ACTIVITY) -> travelmate-backend-api/src/services/navigation/trip-map-service.ts:120 (resolveTripMapActivity); Trip read; provider/cache result, not automatic itinerary identity write
TM-F-061 / TM-FR-061 Selected-day ordered route	travelmate-frontend-new/components/features/traveler/SavedTripMapDialog.tsx:1 -> travelmate-backend-api/src/controllers/trip-navigation-controller.ts:41 (POST_TRIP_DAY_ROUTE) -> travelmate-backend-api/src/services/navigation/navigation-service.ts:75 (fetchNavigationRoute); Read Trip; 5-minute route cache
TM-F-062 / TM-FR-062 Location sharing and route guidance	travelmate-frontend-new/hooks/use-trip-location.ts:1 -> travelmate-frontend-new/hooks/use-route-guidance.ts:1 -> travelmate-frontend-new/components/features/traveler/TripLocationMap.tsx:1 -> travelmate-backend-api/src/controllers/navigation-controller.ts:16 (POST_ROUTE) -> travelmate-backend-api/src/services/navigation/navigation-service.ts:1; Live browser position in memory; no location-trace table
TM-F-063 / TM-FR-063 Activity-specific weather	travelmate-frontend-new/services/trip-navigation.service.ts:1 -> travelmate-backend-api/src/controllers/trip-navigation-controller.ts:66 (POST_TRIP_ACTIVITY_WEATHER) -> travelmate-backend-api/src/services/navigation/activity-weather.ts:1; Read Trip; bounded request/cache data
TM-F-064 / TM-FR-064 Retired GPS URL compatibility	travelmate-frontend-new/app/gps/page.tsx:1 -> travelmate-frontend-new/app/dashboard/gps/page.tsx:1 -> travelmate-frontend-new/lib/retired-navigation.ts:1; No standalone GPS records

SDD 4.7 Managed day activities

Managed day actions operate on an active owned saved trip. Complete/skip/restore and bounded edit/reschedule create audited history and preserve cost evidence as appropriate; completion is separate from visited history. Replan My Day may reorder only the remaining permitted IDs/time durations around completed/skipped work and the local remaining window. A preview is not an itinerary/version write; it performs rate-limit/provider work. Acceptance verifies a signed fifteen-minute owner/trip/revision token, explicit confirmation and a fresh locked revision before saving a new version. Canceling leaves the plan unchanged. [1][2]

Table 19 Managed day activities feature implementation map

Feature and requirement	Source flow and persistence
TM-F-065 / TM-FR-065 Confirm completion, skip or restore activity	travelmate-frontend-new/components/features/traveler/SmartDayManager.tsx:1 -> travelmate-backend-api/src/controllers/smart-itinerary-controller.ts:21 (PATCH_ACTIVITY) -> travelmate-backend-api/src/services/itinerary/smart-itinerary-service.ts:53 (applyActivityMutation); Trip JSON status/history, manual_edit version, AuditEvent
TM-F-066 / TM-FR-066 Confirmed activity edit/reschedule	travelmate-frontend-new/components/features/traveler/SmartDayManager.tsx:1 -> travelmate-backend-api/src/services/itinerary/smart-itinerary-service.ts:79 (saveConfirmed) -> travelmate-backend-api/src/services/itinerary/smart-day-planner.ts:68 (validateChangedSchedule); Locked Trip update, version snapshot, AuditEvent
TM-F-067 / TM-FR-067 AI remaining-day preview	travelmate-frontend-new/components/features/traveler/SmartDayManager.tsx:1 -> travelmate-backend-api/src/services/itinerary/smart-itinerary-service.ts:126 (previewManagedDay) -> travelmate-backend-api/src/services/itinerary/smart-day-planner.ts:84 (generateDayProposal); Read Trip/catalogue; 30-second identical preview cache
TM-F-068 / TM-FR-068 Accept or cancel revised day	travelmate-frontend-new/components/features/traveler/SmartDayManager.tsx:1 -> travelmate-backend-api/src/services/itinerary/smart-itinerary-service.ts:158 (acceptManagedDay) -> travelmate-backend-api/src/services/itinerary/smart-itinerary-service.ts:103 (openDayProposal); Trip row lock, ItineraryVersion(manual_edit), AuditEvent

Feature and requirement	Source flow and persistence
TM-F-069 / TM-FR-069 Activity history and legacy compatibility	travelmate-frontend-new/components/features/traveler/SmartDayManager.tsx:1 -> travelmate-backend-api/src/services/itinerary/activity-management.ts:16 (managedItinerary); Optional Trip.itinerary activity id/status/statusChangedAt/managementHistory

SDD 4.8 Administration

The administrator workspace derives overview metrics from stored records, manages non-admin user state and warnings, resolves reports/appeals, searches feedback, and reads notices/system/audit information. Feedback sentiment remains not_analyzed rather than an AI classification. SystemHealth primarily reports configuration presence plus recent records, not provider uptime or active probes. The SSE stream combines local events with five-second database reconciliation, rotates after five minutes, checks current authorization and cleans up listeners/timers on failure or disconnect. [1][2]

Table 20 Administration feature implementation map

Feature and requirement	Source flow and persistence
TM-F-070 / TM-FR-070 Administrator overview	travelmate-frontend-new/components/features/admin/AdminOverview.tsx:1 -> travelmate-backend-api/src/repositories/admin-overview-repository.ts:11 (readAdminOverview); User, ItineraryGeneration, Feedback aggregate reads
TM-F-071 / TM-FR-071 User search and access state	travelmate-frontend-new/components/features/admin/AdminPanels.tsx:19 (UsersView) -> travelmate-frontend-new/components/features/admin/AccountAccessDialog.tsx:1 -> travelmate-backend-api/src/services/platform/platform-service.ts:399 (adminUserStatus); User, Notification, AuditEvent; pending appeals may resolve on restore
TM-F-072 / TM-FR-072 Send traveler warning	travelmate-frontend-new/components/features/admin/AdminPanels.tsx:19 (UsersView) -> travelmate-backend-api/src/services/platform/user-warning-service.ts:13 (sendUserWarning); Notification, AuditEvent
TM-F-073 / TM-FR-073 Review/resolve problem reports	travelmate-frontend-new/components/features/admin/AdminPanels.tsx:74 (ReportsView) -> travelmate-backend-api/src/services/platform/report-service.ts:28 (resolveReport); Moderation, Notification, AuditEvent

Feature and requirement	Source flow and persistence
TM-F-074 / TM-FR-074 Review suspension appeals	travelmate-frontend-new/components/features/admin/AdminPanels.tsx:74 (ReportsView) -> travelmate-backend-api/src/services/platform/appeal-service.ts:32 (reviewAppeal); User + Moderation + Notification + AuditEvent transaction
TM-F-075 / TM-FR-075 Live reports and reconnect snapshots	travelmate-frontend-new/components/features/admin/AdminDashboard.tsx:79 (EventSource) -> travelmate-backend-api/src/controllers/admin-reports-stream.ts:9 (streamAdminReports) -> travelmate-backend-api/src/services/platform/report-events.ts:1; Read Moderation, reporter User fields and recipient Notification
TM-F-076 / TM-FR-076 System Health and selected audit activity	travelmate-frontend-new/components/features/admin/AdminPanels.tsx:118 (HealthView) -> travelmate-backend-api/src/controllers/platform-controller.ts:48 (integrations); ItineraryGeneration summaries, latest 100 AuditEvent; config presence booleans
TM-F-077 / TM-FR-077 Feedback filters and exact details	travelmate-frontend-new/components/features/admin/UserFeedbackView.tsx:1 -> travelmate-backend-api/src/services/platform/feedback-service.ts:32 (feedbackQuery) -> travelmate-backend-api/src/controllers/feedback-controller.ts:50 (DETAIL); Feedback with optional User/Trip joins; RepeatableRead; 10/page
TM-F-078 / TM-FR-078 Administrator notifications	travelmate-frontend-new/components/features/admin/AdminNotificationsView.tsx:1 -> travelmate-frontend-new/lib/admin-notifications.ts:1 -> travelmate-backend-api/src/repositories/trip-workspace-repository.ts:71 (readNotificationSnapshot) -> travelmate-backend-api/src/services/platform/admin-notification-service.ts:1; Notification; exact unread count and latest 50 entries
TM-F-079 / TM-FR-079 Administrator own profile	travelmate-frontend-new/components/features/admin/AdminPanels.tsx:135 (AccountSettingsView) -> travelmate-backend-api/src/controllers/profile-controller.ts:27 (PATCH) -> travelmate-backend-api/src/services/profile/profile-service.ts:1; User, AuditEvent

SDD 4.9 Technical services

Express readiness/error handling, database connection retry, shared PostgreSQL rate buckets and request-driven expired-record cleanup support all modules. Provider caches coalesce identical requests in memory and apply per-provider TTL/pacing; they are not a persistent job queue. Next/Express production transport/origin headers and paired contract/build/CI definitions supply deployment controls. The route list includes readiness and OAuth/session helpers even when no screen maps one-to-one to the endpoint. [1][2]

Table 21 Technical services feature implementation map

Feature and requirement	Source flow and persistence
TM-F-080 / TM-FR-080 API error contracts and safe recovery	travelmate-frontend-new/services/api-response.ts:1 -> travelmate-frontend-new/components/common/DashboardLoadState.tsx:1 -> travelmate-backend-api/src/middlewares/http-middleware.ts:1 -> travelmate-backend-api/src/middlewares/web-handler-middleware.ts:1 -> travelmate-backend-api/src/contracts/api-error.ts:1; No new business entity
TM-F-081 / TM-FR-081 Database retries and pooled access	travelmate-backend-api/src/lib/prisma.ts:1 -> travelmate-backend-api/src/lib/database-retry.ts:33 (retryDatabaseOperation); PostgreSQL driver adapter; existing constraints/transactions
TM-F-082 / TM-FR-082 Shared rate limits and request-driven cleanup	travelmate-frontend-new/services/api-response.ts:1 -> travelmate-backend-api/src/middlewares/rate-limit-middleware.ts:7 (allowRequest); RateLimitBucket; RevokedSession expiry cleanup
TM-F-083 / TM-FR-083 Provider cache and unavailable disclosures	travelmate-frontend-new/components/common/ProviderFreshness.tsx:1 -> travelmate-frontend-new/services/provider-response.ts:1 -> travelmate-backend-api/src/services/cache/provider-cache.ts:46 (withProviderCache) -> travelmate-backend-api/src/services/cache/policies.ts:1; Bounded in-process cache, not distributed durable data
TM-F-084 / TM-FR-084 Security headers, HTTPS and same-origin writes	travelmate-frontend-new/next.config.ts:1 -> travelmate-frontend-new/proxy.ts:1 -> travelmate-backend-api/src/middlewares/security-middleware.ts:1 -> travelmate-backend-api/src/config/env.ts:3 (validateRuntimeEnvironment); Current User/session checks; no separate authorization table

Feature and requirement	Source flow and persistence
TM-F-085 / TM-FR-085 Operational health and graceful shutdown	travelmate-backend-api/src/controllers/system-controller.ts:1 -> travelmate-backend-api/src/server.ts:1; Read-only database readiness query
TM-F-086 / TM-FR-086 Contracts and paired-revision validation	travelmate-frontend-new/scripts/sync-api-contracts.mjs:1 -> travelmate-frontend-new/lib/contracts.ts:1 -> travelmate-backend-api/openapi/travelmate-api.json:1 -> travelmate-backend-api/src/contracts/api-error.ts:1; No runtime business write
TM-F-087 / TM-FR-087 Container deployment and rollback configuration	travelmate-backend-api/Dockerfile:1 -> travelmate-backend-api/docker/docker-compose.yaml:1 -> travelmate-backend-api/scripts/deploy.sh:1 -> travelmate-backend-api/.github/workflows/deploy.yaml:1; Migrations are deployment mutations; database supplied externally

SDD 4.10 Compatibility and excluded scope

Compatibility endpoints retain flight/activity comparison, hotel quote/revalidation and arrival/departure/rest constraint handling, while the current planner UI does not expose those retired controls. Their capability is partial at product level. Legacy Listing/Booking/PaymentTransaction-related schema is not proof of an operational marketplace, owner role or checkout. Removed preference/comparison tabs normalize to Home and old GPS entrypoints redirect. Guest chat, live crowd occupancy and an automatic daily backup worker were not located. [1][2]

Table 22 Compatibility and excluded scope feature implementation map

Feature and requirement	Source flow and persistence
TM-F-088 / TM-FR-088 Flight and activity comparison API	travelmate-frontend-new/components/features/traveler/TravelComparisonWorkspace.tsx:1 -> travelmate-backend-api/src/controllers/travel-options-controller.ts:1 -> travelmate-backend-api/src/services/travel/amadeus-provider.ts:1; Legacy approved Listing reads; provider caches

Feature and requirement	Source flow and persistence
TM-F-089 / TM-FR-089 Hotel quote/revalidation API	travelmate-frontend-new/components/features/traveler/AccommodationGuests.tsx:1 -> travelmate-backend-api/src/controllers/accommodation-check-controller.ts:1 -> travelmate-backend-api/src/services/travel/hotel-quote.ts:1 -> travelmate-backend-api/src/services/travel/hotel-recommendations.ts:1; HMAC quote tickets; no booking created
TM-F-090 / TM-FR-090 Generation arrival/departure/rest API constraints	travelmate-frontend-new/components/features/traveler/TravelerDashboard.tsx:1 -> travelmate-backend-api/src/services/itinerary/generated-day-scheduler.ts:10 (generationConstraintsSchema) -> travelmate-backend-api/src/controllers/itinerary-controller.ts:629 (constraintsResult); Trip JSON planningStatus/scheduleBlocks
TM-F-091 / TM-FR-091 Removed owner/marketplace/payment workflows	travelmate-frontend-new/docs/Routes.md:1 -> travelmate-backend-api/src/services/platform/platform-service.ts:419 (This workflow is no longer available); Legacy Listing/Booking/Promotion/Review/PaymentTransaction/OwnerDocument tables retained
TM-F-092 / TM-FR-092 Retired preferences/comparison tabs and duplication	travelmate-frontend-new/hooks/use-itinerary-navigation.ts:1 -> travelmate-backend-api/src/services/platform/platform-service.ts:417 (duplicate-trip); Legacy duplicated enum retained only for reading historical versions
TM-F-093 / TM-FR-093 Guest chatbot, live crowd feed and automatic backups	; No Chat/conversation table, live crowd integration or backup job found

SDD 4.11 API contract patterns

Every API operation is listed in the accompanying 50-row route reconciliation with its actual method/path and controller reference. Where OpenAPI is absent or stale, that register identifies the gap; a generic OpenAPI summary is not treated as proof of a runtime payload. Key request patterns below are conceptual field groups from inspected validators, not copy-paste authenticated requests or a replacement for source types. [2]

Table 23 Important request and response contracts

Operation group	Request and validation	Response and state
Account POST /api/auth	Discriminated action; email/code/password/name fields as required by action.	Public user/notice or safe action error; cookies issued/cleared only for appropriate actions.
Generate itinerary	Confirmed destination identity/context/coords; budget/currency; inclusive dates; travelers/party; bounded interests; explicit costs; idempotency key.	Grounded structured itinerary, provider/fallback notices, costs/assumptions, schedule/day status; separate save step.
Platform trips and versions	Action-dispatched owner-scoped payload; trip ID/save key, validated itinerary and grounding; status/restore request.	Owned trip summaries/full snapshot/version metadata; conflicts rather than stale overwrite.
Managed activity	trip ID; stable activity ID; revision; action/edit values and confirm true.	Recomputed saved itinerary, version/history and new revision; invalid/stale/foreign rejected.
Replan preview and accept	Owned trip/day/revision for preview; signed preview token and confirmation to accept.	Preview permitted moves or bounded failure; accepted update is transactional.
Navigation	Owned trip/day; exact-stop context, mode or weather date as applicable.	Manifest/candidates/verified route/weather provenance; unresolved/no-route/unavailable explicit.
Admin moderation	Current admin session; target, command/status and bounded decision/warning.	Updated status/notices/audit or safe validation/permission/conflict response.

SDD 5 User interface design

The public discovery section keeps left-aligned narrative and a country selector alongside an enlarged right-side Earth; small screens stack them. The card uses a consistent photo crop and a compact English-name/flag row. It is an overlay and cannot capture globe pointers. Pointer exit clears card and outline. The latest requirement includes the name/flag row; earlier image-only instructions are superseded. Credits remain in the existing footer. [1]

Traveler navigation separates creating a brief from managing saved trips, budget evidence, actual visits, feedback, notifications and profile. Activity visit controls and map routing remain attached to owned saved trips. Administrator tabs expose overview/users/reports/feedback/notices/system/profile based on the current account role. Current anonymous screenshots are embedded in the manual; authenticated screenshots are explicitly unavailable in this read-only session rather than replaced by fabricated screens. [1]

SDD 6 Algorithms and workflows

SDD 6.1 Grounded generation

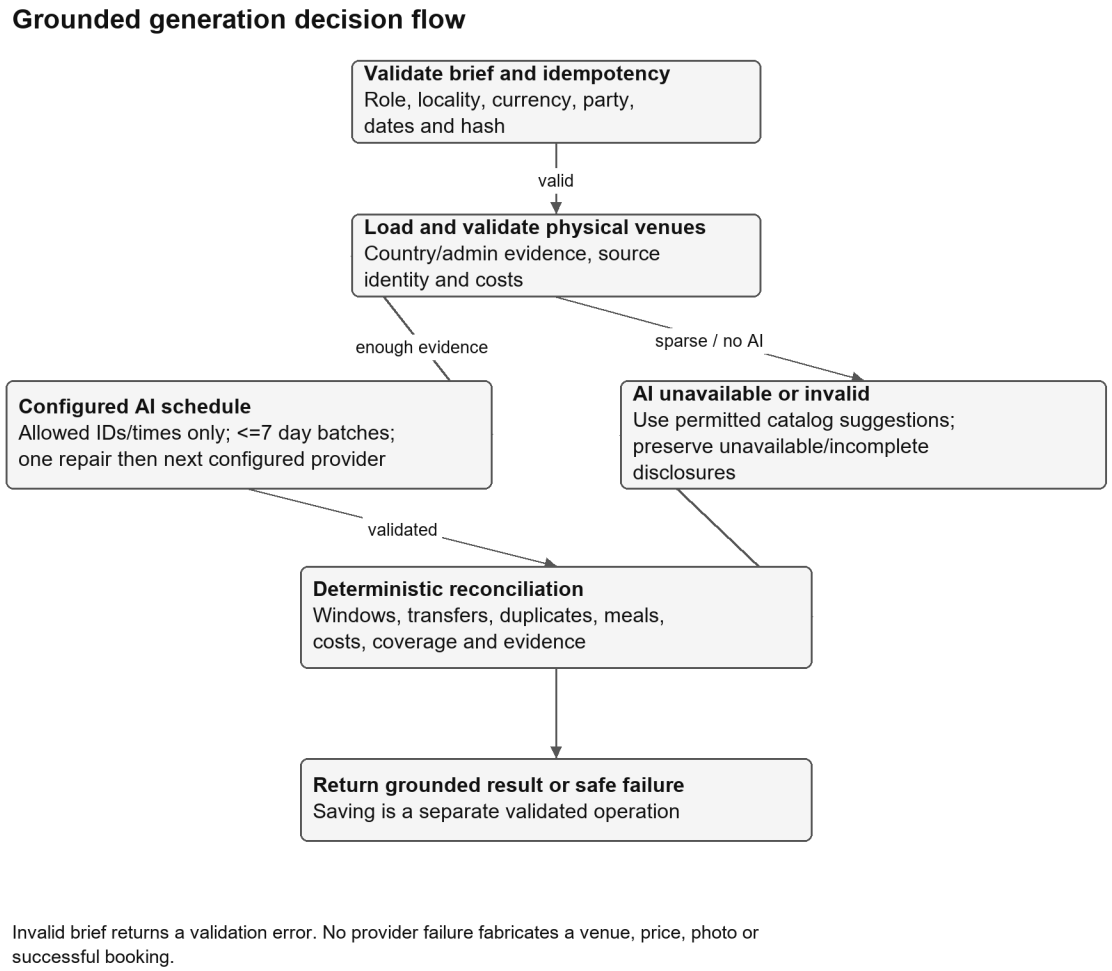


Figure 7 Grounded generation activity flow and honest failure paths [2]

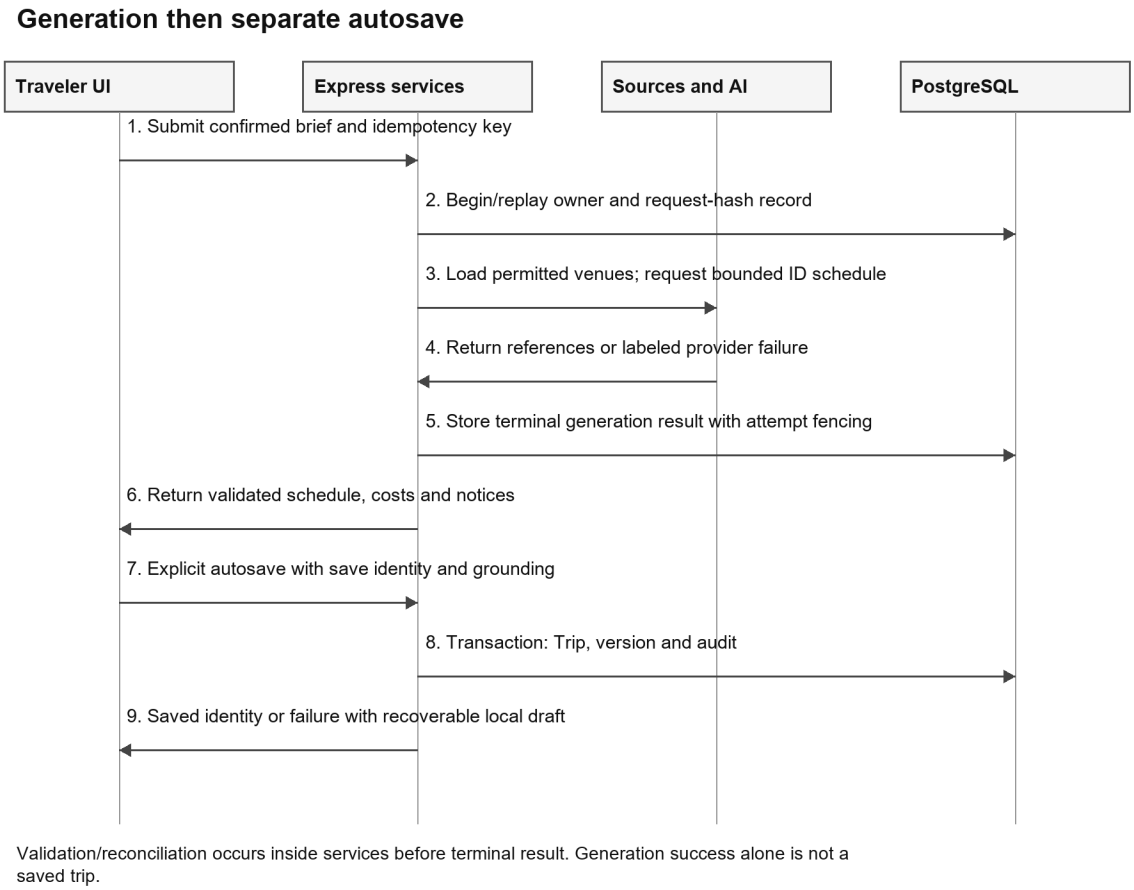


Figure 8 Generation and separate autosave sequence with source and persistence boundaries [1][2]

Grounding obtains Wikipedia/Wikidata physical attractions and OpenStreetMap boundary/membership evidence. It rejects organizations, virtual objects, whole administrative areas and unsupported historical/destroyed/proposed relationships. Coordinates within a maximum thirty-kilometer expansion are only one check; source identity and actual country/admin membership remain necessary. Sparse evidence produces incomplete coverage rather than fabricated attractions. Media uses exact venue/country identity and attribution. [2][8][12]

The configured preferred provider is tried before another configured adapter. OpenAI defaults to gpt-4o-mini with JSON Chat Completions, temperature 0.7 and a lower-temperature repair; Gemini defaults to gemini-3.1-flash-lite with a structured schema. Environment variables can override models. The prompt supplies the allowed place IDs, trip/day windows and constraints, asking only for IDs, start minutes, durations and participants. Trips are processed in consecutive

batches of at most seven days, avoiding previously used places. Malformed/invalid output receives one repair attempt per batch; adapter/provider failure falls through to the next configured provider. Each adapter is bounded to 45 seconds; the overall generation signal is 540 seconds, with the frontend proxy at 580 seconds. These are deadlines, not measured latency promises. [2]

A deterministic reconciler checks dates, identity, duplicates, durations, transfer feasibility, usable hours, budget and weather context. Normal Balanced scheduling caps catalog visits at three per day, with a 09:00-18:00 baseline usable day, a 45-minute meal and buffered transfers (more conservative for families). API-only arrival/departure/rest inputs can narrow windows using 90/120-minute buffers, but those fields are absent from the current planner form. Catalog fallback remains labeled and can include honest allowances/incomplete days. Production provider failure does not turn into a fabricated mock itinerary. [2]

SDD 6.2 Cost and currency calculation

Currency selection is explicit and supports PHP, USD, EUR, JPY, KRW, THB, GBP, AUD, CAD, SGD, CNY, HKD, TWD, MYR, IDR, VND, INR, NZD, CHF and AED. JPY/KRW/VND use zero fractional digits; the others use two. Integer minor-unit arithmetic and exact shares avoid lost cents in group displays. FX is a reference from Frankfurter with pair validation, six-hour cache and a twenty-four-hour stale cutoff, not a currency change to the saved trip. [1][2]

The formulas and PHP worked example in SRS 3.4 specify new grounded plans. A retained legacy domain split of 34/22/20/14/10 percent is not the truth for new expense-evidence generation. The current engine sums known accommodation, food, transport, admissions, explicit fees and pre-trip amounts, avoids exact-name double counting, adds rounded contingency, and shows remaining/shortfall. Unknown costs do not enter known subtotal and remain visibly unpriced. Saving manual edits recalculates and downgrades unsupported stale price evidence. [2]

SDD 6.3 Time maps and managed day changes

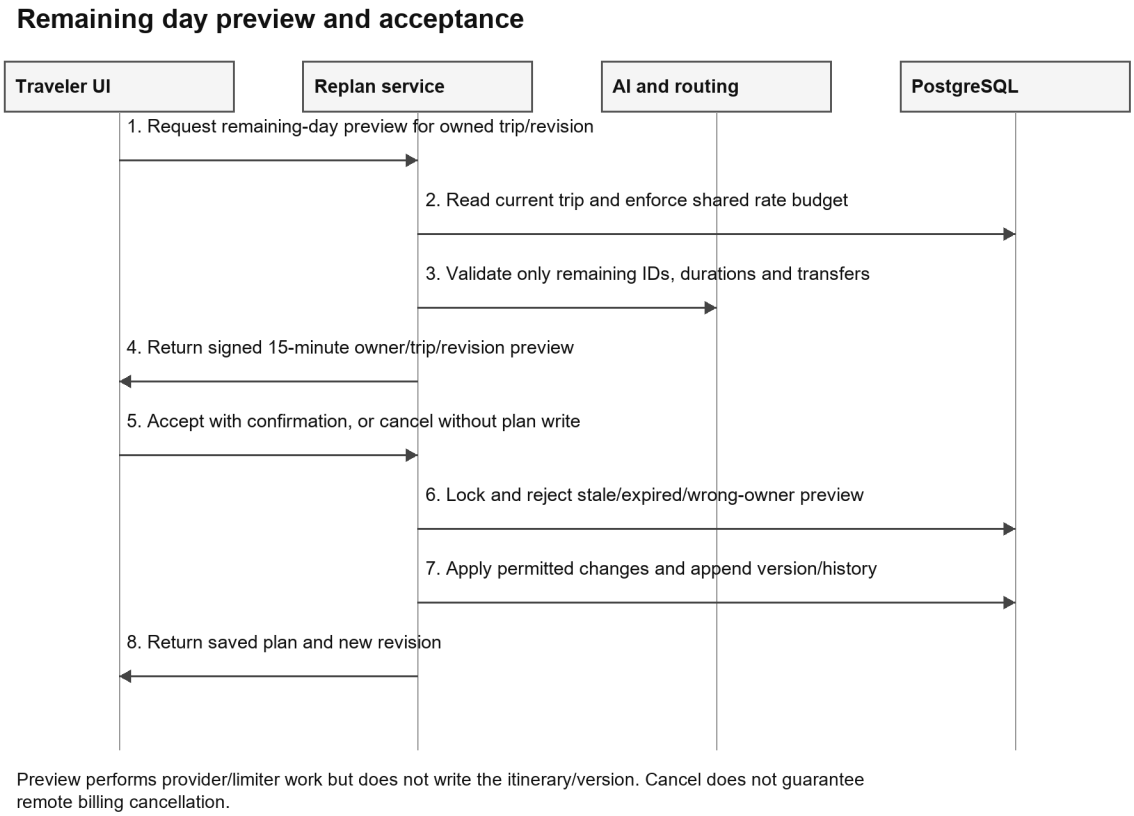


Figure 9 Remaining-day replan preview and revision-checked acceptance [1][2]

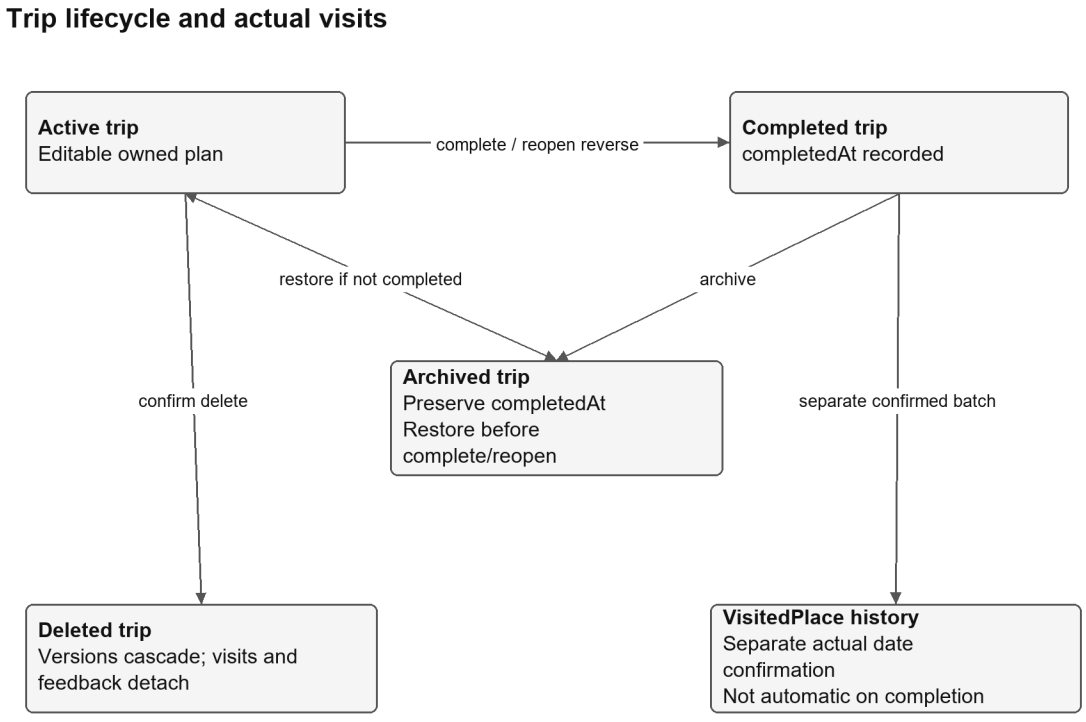


Figure 10 Trip status transitions and separate actual visited-history recording [2]

Dates represent inclusive trip days, while activity clocks are local-day minutes. Weather is limited by provider horizons; future coverage is not current observation. OSRM provides verified route geometry/duration where possible; conservative planning transfer buffers are estimates and not asserted as road navigation. Replan transfer checks use route legs where available or conservative walking at four km/h plus fifteen minutes. Replan uses current allowed activities and a bounded local window, never live traffic, new paid offers or arbitrary invented venues. The local-versus-UTC start-date issue D-03 remains an acceptance concern. [2]

SDD 6.4 Class and component responsibilities

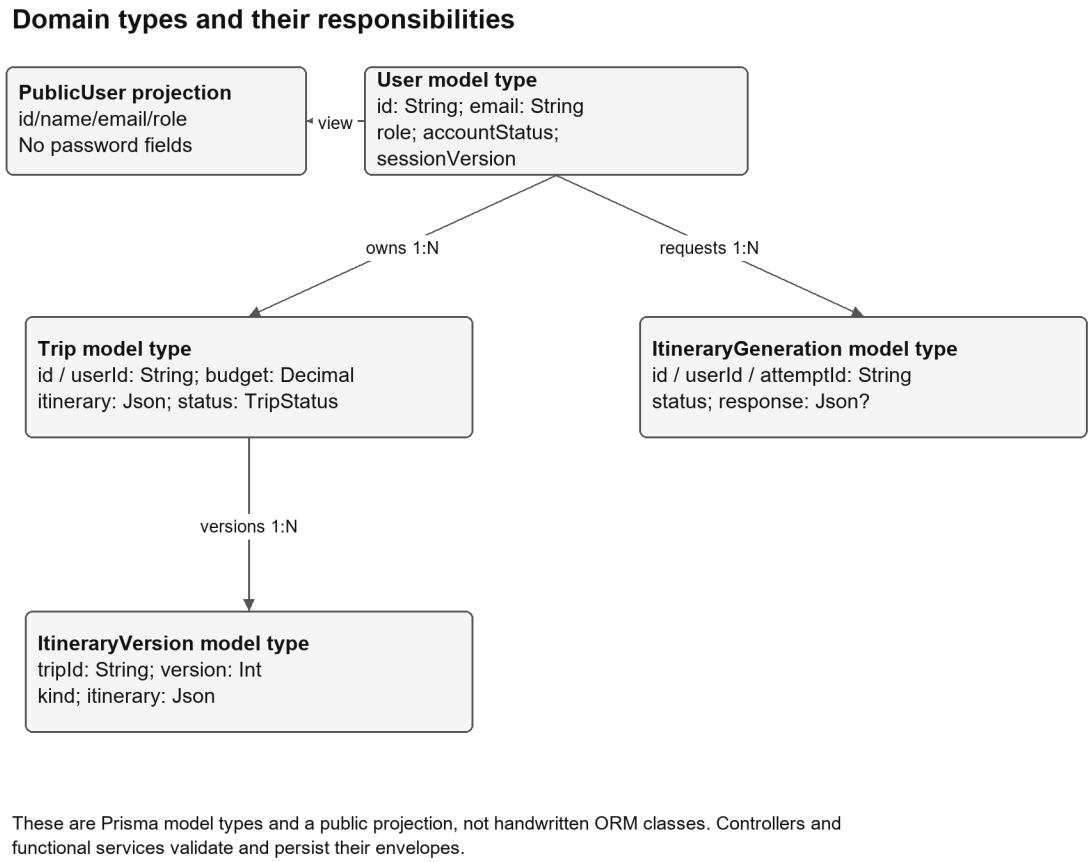


Figure 11 Software responsibilities and data types without inventing ORM service classes [1][2]

This TypeScript implementation primarily uses functional components, exported service functions and Prisma-generated types. The class/component diagram represents those actual software responsibilities rather than claiming a Java-style entity hierarchy. PublicUser is a sanitized account projection, Trip holds the persisted envelope, and ItineraryVersion/ItineraryGeneration represent history and execution state. Controller/service dependencies are module calls, not database foreign keys. [1][2]

SDD 7 Access failures and privacy

SDD 7.1 Access and security

Current-user middleware checks signed payload/expiry, current database role/state/sessionVersion and revocation. Eight-hour HttpOnly/SameSite=Lax sessions become Secure in production. Password salt is sixteen random bytes and scrypt output is sixty-four bytes. Login/register/verify/reset limits differ and use persistent buckets: login 8/15 min, registration 4/hour, verify 8/15 min, forgot 3/hour, reset 6/15 min, with a broader auth IP limit 20/min. Resend requires 60 seconds and no more than three eligible sends/hour. Verification codes expire after five minutes with five attempts; reset after fifteen minutes. [2]

Google state/nonce/PKCE and audience verification protect the callback. Public registration cannot create admin. Production backend transport enforcement returns 426 where appropriate and frontend HTTP redirect uses 308; origin/fetch-site checks protect writes. CSP includes unsafe-inline, so its presence is not a proof that every XSS vector is eliminated. Parameterized ORM queries, allowlisted profile fields and generic production errors are controls, not a completed penetration test. No at-rest AES-256, immutable audit or daily backup claim is supported by code alone. [2]

SDD 7.2 Failure handling and service bounds

Table 24 Failure and recovery design

Boundary	Behavior	User/operator consequence
Mail	Resend production; explicit Gmail development alternative; missing configuration 503/provider failure 502.	Account may exist while the new code is invalidated after send failure; observe cooldown and visible outcome.
AI/catalog	Per-call and whole-generation bounds; schema/identity validation; fallback provider/catalog.	Read source/fallback/incomplete notices; no invented complete schedule.
Search/stay/ photos	Paced providers, caches and exact context; sparse/rights-restricted media can remain unavailable.	Do not infer a hotel rate, a wrong-place fallback or complete catalogue coverage.
Maps/weath er	Stop/size/forecast/quota bounds; no-route and unavailable states.	Use exact-stop/manual alternatives and independent travel checks.

Boundary	Behavior	User/operator consequence
Database	Bounded connection retry and safe errors; transaction/row-state conflicts.	Retry explicitly safe operations; keep draft if save failed; never equate generated with saved.
Logout	Local cookies clear even if revocation persistence fails; warning is explicit.	Do not claim remote token invalidation was confirmed.
SSE/cache/limiter	Cleanup on disconnect; request-driven 1% expired bucket/revocation cleanup; memory TTL/coalescing.	No scheduled distributed worker or persistent cache guarantee.

SDD 7.3 Data handling and provider boundaries

Registration/login/reset exchange email and authentication data with backend identity/mail providers; generation sends a bounded trip brief and approved venue IDs to the configured AI; map/weather/geocoding providers receive necessary coordinates/query context; optional Google photos require explicit server-side opt-in and billing. Secrets remain backend-only. Existing logs, real account records and secrets were not copied into this deliverable. Retention, cross-border disclosures, legal basis and operator contact require institutional review. [2][11]

SDD 8 Deployment and requirement mapping

The backend Dockerfile uses Node 22 bookworm-slim, a non-root user, build-time Prisma/TypeScript generation and a health check. Local Compose exposes the API on loopback and uses development configuration; production Compose uses an external image/env file and no-new-privileges, with explicit migration tooling. The deployment script pulls an image, applies migrations and waits for health, reverting the image on failure. That image rollback does not reverse database migrations and requires a compatible recovery plan. CI definitions include paired-revision frontend/backend checks and disposable PostgreSQL integration jobs; this documentation task did not execute those deployment jobs. [1][2]

The RTM is the authoritative cross-document register: TM-FR/TM-NFR -> TM-F -> SDD section -> actual source and database reference ->

TM-TC/TM-XT/TM-QT -> UM section. Module sections SDD 4.1-4.10 correspond to the feature groups; quality mappings refer to the more detailed security, process, data and operational sections. Complete API/page/process registers resolve entry points with no dedicated screen. The appendices provide setup, editable diagram sources, schema fields and acceptance forms.

Software Test Documentation

Version 2.0. This document contains reproducible procedures and a separate execution record. It does not certify user acceptance or production readiness. The editable case register contains 133 procedures: 93 feature cases, 20 focused boundary/failure cases and 20 quality cases.

STD 1 Scope and evidence rules

Coverage includes guest, traveler, suspended traveler and administrator, normal flows, invalid/boundary inputs, permissions, provider failures, costs, AI validation, persistence and navigation. Static inspection establishes what code does; historical reports describe earlier execution; current isolated regression verifies only its test scope; a manual system case remains Not executed until its own steps and actual results are recorded. Related test filenames are discovery aids, not evidence that a matching manual procedure passed. [1][2][6][7]

STD 2 Environment and baseline

Table 25 Source state inspected

Repository	Commit	Working state
Frontend	1d55ba3c2eb2a9387cb34c0cb6ca5edef8bfbc96	Relevant tracked changes and untracked current files recorded in source-snapshot/status evidence; commit alone is not the inspected state.
Backend	234e08d2317526f772854296bfaf6a33ca5dd0fa	Relevant tracked changes and untracked current files recorded in source-snapshot/status evidence; commit alone is not the inspected state.
Workspace root	97d4749b48c54ec36a60e6c0c869df1016ed7551	Contains baseline/audit and nested repositories; working-tree evidence retained.

The documentation environment was Windows with a Node runtime from the bundled Codex workspace dependencies. Tests ran on copies under work/isolated, not the original source tree. Installed node_modules were

referenced without installation. A preload guard supplied disposable credentials and blocked remote connections and existing local servers; only loopback servers started by a test worker were allowed. Fake provider responses and temporary files were used. Integration tests, email smoke, development watcher and Docker deployment checks were excluded. No migration, seed, live account operation, paid AI request or existing database mutation was performed. [6]

Table 26 Verification execution register

ID and evidence class	Recorded result	Limit
EX-01 Current isolated automated regression	Passed: 152/152 tests; 0 failed, 0 skipped	Copied current source; mocked/test dependencies; external network and existing servers blocked. Not manual TM-TC execution, production build, visual acceptance or live persistence.
EX-02 Current isolated automated regression	Passed: 521/521 tests; 0 failed, 0 skipped	Excluded integration/*, development-environment-watch.test.ts, docker-deployment.test.ts and email-smoke.test.ts. Initial setup failures corrected in harness only. No provider delivery or existing DB calls.
EX-03 Current visual observation	Current screenshots captured; full Earth visible and changing rotation orientation observed; no forms submitted	Single desktop browser viewport; no FPS measurement, authenticated workflow, complete hover sweep or accessibility certification.
HIST-01 Historical report, not rerun as browser/build this task	Report states 152 unit tests, 17 browser tests and isolated production build passed	Prior working state; no new full-country visual certification. Do not sum historical tests with EX-01.
HIST-02 Historical report, not rerun as browser/build/lint this task	Report states 145 unit tests; 28 browser and 19 shared checks with 3 repeated scenarios; 44 distinct checks; build/typecheck passed	Controlled fixtures for role/UI responses; whole-repository ESLint recorded 7 no-explicit-any errors in e2e/travel-preferences.spec.ts. No live persistence acceptance.

ID and evidence class	Recorded result	Limit
HIST-03 Historical report	Report states 135 frontend tests passed and obsolete tab normalization checked	Older snapshot; not additive to later totals.

The final logs are frontend-final-tests.log and travelmate-backend-api-final-tests.log. Frontend: 152 passed, zero failed/skipped, 38 files. Backend: 521 passed, zero failed/skipped, 83 files. Initial harness copy/guard setup errors were corrected in documentation helpers; they are not product defects and were not hidden as final results. Intermediate retest counts overlap the final suite and must not be added. A passing isolated suite does not demonstrate email inbox delivery, live provider availability, real PostgreSQL transaction behavior, accessibility, UAT or performance. [6]

STD 3 Data and execution protocol

Use synthetic accounts A/B/admin/suspended with example.invalid addresses and a test mail sink. Never substitute a real password, user token or confidential itinerary into this register. Every write procedure requires an authorized disposable database/provider stub or separately approved staging environment. Inspect setup scripts first: npm install/postinstall, db:setup, migrations, seeds and email:smoke have side effects. Record tester, date, environment/source hashes, fixture IDs, actual HTTP/UI result, sanitized evidence, defect and cleanup. A blocked or skipped prerequisite is not a pass.

The cases below are specified procedures. Their expected results express intended acceptance even where a static concern exists; the defect register supplies the implementation discrepancy. Numeric examples are synthetic and arithmetically checked, not executed financial transactions. For source paths use the linked requirement row in the RTM/inventory; lengthy implementation paths are not repeated in every case.

STD 4 Individual test cases

STD 4.1 Public discovery

TM-TC-001 Landing and navigation

Requirements and status: TM-FR-001 | Not executed

Prerequisites: Authorized disposable environment; Guest / all. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Anonymous browser; desktop 1440 and mobile 320 CSS pixels; no account data.

Steps: Open the home page. Use the top links or mobile menu to reach features, workflow, budget, product and questions. Use account entry to sign in or register. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: The public site shall present planning guidance, anchored information sections, FAQ, account entry and legal links. Acceptance constraints: Landing samples are illustrative. Some marketing copy still mentions removed travel-style/comparison controls; see discrepancy D-01.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-001; static source references only; no system-case execution evidence. Related isolated test filename: navigation.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-002 Interactive Earth controls

Requirements and status: TM-FR-002 | Not executed

Prerequisites: Authorized disposable environment; Guest / all. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: WebGL browser; reduced-motion off/on; zoom at both allowed limits; country selector.

Steps: Scroll to Discovery. Drag Earth to change the view; use + or - and Pause rotation. Select a country in the dropdown to focus it. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: The globe shall support rotation, drag, zoom, pause and reduced-motion handling, with a country selector fallback. Acceptance constraints: WebGL or dataset failure shows fallback feedback. Selection focuses Earth; it does not create a trip or open a destination-details panel.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-002; static source references only; no system-case execution evidence. Related isolated test filename: discovery-globe.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-003 Country hit detection and border identity

Requirements and status: TM-FR-003 | Not executed

Prerequisites: Authorized disposable environment; Guest / all. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Philippines, Brazil, France, Kenya, Japan, Fiji, Nauru, Vatican City and country edges; stationary pointer while Earth rotates.

Steps: Move the pointer onto a land area. Observe the border around that geographic record. Move to another area or leave Earth to change or clear it. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Hover detection shall use actual country geometry and stable geographic codes, including small-area hit targets, and display that country’s border. Acceptance constraints: Natural Earth geometry has finite resolution. Disputed/non-national areas are data records, not claims of sovereignty. Decorative arcs, dots and rings are absent.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-003; static source references only; no system-case execution evidence. Related isolated test filename: discovery-hover-identity.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-004 Country destination photo and identity card

Requirements and status: TM-FR-004 | Not executed

Prerequisites: Authorized disposable environment; Guest / all. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: The same cross-continent/island set plus Antarctica and unresolved Bir Tawil; rapid pointer transitions with delayed image loads.

Steps: Hover a country; read its compact name-and-flag row below the photo. Hover another country to update all identity elements. Leave the globe to dismiss. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: A hover card shall use the same stable code for the destination photograph, English name, available flag and border, and hide on pointer exit. Acceptance constraints: 264/265 geographic areas have verified photos; Bir Tawil is unresolved and receives no unrelated fallback. All 249 ISO entries plus Kosovo are covered. Antarctica and some non-national areas have no national flag.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-004; static source references only; no system-case execution evidence. Related isolated test filename: discovery-photo-quality.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-005 Image and map credits

Requirements and status: TM-FR-005 | Not executed

Prerequisites: Authorized disposable environment; Guest / all. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Published attribution manifests and public footer; no authenticated state.

Steps: Scroll to the footer. Expand Image and map credits. Follow the relevant source or license link. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: The public footer shall expose image/map source, author, licensing and adaptation credits without a permanent caption under Earth. Acceptance constraints: Credits describe asset rights, not provider availability or travel recommendations.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-005; static source references only; no system-case execution evidence. Related isolated test filename: discovery-countries.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-006 Privacy and terms pages

Requirements and status: TM-FR-006 | Not executed

Prerequisites: Authorized disposable environment; Guest / all. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Public /privacy and /terms pages; no account state.

Steps: Open Privacy policy or Terms of use in the footer. Read the notices and follow the cross-link or TravelMate home link. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Public privacy and terms pages shall describe the planning scope and applicable content handling. Acceptance constraints: Current privacy text emphasizes Google photos although ordinary production enrichment uses reusable providers; notice reconciliation is recommended (D-02).

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-006; static source references only; no system-case execution evidence. Related isolated test filename: not assigned.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

STD 4.2 Accounts and access

TM-TC-007 Traveler registration

Requirements and status: TM-FR-007 | Not executed

Prerequisites: Authorized disposable environment; Guest. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Disposable traveler only: given name Ana, surname Example, ana@example.invalid, optional 09171234567, conforming test-only password.

Steps: Choose Create an account. Enter first and last names, email, optional mobile number, password and confirmation. Submit, then proceed to verification only after observing the outcome. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Registration shall validate names, email, optional Philippine mobile number, strong password and confirmation, and create only a traveler account. Acceptance constraints: Names 2-80 characters combined; password 8-64 with upper/lower/digit/special and no whitespace. Mobile is optional. Email configuration is required; accepted mail is not confirmed inbox delivery.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-007; static source references only; no system-case execution evidence. Related isolated test filename: registration-details.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-008 Email verification

Requirements and status: TM-FR-008 | Not executed

Prerequisites: Authorized disposable environment; Guest with pending account. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Disposable pending account and captured test-mail code; valid/expired/used/wrong 8-character hex code.

Steps: Enter the registered email and received eight-character code in the verification view. Submit before expiry. Sign in after the confirmation. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Verification shall consume a valid account-bound, short-lived code once and permit subsequent password sign-in. Acceptance constraints: Eight hexadecimal characters, 5-minute lifetime and 5 attempts. Verification is separate from legacy profileStatus.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-008; static source references only; no system-case execution evidence. Related isolated test filename: account-token.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-009 Resend verification

Requirements and status: TM-FR-009 | Not executed

Prerequisites: Authorized disposable environment; Guest with pending account. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Disposable pending account, fake clock, test mail sink; attempts at 59 s, 60 s and fourth resend/hour.

Steps: Use Resend verification after waiting 60 seconds. Check the displayed result and inbox/spam. Use only the newest accepted code. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Resending shall replace only an eligible account’s verification code, enforce cooldown/limits and report mail failures honestly. Acceptance constraints: Three eligible resends per hour; failed send invalidates that code while retaining cooldown. No automatic switch from Resend to Gmail.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-009; static source references only; no system-case execution evidence. Related isolated test filename: auth-email-configuration.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-010 Password sign-in and role redirect

Requirements and status: TM-FR-010 | Not executed

Prerequisites: Authorized disposable environment; Guest. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Disposable verified traveler, administrator, suspended and unverified account; valid/wrong password.

Steps: Open Sign in. Enter the registered email and TravelMate password. Submit; proceed to the role-specific workspace or suspension appeal view. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Password sign-in shall validate credentials and email verification, issue a signed session and route traveler/admin/suspended accounts appropriately. Acceptance constraints: Wrong credentials return 401; unverified email returns 403. Suspended accounts can sign in for appeals, but ordinary protected APIs reject them.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-010; static source references only; no system-case execution evidence. Related isolated test filename: account-roles.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-011 Google OAuth

Requirements and status: TM-FR-011 | Not executed

Prerequisites: Authorized disposable environment; Guest / existing account. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Provider stub with matching/mismatched state, PKCE and verified identity; controlled redirect URI.

Steps: Choose Continue with Google. Complete Google’s own authentication. Return to TravelMate and confirm the expected account/role. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Google sign-in shall validate state, PKCE and verified provider identity before resolving the application account and role. Acceptance constraints: Requires provider configuration and authorized redirect origin. Existing verified local accounts require the documented linking-safe behavior; no admin role is granted from public input.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-011; static source references only; no system-case execution evidence. Related isolated test filename: google-oauth-service.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-012 Authoritative sessions and access control

Requirements and status: TM-FR-012 | Not executed

Prerequisites: Authorized disposable environment; Traveler / administrator. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Two traveler accounts A and B, administrator, expired/revoked token and bumped sessionVersion.

Steps: Sign in before opening a protected workspace. If the session is rejected, return to sign-in. Do not rely on a visible menu to establish permission. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Protected API requests shall validate the signed session, expiry, current role, account state, session version and per-session revocation. Acceptance constraints: Custom HMAC-SHA256 session format, not a standard JWT refresh-token system. Eight-hour cookie; proxy checks presence only.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-012; static source references only; no system-case execution evidence. Related isolated test filename: security.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-013 Logout and local cleanup

Requirements and status: TM-FR-013 | Not executed

Prerequisites: Authorized disposable environment; Traveler / administrator / suspended traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Disposable valid session; DB success and revocation failure stub; never use a live user cookie.

Steps: Use Logout or Sign out. Confirm return to the public site. On a shared device, close private views after local sign-out. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Logout shall clear application/OAuth cookies, attempt individual session revocation and disclose unconfirmed revocation. Acceptance constraints: If database revocation fails, local cookies clear but server revocation is explicitly unconfirmed. Other logins are not intentionally revoked.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-013; static source references only; no system-case execution evidence. Related isolated test filename: session-service.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-014 Forgot/reset password

Requirements and status: TM-FR-014 | Not executed

Prerequisites: Authorized disposable environment; Guest with verified active account. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Disposable verified account; fake mail sink; 15-minute token boundary and existing session A.

Steps: Choose Forgot password. Request a code for the verified email. Enter the code and a conforming new password; then sign in again. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Recovery shall issue a short-lived reset code, validate the replacement password, consume the token once and invalidate old sessions. Acceptance constraints: Reset lifetime 15 minutes. Actual email delivery/provider authorization was not tested. Reset is not an in-profile password-change screen.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-014; static source references only; no system-case execution evidence. Related isolated test filename: account-token.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-015 Suspension appeal access

Requirements and status: TM-FR-015 | Not executed

Prerequisites: Authorized disposable environment; Suspended traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Disposable suspended traveler; 10 and 2000 characters, existing pending appeal, parallel duplicate.

Steps: Sign in to open Account appeal. Read the account notices. Enter a 10-2000-character appeal, submit once and check its status. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: A suspended traveler shall read account notices/history and submit one pending appeal with a bounded message. Acceptance constraints: Three attempts/hour; a pending appeal blocks duplicates. Approval/restoration is an administrator decision.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-015; static source references only; no system-case execution evidence. Related isolated test filename: appeal-service.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

STD 4.3 Destination selection

TM-TC-016 Worldwide direct search

Requirements and status: TM-FR-016 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Cebu City PH, Paris FR and same-name towns; empty/no-result/accented/non-Latin queries.

Steps: In Plan a trip, enter Where do you want to go? Read the parent/country/type of each result. Select the intended locality, or use verified manual resolution. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Destination search shall return source-backed place identity, country, hierarchy and coordinates without inventing missing source records. Acceptance constraints: Full imported catalogue records 5,721,309 records/250 source country entries; fallback compressed index is bounded. Coverage is not proof of complete geography.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-016; static source references only; no system-case execution evidence. Related isolated test filename: worldwide-catalogue.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-017 Country and administrative browsing

Requirements and status: TM-FR-017 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: PH region/province/city/barangay and worldwide country/admin hierarchy; pages 1 and next.

Steps: Choose Browse by country. Select a country and available administrative level, use search or Load more, then choose a final city/town. Close or Back to abandon a browsing draft. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: The destination browser shall follow recorded parent IDs/types with pagination and preserve an existing confirmation on cancellation. Acceptance constraints: Available levels/terminology vary. Philippine independent cities retain official parents; coordinates absent from official data require separate resolution.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-017; static source references only; no system-case execution evidence. Related isolated test filename: philippine-catalogue.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-018 Manual resolution and clear

Requirements and status: TM-FR-018 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Ambiguous locality with and without country/parent; geocoder unavailable and provider Retry-After.

Steps: Use the manual-resolution action for a missing direct result. Choose among candidates if ambiguous. Clear or delete the input to remove confirmation before selecting another place. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Manual destination input shall require verified disambiguation; clearing shall remove confirmation and dependent plan/stay state. Acceptance constraints: Country/region/island selections need narrowing before generation. Wrong-country or unresolved coordinates cannot be promoted to confirmed destinations.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-018; static source references only; no system-case execution evidence. Related isolated test filename: manual-destination-resolution.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-019 Area suggestions and comparison

Requirements and status: TM-FR-019 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: 1, 6 and 7 area candidates; omitted category, sparse physical venues and unavailable source.

Steps: Choose an area that needs a locality. Search its cities/towns and use Help me choose where available. Review up to six candidates, then select deliberately. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Area browsing shall expose verified child localities and a limited comparison without implying researched prices. Acceptance constraints: One to six city identifiers; comparison uses entered spending allowance and mapped interest samples, not live tourism popularity or confirmed costs.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-019; static source references only; no system-case execution evidence. Related isolated test filename: destination-areas.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-020 Starting location and optional geolocation

Requirements and status: TM-FR-020 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: GPS allowed, denied and unsupported browser; typed Cebu City origin; optional origin absent.

Steps: Open the optional starting-location control. Type a place or choose Use my current location. Grant permission only if desired, review the resolved area, or enter it manually. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: An origin shall be entered or explicitly obtained through browser location permission, with manual entry after denial/failure. Acceptance constraints: Geolocation requires secure context/permission. Origin selection clears when account changes; it is optional for current generation.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-020; static source references only; no system-case execution evidence. Related isolated test filename: starting-location.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-021 Currency detection and fallback

Requirements and status: TM-FR-021 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: PHP, JPY, KRW, VND and unsupported currency code; party of 3.

Steps: Select a destination. Review the detected destination currency. If unsupported, select a supported planning currency; keep that code through the plan. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Destination country codes shall map to supported planning currency or require an explicit supported fallback. Acceptance constraints: 20 supported codes; JPY/KRW/VND use whole units. Fallback is a planning choice, not currency conversion.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-021; static source references only; no system-case execution evidence. Related isolated test filename: destination-context.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-022 Reference-currency equivalents

Requirements and status: TM-FR-022 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: USD/PHP correct pair, reversed/mismatched pair, stale cache >24 h and 429 response.

Steps: Choose Show equivalents in. Read the reference rate and its date beside relevant amounts. Keep the original currency as the budget authority. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Reference conversion shall preserve the trip currency and label source/date/freshness, retry limits and unavailable rates. Acceptance constraints: Frankfurter mid-market reference, not card/cash conversion. Same-code rate is 1; wrong pair/invalid rate is rejected; Retry-After is honored.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-022; static source references only; no system-case execution evidence. Related isolated test filename: exchange-rate-service.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-023 Destination transport guidance

Requirements and status: TM-FR-023 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Configured transport guidance country and country without guidance; no prices assumed.

Steps: After selecting a locality, review Transportation options. Choose a relevant mode if known or continue without one. Refresh options only when needed. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: The planner shall label curated destination transport modes as guidance and allow progress when no guidance is available. Acceptance constraints: Guidance exists for selected country/city patterns. It is not a live timetable, capacity or fare source.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-023; static source references only; no system-case execution evidence. Related isolated test filename: destination-context.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-024 Destination inspiration carousel

Requirements and status: TM-FR-024 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Verified destination with available photos, empty catalogue and wrong-country candidate.

Steps: Browse destination spot cards. Choose the planning action for a spot. Review the selected destination, dates and budget before generating. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: An inspiration selection shall populate its source destination/preferred activity for review before generation. Acceptance constraints: Inspiration is not a booking or a guarantee the attraction fits the final schedule.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-024; static source references only; no system-case execution evidence. Related isolated test filename: not assigned.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

STD 4.4 Trip planning and enrichment

TM-TC-025 Required trip brief and bounds

Requirements and status: TM-FR-025 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Confirmed city; start tomorrow, 1/31/32 days, solo 1, couple 2, family 20/21, positive budget and zero.

Steps: Open Plan a trip. Confirm destination, dates, Traveling as and whole-party budget. Resolve any field-specific generation message before submitting. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Generation shall require a confirmed locality, positive supported-currency group budget, valid party and 1-31 inclusive future/current dates. Acceptance constraints: Maximum 20 travelers; solo=1, couple=2, family/friends=2-20. Budget ceiling 1,000,000,000. Server start-day check uses UTC while browser uses local date (D-03).

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-025; static source references only; no system-case execution evidence. Related isolated test filename: long-trip-duration.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-026 Preferences with Balanced generation policy

Requirements and status: TM-FR-026 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: New brief and restored legacy style; 20/21 interests and 8/9 stay preferences.

Steps: Set optional comma-separated interests/activities, stay preference and transportation. Generate after reviewing the brief. Historical style metadata may be displayed on reopened plans. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: New/regenerated plans shall use Balanced style/pace while accepting interests, preferred activities, stay preference and transport. Acceptance constraints: No current travel-style or pace selector; old style values remain readable. Arrays max 20; bounded item lengths.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-026; static source references only; no system-case execution evidence. Related isolated test filename: travel-preferences.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-027 Optional accommodation suggestions

Requirements and status: TM-FR-027 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Suggestion-only stay search with exact locality, empty result and LiteAPI/Overpass outage; no paid quotes.

Steps: Browse accommodations after selecting a locality. Inspect name, actual location, photos and amenities where supplied. Select an unpriced stay or choose it later. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: The current planner shall browse relevant property suggestions without requesting room prices or requiring a stay selection. Acceptance constraints: Current UI passes suggestionsOnly. LiteAPI sandbox / Rapid / OSM sources vary; no room reservation. Empty/provider-error stays do not block generation.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-027; static source references only; no system-case execution evidence. Related isolated test filename: accommodation-suggestions-only.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-028 Manual stay and retained property details

Requirements and status: TM-FR-028 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Manual named stay within destination and selected retained stay; edit party/currency/destination context.

Steps: Choose manual accommodation entry if the property is missing. Enter its name and location. Save the generated trip and inspect Saved accommodation after reopening. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: A manual/mapped stay shall persist as an unpriced planning preference and remain recognizable after reopen. Acceptance constraints: Provider identity/allowed photos can be retained. Selection does not establish availability, rate, star quality or a completed booking.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-028; static source references only; no system-case execution evidence. Related isolated test filename: planned-accommodation.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-029 Expense assumptions

Requirements and status: TM-FR-029 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: PHP 100 food/person/day, 2 people, 3 days; 1000 room/night, 1 room, 2 nights; 200 shared fare/day; 10% contingency.

Steps: Expand Budget details. Enter known estimates and their shared/per-person basis. Review the resulting known/estimated/unpriced amounts after generation. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Explicit room, food, fare, fee and contingency assumptions shall be validated and calculated without invented prices. Acceptance constraints: Default contingency 10%; permitted 0-50%. Missing costs are excluded from known subtotal, not free. Currency precision applies.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-029; static source references only; no system-case execution evidence. Related isolated test filename: grounded-itineraries.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-030 Optional pre-trip expenses

Requirements and status: TM-FR-030 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: PHP 1500 whole-party pre-trip total; 2 people and 3 days; currency/context mismatch.

Steps: Choose Add pre-trip expenses. Enter applicable transfers, passport, visa/authorization, travel tax, insurance and other amounts. Review the committed-cost breakdown. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Pre-trip user estimates shall be included once as whole-party costs with matching starting location and currency. Acceptance constraints: No government fee advice is generated. Amounts are entered estimates; verify actual fees separately.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-030; static source references only; no system-case execution evidence. Related isolated test filename: pre-trip-cost-service.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-031 Place grounding and geographic validation

Requirements and status: TM-FR-031 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Physical venue in target city, wrong-country namesake, destroyed venue, company and only coordinates.

Steps: Submit a valid brief. Read the returned place sources and geographic context. Confirm any nearby/outside-city transfer and venue details before travel. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: The backend shall select physical attractions from supported records and reject invented, wrong-country or unsupported administrative/object entries. Acceptance constraints: Wikipedia/Wikidata/OSM references can be incomplete. Max 30 km straight-line is only one filter; membership evidence remains required. Coordinates alone do not prove a venue.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-031; static source references only; no system-case execution evidence. Related isolated test filename: physical-attraction-policy.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-032 AI structured schedule and repair

Requirements and status: TM-FR-032 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: AI stubs: allowed IDs, invented ID, duplicate ID, invalid time, malformed JSON, first provider timeout, second provider success.

Steps: Generate the plan and wait for the request outcome. Review source/provider labels, day order and suggested times. Retry an explicitly retryable failure after it settles. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: AI shall arrange only supplied place IDs/times in batches of at most seven days and pass independent schema, identity, date and transfer checks. Acceptance constraints: OpenAI default gpt-4o-mini / Gemini default gemini-3.1-flash-lite are code defaults and can be configured. One invalid-output repair per batch, sequential provider fallback; 45s per adapter call, 540s outer generation signal.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-032; static source references only; no system-case execution evidence. Related isolated test filename: grounded-generation-http.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-033 Deterministic full-day scheduling

Requirements and status: TM-FR-033 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: 31-day brief with sparse venues; lunch, transfer buffers, no repeated venues; API-only travel-window fixtures separately.

Steps: Review every generated day. Expand day notes for suggested, incomplete, rest or travel status. Check actual routes/hours rather than treating local clocks as reservations. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: The scheduler shall reconcile catalog-backed visits, travel buffers, meal/free-time blocks and honest incomplete days within usable local-day windows. Acceptance constraints: Normal current Balanced cap is three catalog visits/day; no place repetition to fill a long trip. Arrival/departure/rest constraints exist in API but current planner has no input controls (see the API-only constraint inventory entry).

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-033; static source references only; no system-case execution evidence. Related isolated test filename: generated-day-scheduler.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-034 Catalog fallback and incomplete results

Requirements and status: TM-FR-034 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: No configured AI, all provider failures, sparse catalog and explicitly disabled production mock fallback.

Steps: Read a provider/fallback notice after generation. Inspect incomplete days and known costs. Revise the brief or manually refine rather than assuming missing attractions or prices. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: When AI fails or references are sparse, the system shall retain catalog-backed suggestions/allowances with explicit incomplete or unavailable disclosures. Acceptance constraints: Catalog fallback is not a guest chatbot. Mock label is possible only with explicit non-production flag; production does not invent replacement venues.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-034; static source references only; no system-case execution evidence. Related isolated test filename: grounded-generation-http.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-035 Generation idempotency and fencing

Requirements and status: TM-FR-035 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Identical key/body, same key/different body, pending <10 min, stale attempt and delayed old completion.

Steps: Wait for a pending generation. Retry the same unchanged brief only after a visible failure settles. Change the brief deliberately when creating a distinct request. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Repeated identical generation keys shall replay completed results, reject conflicting input and fence stale concurrent attempts. Acceptance constraints: 8-100-character keys. A pending record under 10 minutes conflicts; failed/stale attempts use compare-and-set fencing. This does not guarantee paid remote computation cancellation.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-035; static source references only; no system-case execution evidence. Related isolated test filename: audit-security-regressions.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-036 Automatic save after generation

Requirements and status: TM-FR-036 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Successful generated envelope; duplicate same save key and storage failure after generation.

Steps: Generate the plan. Read whether it was saved automatically. Open Saved trips to confirm; if auto-save failed, keep the draft and use the available save action. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Successful generation shall attempt an owned atomic initial save or regeneration update and preserve a recoverable draft if saving fails. Acceptance constraints: Generation success and save success are separate. Save key is generated by saveIdentity; ordinary API calls without one may have different deduplication behavior.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-036; static source references only; no system-case execution evidence. Related isolated test filename: planning-visited.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-037 Activity photographs and provenance

Requirements and status: TM-FR-037 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Exact venue identity with licensed reusable photo, no reusable media, stale wrong-place photo and provider outage.

Steps: Open a day or activity. Read photo/source credits. If a photo fails, allow recovery or retry without assuming a different image proves the exact venue. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Activity photos shall retain exact place/source identity, reusable rights and credits, with separately labeled illustrative/destination fallbacks. Acceptance constraints: No photo upload endpoint. Stored selection -> matched Wikimedia/Wikipedia -> Unsplash illustration -> trip destination -> neutral placeholder. Ordinary production flow bypasses Google provider.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-037; static source references only; no system-case execution evidence. Related isolated test filename: activity-photo-lifecycle.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-038 Date-matched weather

Requirements and status: TM-FR-038 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Forecast-supported and outside-horizon dates; OpenWeather missing key/failure.

Steps: Read weather beside a trip/day. Check forecast date and freshness. Treat future dates outside provider coverage as unavailable rather than repeated forecasts. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Weather shall distinguish current reference conditions from the dates actually covered by a forecast and disclose missing or stale coverage. Acceptance constraints: OpenWeatherMap when configured, then Open-Meteo (14-day request). Thirty-one trip days do not imply 31 forecast days.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-038; static source references only; no system-case execution evidence. Related isolated test filename: weather-service.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-039 Crowd estimates

Requirements and status: TM-FR-039 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Weather/POI-derived crowd labels and no-data destination; no live occupancy source.

Steps: Read crowd level and its explanatory note. Consider suggested earlier visits, but review the actual venue before changing plans. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Crowd context shall remain an explicitly low-confidence calendar heuristic rather than live congestion data. Acceptance constraints: Weekend and broad Philippines/December heuristics only; no foot-traffic feed or automatic live rerouting.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-039; static source references only; no system-case execution evidence. Related isolated test filename: weather-service.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-040 Advisory refresh without overwriting edits

Requirements and status: TM-FR-040 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Saved trip with selected day; refresh success and provider outage; trip content hash captured beforehand.

Steps: Choose Refresh conditions. Wait for the result. Verify the activities, notes and edits remain while forecast/crowd metadata changes or reports failure. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Refreshing conditions shall update advisory metadata without regenerating or replacing manual activity content. Acceptance constraints: A refresh is not confirmation of unchanged future conditions or automatic rescheduling.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-040; static source references only; no system-case execution evidence. Related isolated test filename: conditions.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

STD 4.5 Saved trips and traveler records

TM-TC-041 Saved-trip list and pagination

Requirements and status: TM-FR-041 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Traveler A with >20 trips and B with different records; active/completed/archived filters and empty list.

Steps: Open Saved trips. Use status filters and inspect a card. Load more when offered; open only the intended plan. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Saved trips shall show owned summary cards with status/source/cost context and load further pages without exposing other accounts. Acceptance constraints: Summary responses omit full itinerary/weather. Internal tab key bookings means Saved trips, not an active booking feature.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-041; static source references only; no system-case execution evidence. Related isolated test filename: trip-workspace.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-042 Reopen, day navigation and return

Requirements and status: TM-FR-042 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Saved 3-day trip with sparse day, selected version and costs; reopen without generating.

Steps: Choose View itinerary. Open View day itinerary, then Back to itinerary. Use Back to saved trips from the overview; native Back/Forward should retain the same trip context. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Reopening and navigating days shall preserve the selected trip/draft and never silently regenerate missing saved content. Acceptance constraints: Opening a sparse day does not repair/populate it. Legacy URL/history normalization is separate from AI generation.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-042; static source references only; no system-case execution evidence. Related isolated test filename: navigation.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-043 Activity editing

Requirements and status: TM-FR-043 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Owned active trip; edit title/time/duration and change place identity; leave before saving.

Steps: Open an active itinerary's Edit activity action. Enter the intended fields. Save the dialog, review changed totals and use Update/Save changes to persist the draft. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: The editor shall permit bounded title, time, description, location, category and estimated-cost edits, recalculating totals as user estimates. Acceptance constraints: Venue identity changes clear stale place/photo evidence. Unsupported changed price claims become estimates; server-owned management history cannot be forged.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-043; static source references only; no system-case execution evidence. Related isolated test filename: itinerary-edit-service.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-044 Add, remove and reorder activities

Requirements and status: TM-FR-044 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Owned draft: add valid venue, remove selected entry and reorder; distinguish budget allowance from visit.

Steps: Use Add activity or a day's edit controls. Reorder with the available controls or confirm removal. Inspect all affected days and save the changes deliberately. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Draft activity changes shall preserve chronological day identity and existing evidence where appropriate, with explicit removal confirmation. Acceptance constraints: At most eight saved activities/day in relevant validation. Completed/skipped management records are protected against silent removal.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-044; static source references only; no system-case execution evidence. Related isolated test filename: itinerary-editor.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-045 Save manual changes and recover drafts

Requirements and status: TM-FR-045 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Draft with unchanged/replaced place identity, currency precision limits, API failure and restored owner-specific local draft.

Steps: Use Update/Save changes after reviewing edits. If a leave/recovery dialog appears, choose deliberately to save, keep/discard or restore the draft. Retry a failed save without navigating away. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Manual edits shall remain a browser draft until saved, with owner-scoped recovery and guards against unintended navigation loss. Acceptance constraints: Browser storage may be blocked/full. A browser draft is not a server backup; invalid/wrong-owner drafts are rejected.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-045; static source references only; no system-case execution evidence. Related isolated test filename: saved-trip-consistency.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-046 Regenerate existing trip

Requirements and status: TM-FR-046 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Owned active trip; modified valid brief; generation failure while existing saved version remains.

Steps: Open an active trip’s regeneration/planning settings. Review its restored brief; move passed dates forward as prompted. Generate deliberately and confirm the saved result/version. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Regeneration shall require deliberate input review and produce a version of the same active trip only after valid generation/save. Acceptance constraints: Current Balanced defaults apply to regenerated plans. Failed generation/save must leave the saved plan available; no automatic regeneration on open.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-046; static source references only; no system-case execution evidence. Related isolated test filename: date.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-047 Version history and whole-version restoration

Requirements and status: TM-FR-047 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Current and older complete version envelopes, legacy incomplete envelope and stale concurrent update.

Steps: Expand a saved trip’s History. Load more metadata if necessary and choose a version. Review and confirm Restore only for an active/reopened trip. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Version restoration shall atomically recover the selected owned snapshot and its matching original trip context as a new version. Acceptance constraints: 50 metadata records/page. Legacy versions can lack party/context and fail safely; unrecorded context is not guessed.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-047; static source references only; no system-case execution evidence. Related isolated test filename: stored-photo-preservation.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-048 Archive and restore trip

Requirements and status: TM-FR-048 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Active and completed owned trips; archive then restore; completedAt preserved.

Steps: Use a card’s More options -> Archive. Inspect Archived trips. Restore when needed before editing or recording a batch of visits. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Archiving shall retain the trip and its prior completion state; restoring shall return it to active/completed accordingly. Acceptance constraints: Archived trips cannot be directly completed/reopened; restore first. Archiving is distinct from permanent deletion.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-048; static source references only; no system-case execution evidence. Related isolated test filename: trip-completion.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-049 Complete and reopen trip

Requirements and status: TM-FR-049 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Active/completed/archived owned trips; separately capture VisitedPlace count.

Steps: Choose Mark completed for a finished plan. Use Reopen to make changes again. Record actual visited places separately. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Trip completion/reopening shall update lifecycle state only and shall not automatically create/remove visit history. Acceptance constraints: Completion does not prove attendance. Completed itineraries remain readable but editing/management requires reopen.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-049; static source references only; no system-case execution evidence. Related isolated test filename: trip-completion.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-050 Delete owned trip

Requirements and status: TM-FR-050 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Disposable trip with versions, linked feedback and linked visited record; audit captured; confirm deletion.

Steps: Open More options -> Delete. Read the confirmation. Confirm only when permanent removal is intended; cancel to retain the trip. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Deletion shall require deliberate confirmation, delete the owned trip/version records and preserve detached visit/feedback history according to foreign keys. Acceptance constraints: No normal UI undelete is implemented. Documentation task did not execute deletion.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-050; static source references only; no system-case execution evidence. Related isolated test filename: trip-workspace.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-051 Saved cover photos and stay details

Requirements and status: TM-FR-051 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Saved trip with exact destination cover and hotel details; media unavailable and retained legacy stay.

Steps: Inspect the Saved trip card cover and selected accommodation. Open its source credits or details. If media is unavailable, retain the plan and its identity. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Saved cards shall preserve relevant destination cover/photo attribution and any selected unpriced stay without altering itinerary content. Acceptance constraints: Cover selection is not attraction verification. Unsplash selection tracking is separate and must not run on every render.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-051; static source references only; no system-case execution evidence. Related isolated test filename: destination-cover-photo.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-052 Manual visited-place recording

Requirements and status: TM-FR-052 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Manual destination/place with actual past date; future date and same identity/date duplicate.

Steps: Open Visited places. Add a place name, destination and actual visit date. Save and inspect the personal history. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: A traveler shall explicitly record a named place and past/present actual visit date under their own account. Acceptance constraints: Future visit dates are rejected using the supplied valid timezone. Manual records are not automatically geographically verified.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-052; static source references only; no system-case execution evidence. Related isolated test filename: planning-visited.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-053 Record places from a saved trip

Requirements and status: TM-FR-053 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Saved trip with valid venue, allowance, rest block and duplicate; stale candidate fingerprints and parallel submission.

Steps: On an active/completed trip choose Record visited places. Select actual places and an actual visit date, then confirm. Reopen to inspect Already recorded states. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: The batch visit modal shall list actual saved place candidates across days, prevent stale/duplicate selections and save selected actual visits atomically. Acceptance constraints: Allowances/rest/placeholders excluded. Archived trip requires restore. Batch duplicate prevention is stronger than the date-specific database unique key.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-053; static source references only; no system-case execution evidence. Related isolated test filename: trip-visit-candidates.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-054 Visited history search and removal

Requirements and status: TM-FR-054 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Owned visited records >20; case-insensitive search, no results and remove only own item.

Steps: Open Visited places. Search a place/destination, load more if offered, inspect retained media and remove a record only if intended. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Visit history shall be owner-scoped, searchable, date-ordered and paginated, with explicit individual removal. Acceptance constraints: Deleting a trip leaves its visited records detached. Removing a visit does not remove the saved itinerary.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-054; static source references only; no system-case execution evidence. Related isolated test filename: trip-visit-candidates.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-055 Itinerary feedback and updates

Requirements and status: TM-FR-055 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Rating 1/5/0/6, valid category, 10/2000-character comment; own trip then repeat feedback.

Steps: Choose Rate trip or Feedback and support. Select the intended saved itinerary, rating/category and comment. Submit; reopen to edit existing feedback. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Feedback shall bind one 1-5 rating, supported category and 10-2000-character comment to an owned saved trip, updating the same record on resubmission. Acceptance constraints: Category budget_accuracy/weather_info/ai_cohesion/route_efficiency. Sentiment remains not_analyzed; no sentiment AI is implemented.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-055; static source references only; no system-case execution evidence. Related isolated test filename: feedback.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-056 Traveler problem reports

Requirements and status: TM-FR-056 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Subject length 5/120; message 10/2000; empty/overlong report; admin notification sink.

Steps: Open Feedback and support or the account report form. Enter a 5-120-character subject and 10-2000-character details. Submit and monitor notifications. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Travelers shall submit bounded problem reports for administrator review and receive resolution notices. Acceptance constraints: Reports are not an itinerary chat command or a guarantee of support response time.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-056; static source references only; no system-case execution evidence. Related isolated test filename: report-service.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-057 Traveler notifications

Requirements and status: TM-FR-057 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Owned read/unread notices; latest 50; mark one/all and attempt B record ID.

Steps: Open the bell or Notifications. Read a notice and follow its available destination. Mark a notice or all notices read as intended. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: The traveler notification view shall display owned notices and permit marking one/all read without changing another recipient’s data. Acceptance constraints: Traveler updates use request/polling behavior; administrator live SSE is a distinct feature. Notices are not push/email delivery guarantees.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-057; static source references only; no system-case execution evidence. Related isolated test filename: admin-notifications.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-058 Traveler profile and contact settings

Requirements and status: TM-FR-058 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Name, optional phone, HTTPS avatar and bio; attempt email/role/status fields and HTTP avatar.

Steps: Open Your account. Edit name, contact number, HTTPS photo URL or Travel profile. Choose Save profile and inspect the result. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Profile editing shall persist only allowed name/contact/avatar/bio fields and preserve account role/email/security state. Acceptance constraints: Name 2-80, phone <= 30, bio <= 500, avatar HTTPS <= 2048. Email is read-only; unknown protected keys do not become writable.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-058; static source references only; no system-case execution evidence. Related isolated test filename: profile-service.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

STD 4.6 Saved trip maps

TM-TC-059 Owned saved-trip map and timeline

Requirements and status: TM-FR-059 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Owned trip selected from Saved trips; verified and unresolved map stops; tile error.

Steps: Choose View map / Navigate on a saved trip. Select the saved day and timeline activity. Read exact/ambiguous/unavailable location states. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: The saved-trip map shall derive days/stops from the selected owned itinerary and separate daily allowances from visit stops. Acceptance constraints: Opening a map does not complete an activity or record a visit. Skipped activities remain visible but are excluded from relevant routing.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-059; static source references only; no system-case execution evidence. Related isolated test filename: trip-navigation.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-060 Exact map-place resolution and photos

Requirements and status: TM-FR-060 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Ambiguous exact stop and another-country namesake; user selects supported exact candidate.

Steps: Select an unresolved stop. Search/choose an exact place candidate and inspect its location and photo/source details. Continue only after confirming the intended venue. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: An uncertain activity shall require an exact source-matched candidate before navigation, without silently promoting a vague name. Acceptance constraints: Vague allowances/activities cannot become precise stops by guess. Location-provider failure leaves unresolved feedback.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-060; static source references only; no system-case execution evidence. Related isolated test filename: trip-navigation.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-061 Selected-day ordered route

Requirements and status: TM-FR-061 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Two/one/50/51 resolved stops, walk/cycle/drive, skipped activity, timeout, no-route and 429.

Steps: Choose a saved day with at least two exact places. Choose the travel mode and request its day route. Review the stop order and any omitted-stop notice. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Day routing shall follow recorded verified-stop order for the selected walking/cycling/driving mode and disclose omitted unresolved stops. Acceptance constraints: OSRM road/path estimates, no live traffic. Public provider quota, no-route, excessive snap distance and unavailable routes are explicit.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-061; static source references only; no system-case execution evidence. Related isolated test filename: trip-navigation.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-062 Location sharing and route guidance

Requirements and status: TM-FR-062 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Explicit GPS start/stop/denial; selected saved day; route off-course and navigation away.

Steps: Choose Navigate/Use current location where offered. Review permission and selected destination/mode. Follow displayed estimated guidance; Stop sharing when finished. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Live route guidance shall begin only after an explicit location-sharing action and permit stopping, manual fallback and mode changes. Acceptance constraints: HTTPS/permission required. Denial, inaccurate/stale position and route failure retain recovery paths. No safety-critical navigation guarantee.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-062; static source references only; no system-case execution evidence. Related isolated test filename: navigation.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-063 Activity-specific weather

Requirements and status: TM-FR-063 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Resolved stop with local date in forecast horizon, outside horizon, missing coordinates and provider budget exhausted.

Steps: Select a mapped activity. Open its weather view and review Current versus scheduled-date weather. Read unavailable/hour-reference messages when applicable. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Activity weather shall keep current and forecast requests distinct and use the exact resolved stop/date/time within provider coverage. Acceptance constraints: Exact location needed; scheduled weather outside actual coverage remains unavailable. Global provider request budgets apply.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-063; static source references only; no system-case execution evidence. Related isolated test filename: activity-weather.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-064 Retired GPS URL compatibility

Requirements and status: TM-FR-064 | Not executed

Prerequisites: Authorized disposable environment; Traveler / guest. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Legacy /gps and /dashboard/gps with valid/invalid trip/day; no new planning form.

Steps: Open an old GPS bookmark. Read the moved-navigation message. Open the selected saved-trip map, signing in if required. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Legacy GPS URLs shall redirect into Saved trips navigation while retaining only valid trip/day identifiers. Acceptance constraints: The standalone GPS workspace is removed; compatibility redirect does not restore removed navigation tabs.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-064; static source references only; no system-case execution evidence. Related isolated test filename: retired-navigation.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

STD 4.7 Managed day activities

TM-TC-065 Confirm completion, skip or restore activity

Requirements and status: TM-FR-065 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Owned active trip, fresh/stale revision, confirm true/false; complete, skip and restore; VisitedPlace count unchanged.

Steps: In the saved-trip map select an activity. Choose Mark as Completed, Skip Activity or restore/undo. Review and confirm; inspect the retained activity/history. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Managed activity status changes shall require owner, active trip, current revision, saved activity identity and explicit confirmation. Acceptance constraints: Completed details are protected until undo. Skip retains cost/history and excludes remaining-day routing/replanning. Status change does not record a VisitedPlace.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-065; static source references only; no system-case execution evidence. Related isolated test filename: smart-itinerary.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-066 Confirmed activity edit/reschedule

Requirements and status: TM-FR-066 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Duration 15/720/14/721, cross-day move, completed entry, duplicate/overlap and wrong owner.

Steps: Select Edit Activity or Reschedule Activity. Enter valid fields/day/local time and duration. Confirm; reload if a conflict reports a changed saved trip. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Managed edit/reschedule shall reject stale revisions, new duplicate places, time overlaps and changes to completed details until undone. Acceptance constraints: Duration 15-720 minutes. Moving to another day requires Reschedule; target max 8 activities. Changed venue/photo evidence is revalidated.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-066; static source references only; no system-case execution evidence. Related isolated test filename: smart-itinerary.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-067 AI remaining-day preview

Requirements and status: TM-FR-067 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Owned day with completed/skipped/remaining stops; stub AI fabricated/new IDs, routing outage and exhausted replan quota.

Steps: Choose Replan My Day for an active saved day. Enter an available destination-local time window. Preview and compare proposed times, Flexible deferrals and warnings. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Replan My Day shall propose times for every remaining saved activity exactly once without writing the itinerary or changing costs/completed history. Acceptance constraints: Three previews per owner/5 minutes. Existing Gemini/OpenAI required; no mock fallback. Missing saved durations must be entered first. Preview can consume provider quota.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-067; static source references only; no system-case execution evidence. Related isolated test filename: smart-itinerary.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-068 Accept or cancel revised day

Requirements and status: TM-FR-068 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Preview token valid/expired >15 min/wrong owner/stale revision; accept confirmed and cancel.

Steps: Inspect the full preview. Choose Accept revised day to persist, or Cancel replanning to retain the original. Reload and preview again after expiry/conflict. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Acceptance shall validate a signed owner/trip/revision-bound preview and persist one reviewed change; cancellation shall leave itinerary/version data unchanged. Acceptance constraints: Preview token expires after 15 minutes. Browser abort does not prove remote model billing stopped. Completed/skipped records and saved costs remain.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-068; static source references only; no system-case execution evidence. Related isolated test filename: smart-itinerary.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-069 Activity history and legacy compatibility

Requirements and status: TM-FR-069 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Legacy item without stable ID and modern persisted action history; read-only derivation then first explicit edit.

Steps: Expand Activity history after a confirmed management change. Review recorded action/date/time. Use version restore when a whole prior plan is needed. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Management shall preserve stable activity identity/history and default legacy records to Planned without writing on open. Acceptance constraints: Legacy IDs are derived read-only until confirmed persistence. Management metadata is server-owned, separate from a free-form manual edit.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-069; static source references only; no system-case execution evidence. Related isolated test filename: smart-itinerary.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

STD 4.8 Administration

TM-TC-070 Administrator overview

Requirements and status: TM-FR-070 | Not executed

Prerequisites: Authorized disposable environment; Administrator. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Administrator with known disposable generations/feedback; no feedback; UTC+8 day boundary.

Steps: Sign in as an authorized administrator. Open Overview. Read all-time totals, recent generation chart and absent/available average rating. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: The overview shall show stored aggregate account/generation/feedback counts and seven calendar days without invented activity. Acceptance constraints: Calendar uses UTC+8/Asia/Shanghai. Counts describe stored records, not active concurrent users or measured satisfaction.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-070; static source references only; no system-case execution evidence. Related isolated test filename: admin-overview.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-071 User search and access state

Requirements and status: TM-FR-071 | Not executed

Prerequisites: Authorized disposable environment; Administrator. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Administrator and traveler; search/filter user, admin target, explicit suspension/restore and unknown action.

Steps: Open Users. Search name/email/status and select filters. Choose a traveler’s suspend/restore action, inspect the confirmation and submit deliberately. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: User management shall search/filter permitted directory fields and suspend/restore non-admin accounts with confirmation and notifications. Acceptance constraints: Admin accounts cannot be changed through this action. No public admin creation/role reassignment is exposed. Unknown status currently normalizes to active (D-04).

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-071; static source references only; no system-case execution evidence. Related isolated test filename: account-roles.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-072 Send traveler warning

Requirements and status: TM-FR-072 | Not executed

Prerequisites: Authorized disposable environment; Administrator. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Administrator warning text bounds and selected non-admin; notification/audit transaction failure.

Steps: Open Users and the selected traveler’s warning action. Enter a factual bounded warning. Confirm and inspect the outcome. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: A warning shall target an existing traveler with a 10-2000-character message and persist a recipient-scoped notice/audit event. Acceptance constraints: In-app notice only; not automatic email/SMS. The documentation task sent no warning.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-072; static source references only; no system-case execution evidence. Related isolated test filename: user-warning.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-073 Review/resolve problem reports

Requirements and status: TM-FR-073 | Not executed

Prerequisites: Authorized disposable environment; Administrator. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Pending/closed report and wrong kind; resolve once, duplicate and permission denied.

Steps: Open Reports. Read the selected report’s full content and reporter context. Choose Mark resolved after review; observe the resulting status. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Administrators shall inspect full traveler reports and resolve only pending problem reports with an audited recipient notification. Acceptance constraints: Resolution is a recorded decision, not technical proof the underlying defect is fixed. Already resolved/wrong-kind requests are rejected.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-073; static source references only; no system-case execution evidence. Related isolated test filename: report-service.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-074 Review suspension appeals

Requirements and status: TM-FR-074 | Not executed

Prerequisites: Authorized disposable environment; Administrator. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Pending appeal with suspended account; approve/reject plus 10/2000-character decision and concurrent reviewer.

Steps: Open Reports and the appeal category. Read the appeal. Enter a 10-2000-character decision message, select the decision and confirm. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Appeal review shall validate a pending appeal and current suspended traveler state before approved/rejected decision and bounded explanation. Acceptance constraints: Approve restores account access; reject retains suspension. Stale/double review and already-restored account are refused.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-074; static source references only; no system-case execution evidence. Related isolated test filename: appeal-service.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-075 Live reports and reconnect snapshots

Requirements and status: TM-FR-075 | Not executed

Prerequisites: Authorized disposable environment; Administrator. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Admin SSE connection, state change on another process, session invalidation and disconnect/backpressure.

Steps: Open the admin workspace. Read Live reports connected or reconnect feedback. On a connection failure, retain the view and allow reconnection/refresh. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: The live report stream shall authenticate repeatedly, send scoped snapshots, reconcile cross-instance changes and clean up closed/stalled connections. Acceptance constraints: SSE (not Socket.IO): 5-second reconciliation, 5-minute connection rotation, event-triggered local refresh and bounded write buffer.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-075; static source references only; no system-case execution evidence. Related isolated test filename: admin-reports-stream.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-076 System Health and selected audit activity

Requirements and status: TM-FR-076 | Not executed

Prerequisites: Authorized disposable environment; Administrator. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Provider configured/unconfigured without calling it; known 25 generation/100 audit records; non-admin denied.

Steps: Open System Health. Inspect configuration indicators, failed generations and available audit search. Report a provider outage using its actual evidence. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: System Health shall distinguish service configuration from verified availability and show recent generation failures and selected account/audit activity. Acceptance constraints: No live uptime/latency dashboard, AI prompt editor or automated model curation. Configured means key presence, not a successful provider call.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-076; static source references only; no system-case execution evidence. Related isolated test filename: admin-overview.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-077 Feedback filters and exact details

Requirements and status: TM-FR-077 | Not executed

Prerequisites: Authorized disposable environment; Administrator. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Admin feedback search, rating/sentiment/date range, pagination and deleted linked trip.

Steps: Open User Feedback. Search or filter rating/sentiment/date, page through results and select a record. Open its related itinerary if retained. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Feedback review shall paginate/search/filter stored feedback and open exact linked feedback/related itinerary without inferring sentiment. Acceptance constraints: Deleted trip/user links can be null. Sentiment defaults not_analyzed; filters do not prove automated analysis.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-077; static source references only; no system-case execution evidence. Related isolated test filename: feedback-notification-link.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-078 Administrator notifications

Requirements and status: TM-FR-078 | Not executed

Prerequisites: Authorized disposable environment; Administrator. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Admin unread and read notifications, deep links, latest 50 and selected/all marking.

Steps: Open Notifications or a bell notice. Follow its report/feedback/system destination. Mark read and inspect the unread badge. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Admin notifications shall retain recipient isolation, meaningful category/deep link and accurate unread count while allowing one/all read. Acceptance constraints: Category is encoded in link metadata for current notices; compatibility handles older notices. Only that admin’s read state changes.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-078; static source references only; no system-case execution evidence. Related isolated test filename: admin-notifications.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-079 Administrator own profile

Requirements and status: TM-FR-079 | Not executed

Prerequisites: Authorized disposable environment; Administrator. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Administrator own profile only; same bounded fields as traveler; attempted role mutation.

Steps: Open Account Settings. Edit allowed name/contact/profile fields and save. Verify the returned profile before leaving. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Account Settings shall update the administrator’s own permitted contact/profile fields without exposing password/role mutation. Acceptance constraints: Same allowed-field and length constraints as the traveler profile; no email/role/security-field editing.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-079; static source references only; no system-case execution evidence. Related isolated test filename: profile-service.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

STD 4.9 Technical services

TM-TC-080 API error contracts and safe recovery

Requirements and status: TM-FR-080 | Not executed

Prerequisites: Authorized disposable environment; All callers. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Validation/auth/permission/conflict/rate/provider/database failures; malformed JSON and unexpected exception.

Steps: If a request fails, read the visible message. Retry only after a retryable failure/cooldown; retain unsaved work. Provide time and non-sensitive context to support. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: The API shall return stable safe error/code/retryability metadata, sanitized malformed-JSON feedback and distinct dependency failures. Acceptance constraints: Database outages can return retryable 503 rather than false logout. No exhaustive penetration/security certification is asserted.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-080; static source references only; no system-case execution evidence. Related isolated test filename: api-error-contract.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-081 Database retries and pooled access

Requirements and status: TM-FR-081 | Not executed

Prerequisites: Authorized disposable environment; API process. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Transient database connection failure, bounded retry and terminal outage; no production connection.

Steps: Operators configure the backend database URL securely. On an outage, inspect safe health/error feedback and restore connectivity before retrying affected operations. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: The API shall use PostgreSQL/Prisma pooled TCP access with bounded retries only for recognized connection failures. Acceptance constraints: An ambiguous limiter retry may count twice (fail-closed). Not a blanket retry of every business write or a proven backup/restore service.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-081; static source references only; no system-case execution evidence. Related isolated test filename: oauth-database-retry.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-082 Shared rate limits and request-driven cleanup

Requirements and status: TM-FR-082 | Not executed

Prerequisites: Authorized disposable environment; API process. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Two test processes sharing disposable limiter DB; boundary attempts and expired buckets/revocations.

Steps: Operators monitor 429 outcomes and database health. Respect Retry-After and avoid rapid repeat submissions. Review provider quotas before scaling. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Rate limits shall use shared fixed-window PostgreSQL buckets and opportunistically prune old buckets/revocations. Acceptance constraints: Cleanup is a 1% request-triggered chance, not a scheduled worker. Authentication itself depends on database availability for rate limiting.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-082; static source references only; no system-case execution evidence. Related isolated test filename: security.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-083 Provider cache and unavailable disclosures

Requirements and status: TM-FR-083 | Not executed

Prerequisites: Authorized disposable environment; API process / traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Fake clock and provider stubs; identical concurrent request, expiry, failure, Retry-After and cleanup.

Steps: Read source/freshness labels. Refresh only when needed. Treat stale data as a fallback and confirm current prices/availability with the original provider. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Provider caching shall coalesce identical in-flight requests, bound entries, honor freshness/stale cutoffs and expose unavailable results honestly. Acceptance constraints: General cache max 300 entries. Flights/hotels 5m fresh/15m cutoff; activities 30m/2h; weather 15m/1h; FX 6h/24h.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-083; static source references only; no system-case execution evidence. Related isolated test filename: provider-cache.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-084 Security headers, HTTPS and same-origin writes

Requirements and status: TM-FR-084 | Not executed

Prerequisites: Authorized disposable environment; All callers / operator. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Production-like HTTPS/proxy stub; HTTP, foreign Origin, missing/safe fetch metadata and CSP.

Steps: Operators configure trusted frontend HTTPS origins and independent secrets. Clients access the normal frontend; an origin or access denial is not solved by bypassing security controls. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Production shall enforce HTTPS/security headers and reject untrusted cross-site browser writes with server-side authorization. Acceptance constraints: No full encryption-at-rest/legal compliance or exhaustive XSS/CSRF proof. Frontend CSP includes unsafe-inline and is an explicit review item.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-084; static source references only; no system-case execution evidence. Related isolated test filename: security.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-085 Operational health and graceful shutdown

Requirements and status: TM-FR-085 | Not executed

Prerequisites: Authorized disposable environment; Operator. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Readiness with reachable/unreachable disposable DB and graceful signal; no live process terminated.

Steps: Inspect GET / and /health in an authorized environment. Check dependency state as well as HTTP status; use the documented healthcheck on a release candidate. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Operational endpoints shall expose basic API identity/database readiness and the process shall close HTTP/database resources on termination. Acceptance constraints: Health is a point observation, not measured uptime or a restore drill. The healthcheck script forwards HTTPS scheme for container checks.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-085; static source references only; no system-case execution evidence. Related isolated test filename: container-healthcheck.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-086 Contracts and paired-revision validation

Requirements and status: TM-FR-086 | Not executed

Prerequisites: Authorized disposable environment; Maintainer. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Copied source with matching generated contracts and deliberately drifted copied schema; no original writes.

Steps: Use contracts:check in an isolated candidate. Compare schema differences before authorizing contracts:sync or a release. Record both repository revisions. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: API contracts shall remain aligned with generated frontend snapshots and be checked against a pinned compatible backend revision. Acceptance constraints: contracts:sync writes files; documentation task uses copies/check mode only. CI definitions are configuration evidence, not proof a run passed.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-086; static source references only; no system-case execution evidence. Related isolated test filename: api-error-contract.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-087 Container deployment and rollback configuration

Requirements and status: TM-FR-087 | Not executed

Prerequisites: Authorized disposable environment; Maintainer. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Disposable containers/database and fake release image; health success/failure and additive migration rollback plan.

Steps: Prepare an isolated release and required configuration. Review backup/migration/rollback plans. Only after separate release authorization run the documented deployment procedure and verify health. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: A proposed release shall build a non-root image, apply reviewed migrations explicitly and require health before retaining the new app image. Acceptance constraints: Configured pipeline, not executed here. Image rollback does not undo applied database migrations. Domain/hosting/backup acceptance needs actual operational evidence.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-087; static source references only; no system-case execution evidence. Related isolated test filename: docker-deployment.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

STD 4.10 Compatibility and excluded scope

TM-TC-088 Flight and activity comparison API

Requirements and status: TM-FR-088 | Not executed

Prerequisites: Authorized disposable environment; Traveler API consumer; UI removed. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Retained comparison API, configured stub provider, no frontend tab; expired/invalid offer.

Steps: There is no current Travel options tab. Maintainers can inspect the API contract in an isolated integration environment; ordinary travelers use the retained planner. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Existing authenticated comparison endpoints shall normalize source-backed flight/activity results and unavailable states, without claiming a current traveler comparison screen. Acceptance constraints: Unreferenced comparison component retained in source. Amadeus 1-9 travelers; configured production/test distinction. Mark partial because API exists but UI is retired.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-088; static source references only; no system-case execution evidence. Related isolated test filename: amadeus-travel-provider.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-089 Hotel quote/revalidation API

Requirements and status: TM-FR-089 | Not executed

Prerequisites: Authorized disposable environment; Traveler API consumer; current UI suggestions only. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Retained quote API with signed offer, context change/expiry and current suggestion-only UI.

Steps: The main planner shows suggestions only. In an isolated API test, request a supported quote, revalidate it and inspect changed/expired/provider-error outcomes. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Existing quote APIs shall bind owner/context/occupancy/expiry and reject changed/sold-out/invalid quotes without asserting current UI booking. Acceptance constraints: API occupancy 1-8 rooms, adults/child ages 0-17 and total <= 20. LiteAPI remains sandbox; Rapid rights gated. No current price/occupancy controls in main planner.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-089; static source references only; no system-case execution evidence. Related isolated test filename: accommodation-check.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-090 Generation arrival/departure/rest API constraints

Requirements and status: TM-FR-090 | Not executed

Prerequisites: Authorized disposable environment; Traveler API consumer; no current form controls. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: API-only arrival/departure/rest constraints valid/invalid; verify current planner lacks those controls.

Steps: No current planner controls set these fields. Maintainers may specify them in an isolated API case and inspect rest/travel/incomplete day output. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: The generation API shall validate optional destination-local arrival/departure clocks and in-range rest dates without implying an exposed UI. Acceptance constraints: HH:MM clocks; <= 31 in-range rest dates. Arrival adds 90 minutes; departure reserves 120. API-only capability is partial for normal users.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-090; static source references only; no system-case execution evidence. Related isolated test filename: generated-day-scheduler.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-091 Removed owner/marketplace/payment workflows

Requirements and status: TM-FR-091 | Not executed

Prerequisites: Authorized disposable environment; No active user role. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Current guest/traveler/admin menus, route register and legacy schema; no payment credentials.

Steps: Do not look for an Owner workspace, checkout or payment action. Use the retained trip planner and obtain actual reservations from external providers. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Removed owner registration, booking, payment, promotion and trust/profile-moderation actions shall remain unavailable rather than be documented as working product features. Acceptance constraints: Schema presence is not feature availability. PayMongo is not integrated; owner role no longer exists. Legacy data is not deleted by this documentation work.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-091; static source references only; no system-case execution evidence. Related isolated test filename: account-roles.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-092 Retired preferences/comparison tabs and duplication

Requirements and status: TM-FR-092 | Not executed

Prerequisites: Authorized disposable environment; Traveler. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Old preferences/compare navigation identifiers and direct obsolete GPS URLs.

Steps: Use current sidebar entries. An old preferences/compare bookmark returns to the supported workspace; use new trip planning when a distinct plan is needed. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Retired Preferences/Travel options tab URLs shall normalize safely, and the removed duplicate-trip action shall reject requests. Acceptance constraints: Unreferenced TripPlannerForm still contains a travel-style prop; current dashboard does not render it. Do not infer a current style selector from that file.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-092; static source references only; no system-case execution evidence. Related isolated test filename: retired-navigation.test.ts.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

TM-TC-093 Guest chatbot, live crowd feed and automatic backups

Requirements and status: TM-FR-093 | Not executed

Prerequisites: Authorized disposable environment; Unavailable / proposed. For state-changing workflows, create only test records and reset the disposable database after evidence capture.

Test data: Public site, current route/process register, missing AI/backup setup; no guest chat or live crowd source.

Steps: Guests can explore the public site and account dialogs. Use authenticated structured planning; do not expect a guest chat, live crowd telemetry or an application-managed daily backup interface. Repeat with the stated boundary/failure fixture; inspect the network result and reload where persistence applies. Compare records only in the disposable database.

Expected: Unimplemented historical requirements shall be identified explicitly and must not be presented as observed system capabilities. Acceptance constraints: Future work requires separate requirements, provider/security design and execution evidence. Old 99.9% uptime, 500-user capacity and automatic daily backup statements remain unverified targets.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-093; static source references only; no system-case execution evidence. Related isolated test filename: not assigned.

Cleanup: Delete/reset disposable fixtures only; never remove existing application records.

STD 4.11 Focused boundary and failure procedures

TM-XT-001 Date boundary consistency

Requirements and status: TM-FR-025, TM-NFR-020 | Not executed

Prerequisites: Authorized disposable fixtures, stubs and isolated PostgreSQL where persistence is involved. No existing records or external paid services.

Test data: Fake browser at UTC+8 00:30 while server UTC date is previous day; then inverse offset.

Steps: Submit today/yesterday/tomorrow in client and API with identical clock.

Expected: Client and server apply the approved same policy; discrepancies fail intended acceptance and link D-03.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-025, TM-NFR-020; static source references only; no system-case execution evidence.

Cleanup: Reset disposable fixtures; retain sanitized results only.

TM-XT-002 Known cost arithmetic

Requirements and status: TM-FR-029, TM-NFR-004 | Not executed

Prerequisites: Authorized disposable fixtures, stubs and isolated PostgreSQL where persistence is involved. No existing records or external paid services.

Test data: PHP 100 food x2 people x3 days; room 1000 x1 x2 nights; shared fare 200 x3 days; pre-trip 1500; activities/other 0; contingency 10%; budget 10000.

Steps: Generate from deterministic grounded fixture, save, reopen and compare components.

Expected: Food 600; room 2000; fare 600; pre-trip 1500; subtotal 4700; contingency 470; total 5170; remainder 4830; no additional charge per person for shared fare or pre-trip.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-029, TM-NFR-004; static source references only; no system-case execution evidence.

Cleanup: Reset disposable fixtures; retain sanitized results only.

TM-XT-003 Zero decimal and exact shares

Requirements and status: TM-FR-021, TM-NFR-004 | Not executed

Prerequisites: Authorized disposable fixtures, stubs and isolated PostgreSQL where persistence is involved. No existing records or external paid services.

Test data: JPY 10000 group of 3; JPY 1.5 invalid raw input; PHP 0.01 boundary.

Steps: Validate brief and save; calculate per-person integer shares and compare summed shares.

Expected: Reject fractional JPY; distribute total 10000 as 3334/3333/3333; shares sum exactly to total. Do not claim raw save precision passes until D-07 is resolved.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-021, TM-NFR-004; static source references only; no system-case execution evidence.

Cleanup: Reset disposable fixtures; retain sanitized results only.

TM-XT-004 Save currency precision guard

Requirements and status: TM-FR-036, TM-FR-045, TM-NFR-007 | Not executed

Prerequisites: Authorized disposable fixtures, stubs and isolated PostgreSQL where persistence is involved. No existing records or external paid services.

Test data: PHP 1.001 and JPY 1.5 in direct save payload, then conforming values.

Steps: Submit via disposable API with a valid signed grounding envelope and inspect validation/storage.

Expected: Reject unsupported precision before normalization; no silent rounding of unauthorized raw amounts. D-07 remains a static concern, not a passing intended behavior.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-036, TM-FR-045, TM-NFR-007; static source references only; no system-case execution evidence.

Cleanup: Reset disposable fixtures; retain sanitized results only.

TM-XT-005 Long destination identity persistence

Requirements and status: TM-FR-025, TM-FR-035 | Not executed

Prerequisites: Authorized disposable fixtures, stubs and isolated PostgreSQL where persistence is involved. No existing records or external paid services.

Test data: Supported selected destination label 120/121/2000 characters and stable source identity.

Steps: Submit valid brief, observe generation-record creation, terminal response and no sensitive server details.

Expected: Documented accepted field lengths are consistent with DB storage; reject overlength with actionable 400 rather than internal failure. Pending D-06.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-025, TM-FR-035; static source references only; no system-case execution evidence.

Cleanup: Reset disposable fixtures; retain sanitized results only.

TM-XT-006 Unknown administrator status command

Requirements and status: TM-FR-071, TM-NFR-001 | Not executed

Prerequisites: Authorized disposable fixtures, stubs and isolated PostgreSQL where persistence is involved. No existing records or external paid services.

Test data: Administrator with target traveler; unknown status string, missing status, active and suspended.

Steps: Send each command on a disposable account and inspect status/audit.

Expected: Unknown/missing commands are rejected without restoring the account. Static normalization concern D-04 must not be adopted as intended behavior.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-071, TM-NFR-001; static source references only; no system-case execution evidence.

Cleanup: Reset disposable fixtures; retain sanitized results only.

TM-XT-007 Cross-owner identifiers

Requirements and status: TM-FR-041, TM-FR-047, TM-FR-059, TM-FR-065 | Not executed

Prerequisites: Authorized disposable fixtures, stubs and isolated PostgreSQL where persistence is involved. No existing records or external paid services.

Test data: Travelers A and B; B trip/version/visit/map/activity IDs.

Steps: As A, read/update/delete/restore/map/replan B identifiers; then repeat as guest.

Expected: No B data disclosed or mutated; consistent unauthorized/not-found response; state unchanged.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-041, TM-FR-047, TM-FR-059, TM-FR-065; static source references only; no system-case execution evidence.

Cleanup: Reset disposable fixtures; retain sanitized results only.

TM-XT-008 Moderation race

Requirements and status: TM-FR-015, TM-FR-073, TM-FR-074, TM-NFR-008 | Not executed

Prerequisites: Authorized disposable fixtures, stubs and isolated PostgreSQL where persistence is involved. No existing records or external paid services.

Test data: Suspended disposable traveler; two simultaneous appeals or two simultaneous reviewers.

Steps: Synchronize submissions, capture committed rows and notifications.

Expected: At most one pending appeal and one terminal decision; no duplicate restoration/notification; transaction rollback is atomic.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-015, TM-FR-073, TM-FR-074, TM-NFR-008; static source references only; no system-case execution evidence.

Cleanup: Reset disposable fixtures; retain sanitized results only.

TM-XT-009 Visit fingerprint and concurrency

Requirements and status: TM-FR-053, TM-NFR-008 | Not executed

Prerequisites: Authorized disposable fixtures, stubs and isolated PostgreSQL where persistence is involved. No existing records or external paid services.

Test data: Same confirmed visit batch twice concurrently; then change saved itinerary.

Steps: Submit same batch then stale candidate batch; inspect visited rows and feedback.

Expected: No duplicate selected visit; stale identities reject; rest/allowance never recorded as real visit.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-053, TM-NFR-008; static source references only; no system-case execution evidence.

Cleanup: Reset disposable fixtures; retain sanitized results only.

TM-XT-010 Untrusted AI response

Requirements and status: TM-FR-032, TM-FR-034, TM-NFR-005 | Not executed

Prerequisites: Authorized disposable fixtures, stubs and isolated PostgreSQL where persistence is involved. No existing records or external paid services.

Test data: Stub output with invented IDs, prompt-injection text, executable markup, invalid date/time and unapproved price/photo keys.

Steps: Run each response through schedule validation, single repair and provider fallback.

Expected: No invented identity/media/price persists; invalid schedule is repaired once or honest fallback returned; no raw markup execution.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-032, TM-FR-034, TM-NFR-005; static source references only; no system-case execution evidence.

Cleanup: Reset disposable fixtures; retain sanitized results only.

TM-XT-011 Replan stale acceptance

Requirements and status: TM-FR-067, TM-FR-068, TM-NFR-007 | Not executed

Prerequisites: Authorized disposable fixtures, stubs and isolated PostgreSQL where persistence is involved. No existing records or external paid services.

Test data: Preview revision R; manual edit to R2; then accept R token, expired token and wrong-owner token.

Steps: Request preview in disposable environment, change trip, confirm acceptance and cancel a fresh preview.

Expected: Stale/expired/wrong-owner preview rejects; cancel does not write itinerary/version; fresh accepted preview changes only allowed remaining activities.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-067, TM-FR-068, TM-NFR-007; static source references only; no system-case execution evidence.

Cleanup: Reset disposable fixtures; retain sanitized results only.

TM-XT-012 Mail failure is visible

Requirements and status: TM-FR-009, TM-FR-014, TM-NFR-006 | Not executed

Prerequisites: Authorized disposable fixtures, stubs and isolated PostgreSQL where persistence is involved. No existing records or external paid services.

Test data: Test mail adapter accepted/rejected/config missing; disposable account tokens.

Steps: Register/resend/reset without contacting mail service; inspect token lifetime and UI error.

Expected: Failure discloses mail unavailability, invalidates unusable issued code as implemented, respects cooldown, and does not claim inbox delivery.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-009, TM-FR-014, TM-NFR-006; static source references only; no system-case execution evidence.

Cleanup: Reset disposable fixtures; retain sanitized results only.

TM-XT-013 Wrong pair and stale FX

Requirements and status: TM-FR-022, TM-NFR-006 | Not executed

Prerequisites: Authorized disposable fixtures, stubs and isolated PostgreSQL where persistence is involved. No existing records or external paid services.

Test data: USD/PHP request with returned PHP/USD, stale >24-hour cache and provider 429.

Steps: Refresh estimate through stub provider and inspect output provenance.

Expected: Reject wrong pair and stale-as-current result; show unavailable/reference status and bounded Retry-After; trip currency remains unchanged.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-022, TM-NFR-006; static source references only; no system-case execution evidence.

Cleanup: Reset disposable fixtures; retain sanitized results only.

TM-XT-014 All-code media reconciliation

Requirements and status: TM-FR-004, TM-NFR-013 | Not executed

Prerequisites: Authorized disposable fixtures, stubs and isolated PostgreSQL where persistence is involved. No existing records or external paid services.

Test data: Every local geometry/selector/photo/flag key; all 249 ISO codes plus Kosovo; Bir Tawil exception.

Steps: Join manifests by stable code, decode all local photos, inspect focal crops and sample all continents/islands; delay asynchronous loads.

Expected: Country/flag/border/photo owners match; no stale image flash; all country codes covered; one unresolved non-national area explicitly reported; no unrelated fallback.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-004, TM-NFR-013; static source references only; no system-case execution evidence.

Cleanup: Reset disposable fixtures; retain sanitized results only.

TM-XT-015 Rotating pointer identity

Requirements and status: TM-FR-002, TM-FR-003, TM-FR-004 | Not executed

Prerequisites: Authorized disposable fixtures, stubs and isolated PostgreSQL where persistence is involved. No existing records or external paid services.

Test data: Pointer stationary over land then water while rotation continues; zoom/drag and pointer leave.

Steps: Observe hit identity on successive animation frames and card/outline cleanup.

Expected: Photo, name, flag and outline update from the same current hit record; ocean/exit hides overlay; card has no pointer capture or layout shift.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-002, TM-FR-003, TM-FR-004; static source references only; no system-case execution evidence.

Cleanup: Reset disposable fixtures; retain sanitized results only.

TM-XT-016 No fabricated route or weather

Requirements and status: TM-FR-061, TM-FR-063, TM-NFR-006 | Not executed

Prerequisites: Authorized disposable fixtures, stubs and isolated PostgreSQL where persistence is involved. No existing records or external paid services.

Test data: Unresolved coordinates, OSRM no route/429, outside forecast horizon, exhausted provider budget.

Steps: Open saved day map and request guidance/stop weather with stub failures.

Expected: Unresolved stops remain explicit; no straight-line route claimed as navigable OSRM result; unavailable weather is not current observation.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-061, TM-FR-063, TM-NFR-006; static source references only; no system-case execution evidence.

Cleanup: Reset disposable fixtures; retain sanitized results only.

TM-XT-017 Lifecycle and deletion isolation

Requirements and status: TM-FR-048, TM-FR-049, TM-FR-050 | Not executed

Prerequisites: Authorized disposable fixtures, stubs and isolated PostgreSQL where persistence is involved. No existing records or external paid services.

Test data: Completed trip archived/restored; linked visits and feedback; owned and foreign trip.

Steps: Archive/restore/reopen/complete/delete only in disposable DB; compare all affected entities.

Expected: Completion retained through archive; manual visited history not auto-created; deletion cascades versions, detaches visit/feedback references and retains applicable audit.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-048, TM-FR-049, TM-FR-050; static source references only; no system-case execution evidence.

Cleanup: Reset disposable fixtures; retain sanitized results only.

TM-XT-018 Profile mass assignment and media URL

Requirements and status: TM-FR-058, TM-FR-079, TM-NFR-012 | Not executed

Prerequisites: Authorized disposable fixtures, stubs and isolated PostgreSQL where persistence is involved. No existing records or external paid services.

Test data: Valid own profile plus attempted role/email/sessionVersion/trustScore and javascript/http avatar.

Steps: Submit through both role-specific profile views and direct API; reload.

Expected: Only allowlisted fields change; HTTPS avatar validation applies; role/email/security state unchanged.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-058, TM-FR-079, TM-NFR-012; static source references only; no system-case execution evidence.

Cleanup: Reset disposable fixtures; retain sanitized results only.

TM-XT-019 SSE auth and resource cleanup

Requirements and status: TM-FR-075, TM-NFR-001 | Not executed

Prerequisites: Authorized disposable fixtures, stubs and isolated PostgreSQL where persistence is involved. No existing records or external paid services.

Test data: Guest/traveler/admin requests; suspended/revoked admin session; slow client >1 MB buffer.

Steps: Open streams, change current DB session state and disconnect; observe timers/listeners.

Expected: Only authorized admin receives changes; periodic recheck closes invalid sessions; rotation/backpressure/disconnect removes resources.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-075, TM-NFR-001; static source references only; no system-case execution evidence.

Cleanup: Reset disposable fixtures; retain sanitized results only.

TM-XT-020 Migration and restore rehearsal

Requirements and status: TM-FR-087, TM-NFR-014 | Not executed

Prerequisites: Authorized disposable fixtures, stubs and isolated PostgreSQL where persistence is involved. No existing records or external paid services.

Test data: Disposable Postgres clone with synthetic data, backup file and old/new compatible images.

Steps: Backup, validate restore checksum, run migration, fail new health check, restore image and inspect schema compatibility.

Expected: Recovery is measured and documented; image rollback never claims to undo DB migration; retain recovery evidence before production approval.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-FR-087, TM-NFR-014; static source references only; no system-case execution evidence.

Cleanup: Reset disposable fixtures; retain sanitized results only.

STD 4.12 Quality acceptance procedures

TM-QT-001 Authoritative access

Requirements and status: TM-NFR-001 | Not executed

Prerequisites: Controlled acceptance environment with authorized tester, representative synthetic data, and approved target for any measurable service objective.

Test data: Implemented control; live multi-account acceptance pending.

Steps: Locate the linked feature cases in the RTM. Execute relevant positive, negative and failure variants. For measured targets record environment, repetitions, denominators and raw sanitized evidence; request reviewer approval of the target before scoring.

Expected: Protected requests shall recheck role, ownership, suspension, expiry, session version and token revocation on the server.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-NFR-001; static source references only; no system-case execution evidence.

Cleanup: Restore disposable configuration and fixtures; no existing application changes.

TM-QT-002 Credential protection

Requirements and status: TM-NFR-002 | Not executed

Prerequisites: Controlled acceptance environment with authorized tester, representative synthetic data, and approved target for any measurable service objective.

Test data: Implemented control; production cookie inspection pending.

Steps: Locate the linked feature cases in the RTM. Execute relevant positive, negative and failure variants. For measured targets record environment, repetitions, denominators and raw sanitized evidence; request reviewer approval of the target before scoring.

Expected: Passwords shall use per-account random salts and scrypt; browser cookies shall be HttpOnly, SameSite Lax and Secure in production.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-NFR-002; static source references only; no system-case execution evidence.

Cleanup: Restore disposable configuration and fixtures; no existing application changes.

TM-QT-003 Request origin and transport

Requirements and status: TM-NFR-003 | Not executed

Prerequisites: Controlled acceptance environment with authorized tester, representative synthetic data, and approved target for any measurable service objective.

Test data: Implemented control; deployed reverse-proxy acceptance pending.

Steps: Locate the linked feature cases in the RTM. Execute relevant positive, negative and failure variants. For measured targets record environment, repetitions, denominators and raw sanitized evidence; request reviewer approval of the target before scoring.

Expected: Production shall reject insecure backend transport and unexpected cross-origin writes; browser security headers shall be applied.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-NFR-003; static source references only; no system-case execution evidence.

Cleanup: Restore disposable configuration and fixtures; no existing application changes.

TM-QT-004 Financial precision

Requirements and status: TM-NFR-004 | Not executed

Prerequisites: Controlled acceptance environment with authorized tester, representative synthetic data, and approved target for any measurable service objective.

Test data: Intended acceptance rule; inspect D-07 before sign-off.

Steps: Locate the linked feature cases in the RTM. Execute relevant positive, negative and failure variants. For measured targets record environment, repetitions, denominators and raw sanitized evidence; request reviewer approval of the target before scoring.

Expected: Amounts shall respect the selected supported currency and its minor units; explicit costs shall not be charged twice.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-NFR-004; static source references only; no system-case execution evidence.

Cleanup: Restore disposable configuration and fixtures; no existing application changes.

TM-QT-005 Grounded identities

Requirements and status: TM-NFR-005 | Not executed

Prerequisites: Controlled acceptance environment with authorized tester, representative synthetic data, and approved target for any measurable service objective.

Test data: Implemented validation; reference completeness and provider behavior not guaranteed.

Steps: Locate the linked feature cases in the RTM. Execute relevant positive, negative and failure variants. For measured targets record environment, repetitions, denominators and raw sanitized evidence; request reviewer approval of the target before scoring.

Expected: AI shall not introduce unapproved place identities, prices or media; country and venue evidence shall be validated separately from coordinates.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-NFR-005; static source references only; no system-case execution evidence.

Cleanup: Restore disposable configuration and fixtures; no existing application changes.

TM-QT-006 Honest degraded service

Requirements and status: TM-NFR-006 | Not executed

Prerequisites: Controlled acceptance environment with authorized tester, representative synthetic data, and approved target for any measurable service objective.

Test data: Implemented control; external outage rehearsals pending.

Steps: Locate the linked feature cases in the RTM. Execute relevant positive, negative and failure variants. For measured targets record environment, repetitions, denominators and raw sanitized evidence; request reviewer approval of the target before scoring.

Expected: Failures shall produce a labeled unavailable, retryable or catalog fallback result without fabricating successful delivery, pricing or reservations.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-NFR-006; static source references only; no system-case execution evidence.

Cleanup: Restore disposable configuration and fixtures; no existing application changes.

TM-QT-007 Atomic persistence

Requirements and status: TM-NFR-007 | Not executed

Prerequisites: Controlled acceptance environment with authorized tester, representative synthetic data, and approved target for any measurable service objective.

Test data: Implemented control; live PostgreSQL contention tests pending.

Steps: Locate the linked feature cases in the RTM. Execute relevant positive, negative and failure variants. For measured targets record environment, repetitions, denominators and raw sanitized evidence; request reviewer approval of the target before scoring.

Expected: Multi-record changes shall use transactions and concurrency checks; stale generation, versions and activity previews shall not overwrite newer state.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-NFR-007; static source references only; no system-case execution evidence.

Cleanup: Restore disposable configuration and fixtures; no existing application changes.

TM-QT-008 Duplicate prevention

Requirements and status: TM-NFR-008 | Not executed

Prerequisites: Controlled acceptance environment with authorized tester, representative synthetic data, and approved target for any measurable service objective.

Test data: Implemented control; isolated service evidence only.

Steps: Locate the linked feature cases in the RTM. Execute relevant positive, negative and failure variants. For measured targets record environment, repetitions, denominators and raw sanitized evidence; request reviewer approval of the target before scoring.

Expected: Generation, save, batch visits, feedback and pending appeals shall apply their documented identity keys and duplicate semantics.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-NFR-008; static source references only; no system-case execution evidence.

Cleanup: Restore disposable configuration and fixtures; no existing application changes.

TM-QT-009 Responsive layout

Requirements and status: TM-NFR-009 | Not executed

Prerequisites: Controlled acceptance environment with authorized tester, representative synthetic data, and approved target for any measurable service objective.

Test data: Proposed release acceptance set; historical emulated evidence is limited to named reports.

Steps: Locate the linked feature cases in the RTM. Execute relevant positive, negative and failure variants. For measured targets record environment, repetitions, denominators and raw sanitized evidence; request reviewer approval of the target before scoring.

Expected: Public and role workspaces shall remain usable without unintended horizontal overflow at 320, 768, 1024 and 1440 CSS pixels.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-NFR-009; static source references only; no system-case execution evidence.

Cleanup: Restore disposable configuration and fixtures; no existing application changes.

TM-QT-010 Keyboard and motion access

Requirements and status: TM-NFR-010 | Not executed

Prerequisites: Controlled acceptance environment with authorized tester, representative synthetic data, and approved target for any measurable service objective.

Test data: Partly inspected; complete keyboard, contrast and assistive-technology audit not executed.

Steps: Locate the linked feature cases in the RTM. Execute relevant positive, negative and failure variants. For measured targets record environment, repetitions, denominators and raw sanitized evidence; request reviewer approval of the target before scoring.

Expected: Core actions shall be keyboard reachable, inputs shall have meaningful labels, images shall have useful alt text, and Earth shall respect reduced motion.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-NFR-010; static source references only; no system-case execution evidence.

Cleanup: Restore disposable configuration and fixtures; no existing application changes.

TM-QT-011 Timeout and cancellation

Requirements and status: TM-NFR-011 | Not executed

Prerequisites: Controlled acceptance environment with authorized tester, representative synthetic data, and approved target for any measurable service objective.

Test data: Implemented deadlines; 45 s AI adapter, 540 s outer generation and 580 s proxy are ceilings, not performance measurements.

Steps: Locate the linked feature cases in the RTM. Execute relevant positive, negative and failure variants. For measured targets record environment, repetitions, denominators and raw sanitized evidence; request reviewer approval of the target before scoring.

Expected: Provider adapters and generation shall use bounded deadlines and cleanup; cancellation shall not be represented as guaranteed provider billing cancellation.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-NFR-011; static source references only; no system-case execution evidence.

Cleanup: Restore disposable configuration and fixtures; no existing application changes.

TM-QT-012 Data minimization

Requirements and status: TM-NFR-012 | Not executed

Prerequisites: Controlled acceptance environment with authorized tester, representative synthetic data, and approved target for any measurable service objective.

Test data: Implemented controls inspected; retention policy and privacy notice reconciliation pending.

Steps: Locate the linked feature cases in the RTM. Execute relevant positive, negative and failure variants. For measured targets record environment, repetitions, denominators and raw sanitized evidence; request reviewer approval of the target before scoring.

Expected: Application outputs and documentation shall not expose credentials, tokens or confidential user records; optional GPS shall require explicit permission and stop on cleanup.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-NFR-012; static source references only; no system-case execution evidence.

Cleanup: Restore disposable configuration and fixtures; no existing application changes.

TM-QT-013 Attribution and provenance

Requirements and status: TM-NFR-013 | Not executed

Prerequisites: Controlled acceptance environment with authorized tester, representative synthetic data, and approved target for any measurable service objective.

Test data: Implemented manifest/credits; Bir Tawil geographic-area photo remains unresolved.

Steps: Locate the linked feature cases in the RTM. Execute relevant positive, negative and failure variants. For measured targets record environment, repetitions, denominators and raw sanitized evidence; request reviewer approval of the target before scoring.

Expected: Photos, map tiles and geographic assets shall retain required attribution and source metadata; unresolved media shall not borrow another place identity.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-NFR-013; static source references only; no system-case execution evidence.

Cleanup: Restore disposable configuration and fixtures; no existing application changes.

TM-QT-014 Operational recovery

Requirements and status: TM-NFR-014 | Not executed

Prerequisites: Controlled acceptance environment with authorized tester, representative synthetic data, and approved target for any measurable service objective.

Test data: Proposed operational acceptance; no automatic backup worker or executed recovery evidence located.

Steps: Locate the linked feature cases in the RTM. Execute relevant positive, negative and failure variants. For measured targets record environment, repetitions, denominators and raw sanitized evidence; request reviewer approval of the target before scoring.

Expected: A release shall have an approved backup, restore, migration and rollback procedure tested against disposable data before production use.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-NFR-014; static source references only; no system-case execution evidence.

Cleanup: Restore disposable configuration and fixtures; no existing application changes.

TM-QT-015 Contract consistency

Requirements and status: TM-NFR-015 | Not executed

Prerequisites: Controlled acceptance environment with authorized tester, representative synthetic data, and approved target for any measurable service objective.

Test data: Partial; generated checks exist but 10 actual route operations lack exact OpenAPI entries.

Steps: Locate the linked feature cases in the RTM. Execute relevant positive, negative and failure variants. For measured targets record environment, repetitions, denominators and raw sanitized evidence; request reviewer approval of the target before scoring.

Expected: Generated frontend API contracts shall match the authoritative backend schemas; published OpenAPI shall cover actual routes.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-NFR-015; static source references only; no system-case execution evidence.

Cleanup: Restore disposable configuration and fixtures; no existing application changes.

TM-QT-016 Traceable changes

Requirements and status: TM-NFR-016 | Not executed

Prerequisites: Controlled acceptance environment with authorized tester, representative synthetic data, and approved target for any measurable service objective.

Test data: Implemented records; role-restricted read access, not a tamper-proof ledger.

Steps: Locate the linked feature cases in the RTM. Execute relevant positive, negative and failure variants. For measured targets record environment, repetitions, denominators and raw sanitized evidence; request reviewer approval of the target before scoring.

Expected: Version changes, managed activity actions and moderation shall preserve attributable history without claiming cryptographic audit immutability.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-NFR-016; static source references only; no system-case execution evidence.

Cleanup: Restore disposable configuration and fixtures; no existing application changes.

TM-QT-017 Shared rate limits

Requirements and status: TM-NFR-017 | Not executed

Prerequisites: Controlled acceptance environment with authorized tester, representative synthetic data, and approved target for any measurable service objective.

Test data: Implemented PostgreSQL buckets and in-memory provider pacing; distributed provider quotas not proven.

Steps: Locate the linked feature cases in the RTM. Execute relevant positive, negative and failure variants. For measured targets record environment, repetitions, denominators and raw sanitized evidence; request reviewer approval of the target before scoring.

Expected: Rate limits shall persist across instances where configured and provider requests shall honor bounded request pacing and retry advice.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-NFR-017; static source references only; no system-case execution evidence.

Cleanup: Restore disposable configuration and fixtures; no existing application changes.

TM-QT-018 Reproducible verification

Requirements and status: TM-NFR-018 | Not executed

Prerequisites: Controlled acceptance environment with authorized tester, representative synthetic data, and approved target for any measurable service objective.

Test data: Satisfied for this documentation evidence boundary; release approval pending.

Steps: Locate the linked feature cases in the RTM. Execute relevant positive, negative and failure variants. For measured targets record environment, repetitions, denominators and raw sanitized evidence; request reviewer approval of the target before scoring.

Expected: A release decision shall separate static inspection, historical evidence, newly executed checks and unexecuted acceptance procedures.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-NFR-018; static source references only; no system-case execution evidence.

Cleanup: Restore disposable configuration and fixtures; no existing application changes.

TM-QT-019 Measured performance

Requirements and status: TM-NFR-019 | Not executed

Prerequisites: Controlled acceptance environment with authorized tester, representative synthetic data, and approved target for any measurable service objective.

Test data: Proposed acceptance method; legacy sub-2-second/500-user/99.9-percent targets are unapproved and unmeasured.

Steps: Locate the linked feature cases in the RTM. Execute relevant positive, negative and failure variants. For measured targets record environment, repetitions, denominators and raw sanitized evidence; request reviewer approval of the target before scoring.

Expected: Performance acceptance shall record hardware, data volume, network, load, p50/p95/p99 and error rate against an approved target.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-NFR-019; static source references only; no system-case execution evidence.

Cleanup: Restore disposable configuration and fixtures; no existing application changes.

TM-QT-020 Date and time consistency

Requirements and status: TM-NFR-020 | Not executed

Prerequisites: Controlled acceptance environment with authorized tester, representative synthetic data, and approved target for any measurable service objective.

Test data: Intended rule; browser-local versus server-UTC past-date edge remains D-03.

Steps: Locate the linked feature cases in the RTM. Execute relevant positive, negative and failure variants. For measured targets record environment, repetitions, denominators and raw sanitized evidence; request reviewer approval of the target before scoring.

Expected: Trip date eligibility and activity-local clocks shall use an explicit, consistent time-zone policy across client and server.

Actual: Not observed in this procedure; isolated/historical tests are separate.

Evidence: RTM row TM-NFR-020; static source references only; no system-case execution evidence.

Cleanup: Restore disposable configuration and fixtures; no existing application changes.

STD 5 Performance security and operational limits

No load test, availability measurement, physical-device study or end-user satisfaction survey was executed. Old sub-two-second, 500-user, 5,000-user, 100,000-record and 99.9 percent statements remain historical targets without current approval/evidence. TM-NFR-019 requires approved workloads and reporting of hardware/data/network, concurrency, warm/cold state, request count, p50/p95/p99 and error rate. Separate generation latency from standard CRUD and do not mistake timeout ceilings for service-level objectives. [3][4]

A production-like verification plan should include: two-owner authorization and role negatives; session revocation and reset; independent provider/mail credentials and delivery; PostgreSQL constraint/concurrency cases; security headers under the actual proxy; keyboard/screen-reader/contrast and small-screen tasks; provider rate/rights review; and backup/restore/migration rollback rehearsal. These remain Not executed, with no fabricated pass rate. Staging requests with billing, messages or data writes require explicit authorization.

STD 6 Defect and retest register

Table 27 Discrepancies and acceptance concerns

ID and class	Observed evidence	Disposition
D-01 Content discrepancy	Landing copy still mentions removed travel style and comparison controls.	Current dashboard uses Balanced defaults; reconcile marketing with accessible UI. Static inspection; no application change.
D-02 Privacy notice discrepancy	Privacy text emphasizes Google photo handling although ordinary production enrichment also uses reusable Wikimedia/other sources.	Review the actual provider/data-flow disclosures and retention/contact details before publication. Static inspection; legal/compliance approval not established.
D-03 Potential date boundary defect	Browser uses local today; server start-date eligibility uses UTC today.	Adopt a documented date policy and run TM-XT-001 at offset boundaries. Static concern; not reproduced as a live system failure.
D-04 Potential ad ministrative validation defect	Unknown status input can normalize to active in the user-management path.	Reject unknown/missing commands; test TM-XT-006 before acceptance. Static concern; no live account mutation.
D-05 Confirmed contract docu mentation gap	Ten actual method/path operations lack exact OpenAPI entries; exchange rate is documented at the wrong path.	Document actual /api/destination-context/exchange-rate and reconcile the route register. Static route/spec comparison; not an executed API failure.
D-06 Potential storage length defect	Generation accepts destination labels up to 2000 characters at the controller boundary; ItineraryGeneration.destination is varchar(120).	Align accepted lengths and run TM-XT-005 using disposable data. Static concern; no live database reproduction.
D-07 Potential save precision defect	Platform positiveMoney rounds to two decimals before precision validation.	Validate raw currency precision first and run TM-XT-004; silent rounding is not intended acceptance. Static concern; not a passing business rule.

ID and class	Observed evidence	Disposition
D-08 Historical code quality failure	15 October itinerary-overview report records seven no-explicit-any ESLint errors in e2e/travel-preferences.spec.ts.	Correct and rerun whole-repository lint separately. Historical report only; lint not rerun this task.
D-09 Explicit non-national media gap	264 of 265 geographic area records have photos; Bir Tawil has no verified suitable photo.	Keep an honest missing photo state; curate only a verified exact-area image. Manifest/static evidence and related current isolated tests; all 249 ISO entries plus Kosovo are covered.

Severity and release disposition require team review. Critical issues are those allowing security compromise/data loss or making the core system unusable; major issues prevent a required workflow; minor issues have a safe workaround. This document does not assign an unapproved organizational severity to static concerns. No defect was corrected in application code during this documentation task; consequently no correction retest or closure is asserted.

STD 7 Traceability and evidence custody

The RTM maps all 113 requirements to design, implementation, cases and manual sections. The feature, page, API and process reconciliation registers explain the complete discovered entry-point set. Evidence hashes and original-source comparisons are preserved in the audit package. Raw secrets, authenticated browser sessions and copied application databases are excluded from the delivered evidence bundle. Use the final suite logs, not initial harness output, to cite current execution.

STD 8 Acceptance and sign off

Acceptance remains pending. Before release: execute required system/permission/persistence/provider cases; resolve or formally accept defects; approve performance and recovery targets; run an authorized UAT with approved tasks and consent; complete provider/privacy review; confirm final administrative details; and review Word pagination and field updates using a working renderer. A proposed UAT form is supplied in Appendix E. No participants, survey scores, signatures or approval dates have been invented.

Table 28 Acceptance record for later completion

Decision field	Record
Test lead and date	[TO BE CONFIRMED]
Environment and approved baseline	[TO BE CONFIRMED]
Executed cases and evidence	[TO BE COMPLETED AFTER EXECUTION]
Open defects and disposition	D-01-D-09 pending review; update with actual retest evidence.
UAT and operational approvals	[TO BE COMPLETED AFTER AUTHORIZED EXECUTION]
Adviser/panel approval	[TO BE CONFIRMED]

User Manual

This manual explains the current accessible TravelMate interfaces. Guest users need no account for the public globe. Planning and saved records require an active authorized traveler account; administrative functions require a provisioned administrator. Use only your own records. Estimates, forecasts and suggested routes require independent travel checks. [1][2]

The screenshots labeled current were captured from the anonymous localhost application on 15 October 2026 without submitting forms. Authenticated traveler, map, activity and administrator screens are not captured in this task because opening/using them could require account and audit writes. Those missing screenshots are stated explicitly. No synthetic UI mockup is presented as evidence.

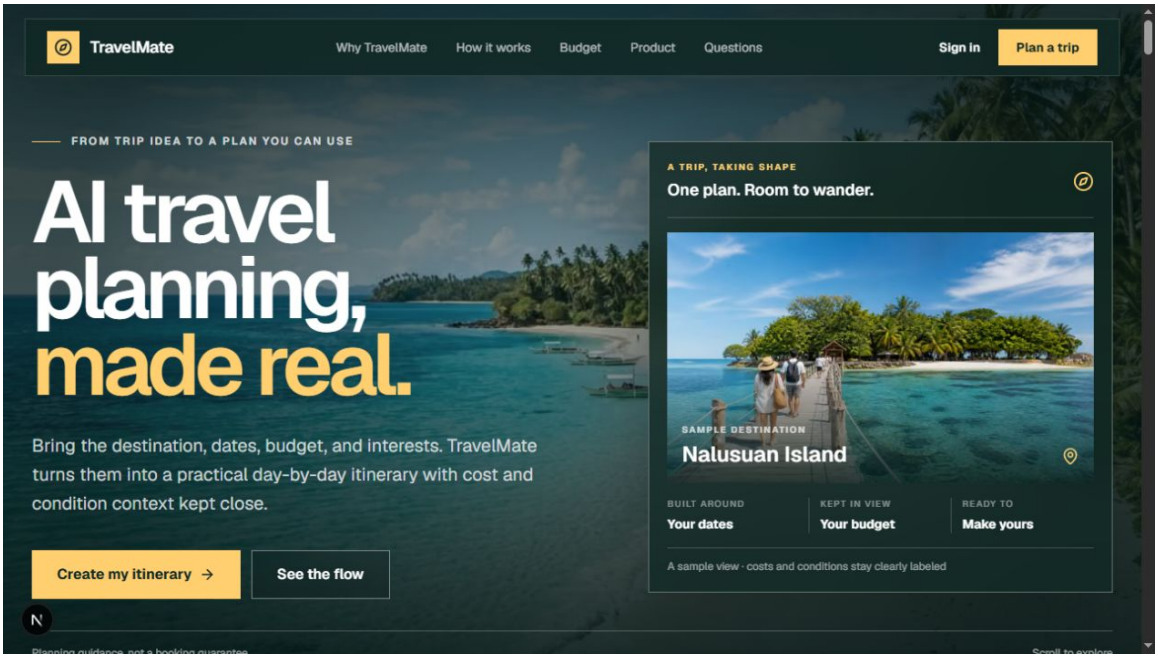


Figure 12 Current anonymous home page at localhost on 15 October 2026; desktop viewport capture, no form submission

UM 1 Public discovery

Open the public home page and scroll to Discovery. The country selector offers a non-pointer route to choosing a country; hover behavior uses the pointer. On touch devices use the selector and controls rather than assuming pointer hover. If WebGL fails, read the fallback message and keep using the text/navigation. Missing screenshot: responsive/mobile and exhaustive hover states were not captured this task.

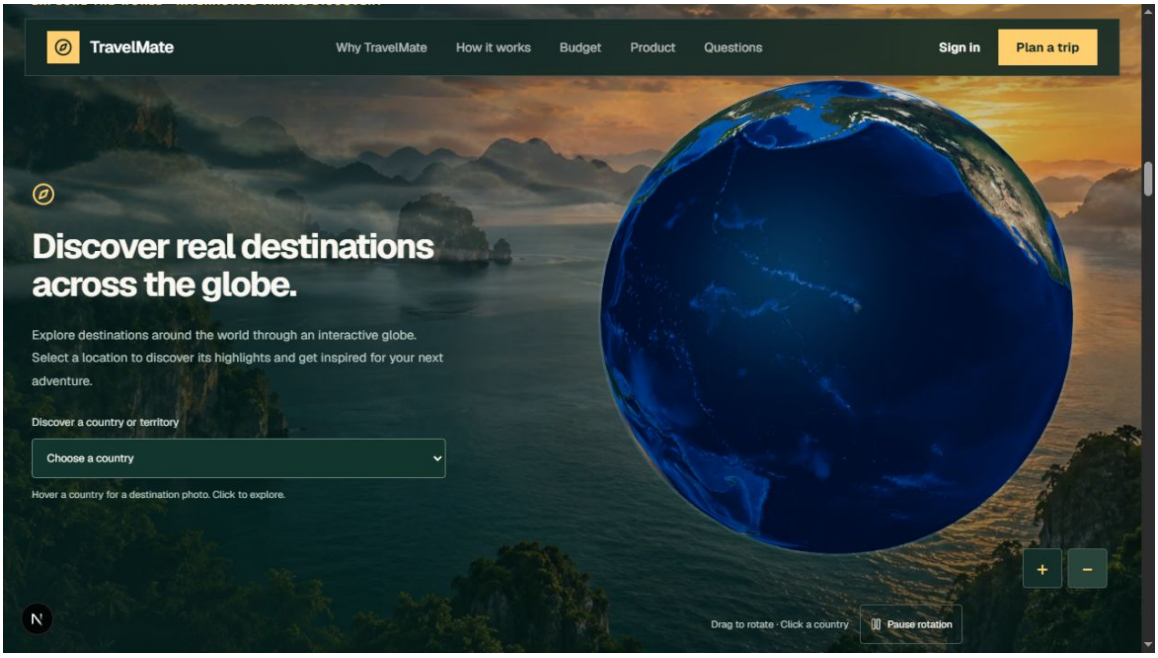


Figure 13 Current discovery section with full Earth and controls; rotating orientation observed, not a full-country hover certification

UM 1.001 Landing and navigation

Access and purpose: Guest / all. The public site shall present planning guidance, anchored information sections, FAQ, account entry and legal links.

1. Open the home page.

2. Use the top links or mobile menu to reach features, workflow, budget, product and questions.

3. Use account entry to sign in or register.

Expected outcome and limits: Landing samples are illustrative. Some marketing copy still mentions removed travel-style/comparison controls; see discrepancy D-01.

UM 1.002 Interactive Earth controls

Access and purpose: Guest / all. The globe shall support rotation, drag, zoom, pause and reduced-motion handling, with a country selector fallback.

1. Scroll to Discovery.

2. Drag Earth to change the view; use + or - and Pause rotation.

3. Select a country in the dropdown to focus it.

Expected outcome and limits: WebGL or dataset failure shows fallback feedback. Selection focuses Earth; it does not create a trip or open a destination-details panel.

UM 1.003 Country hit detection and border identity

Access and purpose: Guest / all. Hover detection shall use actual country geometry and stable geographic codes, including small-area hit targets, and display that country’s border.

1. Move the pointer onto a land area.

2. Observe the border around that geographic record.

3. Move to another area or leave Earth to change or clear it.

Expected outcome and limits: Natural Earth geometry has finite resolution. Disputed/non-national areas are data records, not claims of sovereignty. Decorative arcs, dots and rings are absent.

UM 1.004 Country destination photo and identity card

Access and purpose: Guest / all. A hover card shall use the same stable code for the destination photograph, English name, available flag and border, and hide on pointer exit.

1. Hover a country; read its compact name-and-flag row below the photo.

2. Hover another country to update all identity elements.

3. Leave the globe to dismiss.

Expected outcome and limits: 264/265 geographic areas have verified photos; Bir Tawil is unresolved and receives no unrelated fallback. All 249 ISO entries plus

Kosovo are covered. Antarctica and some non-national areas have no national flag.

UM 1.005 Image and map credits

Access and purpose: Guest / all. The public footer shall expose image/map source, author, licensing and adaptation credits without a permanent caption under Earth.

1. Scroll to the footer.

2. Expand Image and map credits.

3. Follow the relevant source or license link.

Expected outcome and limits: Credits describe asset rights, not provider availability or travel recommendations.

UM 1.006 Privacy and terms pages

Access and purpose: Guest / all. Public privacy and terms pages shall describe the planning scope and applicable content handling.

1. Open Privacy policy or Terms of use in the footer.

2. Read the notices and follow the cross-link or TravelMate home link.

Expected outcome and limits: Current privacy text emphasizes Google photos although ordinary production enrichment uses reusable providers; notice reconciliation is recommended (D-02).

UM 2 Accounts and access

Use the header Sign in control. Switch between sign-in, registration, verification and Forgot password views as needed. Use an email you control only in normal authorized operation; documentation fixtures use a test mail sink. Google is optional and may be unavailable when configuration is absent. Missing screenshots: verified account, appeal and authenticated profile/logout states.

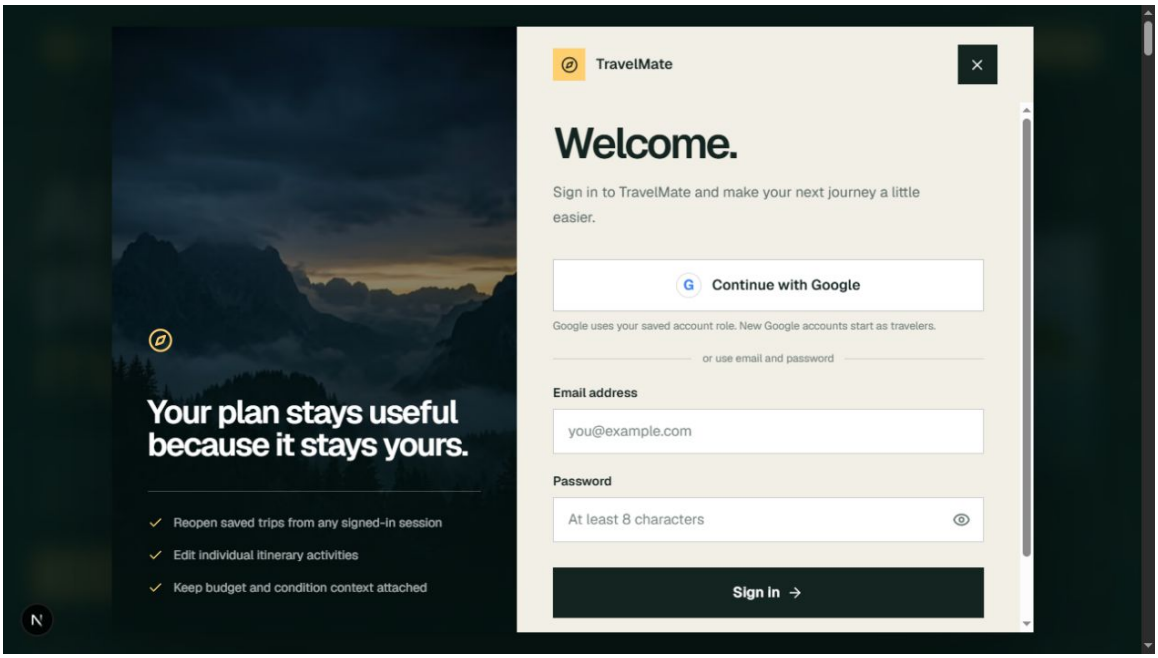


Figure 14 Current sign-in dialog; anonymous capture with no entered credentials

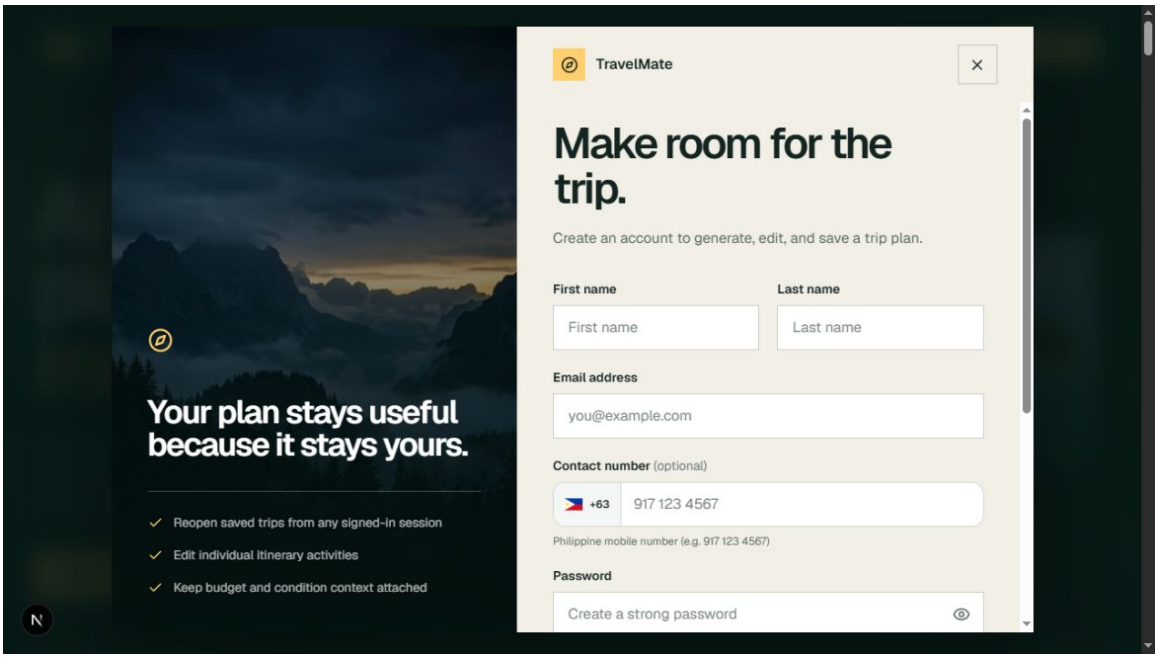


Figure 15 Current registration dialog; no account created

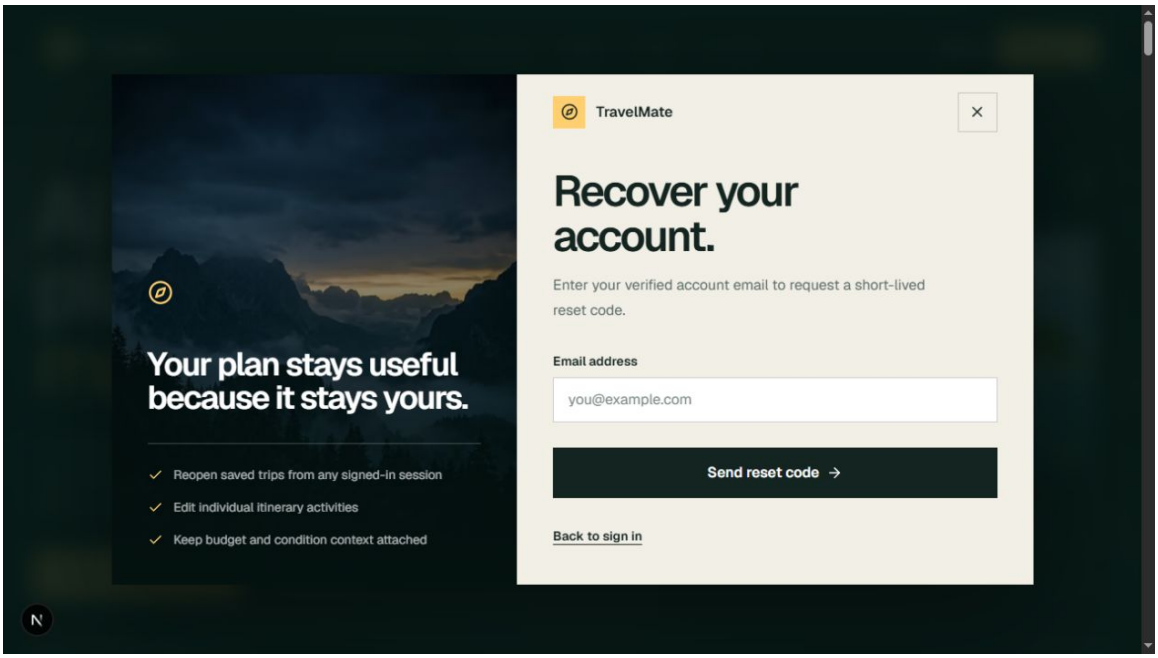


Figure 16 Current recovery entry; no code sent

UM 2.007 Traveler registration

Access and purpose: Guest. Registration shall validate names, email, optional Philippine mobile number, strong password and confirmation, and create only a traveler account.

1. Choose Create an account.

2. Enter first and last names, email, optional mobile number, password and confirmation.

3. Submit, then proceed to verification only after observing the outcome.

Expected outcome and limits: Names 2-80 characters combined; password 8-64 with upper/lower/digit/special and no whitespace. Mobile is optional. Email configuration is required; accepted mail is not confirmed inbox delivery.

UM 2.008 Email verification

Access and purpose: Guest with pending account. Verification shall consume a valid account-bound, short-lived code once and permit subsequent password sign-in.

1. Enter the registered email and received eight-character code in the verification view.

2. Submit before expiry.

3. Sign in after the confirmation.

Expected outcome and limits: Eight hexadecimal characters, 5-minute lifetime and 5 attempts. Verification is separate from legacy profileStatus.

UM 2.009 Resend verification

Access and purpose: Guest with pending account. Resending shall replace only an eligible account's verification code, enforce cooldown/limits and report mail failures honestly.

1. Use Resend verification after waiting 60 seconds.

2. Check the displayed result and inbox/spam.

3. Use only the newest accepted code.

Expected outcome and limits: Three eligible resends per hour; failed send invalidates that code while retaining cooldown. No automatic switch from Resend to Gmail.

UM 2.010 Password sign-in and role redirect

Access and purpose: Guest. Password sign-in shall validate credentials and email verification, issue a signed session and route traveler/admin/suspended accounts appropriately.

1. Open Sign in.

2. Enter the registered email and TravelMate password.

3. Submit; proceed to the role-specific workspace or suspension appeal view.

Expected outcome and limits: Wrong credentials return 401; unverified email returns 403. Suspended accounts can sign in for appeals, but ordinary protected APIs reject them.

UM 2.011 Google OAuth

Access and purpose: Guest / existing account. Google sign-in shall validate state, PKCE and verified provider identity before resolving the application account and role.

1. Choose Continue with Google.

2. Complete Google’s own authentication.

3. Return to TravelMate and confirm the expected account/role.

Expected outcome and limits: Requires provider configuration and authorized redirect origin. Existing verified local accounts require the documented linking-safe behavior; no admin role is granted from public input.

UM 2.012 Authoritative sessions and access control

Access and purpose: Traveler / administrator. Protected API requests shall validate the signed session, expiry, current role, account state, session version and per-session revocation.

1. Sign in before opening a protected workspace.

2. If the session is rejected, return to sign-in.

3. Do not rely on a visible menu to establish permission.

Expected outcome and limits: Custom HMAC-SHA256 session format, not a standard JWT refresh-token system. Eight-hour cookie; proxy checks presence only.

UM 2.013 Logout and local cleanup

Access and purpose: Traveler / administrator / suspended traveler. Logout shall clear application/OAuth cookies, attempt individual session revocation and disclose unconfirmed revocation.

1. Use Logout or Sign out.

2. Confirm return to the public site.

3. On a shared device, close private views after local sign-out.

Expected outcome and limits: If database revocation fails, local cookies clear but server revocation is explicitly unconfirmed. Other logins are not intentionally revoked.

UM 2.014 Forgot/reset password

Access and purpose: Guest with verified active account. Recovery shall issue a short-lived reset code, validate the replacement password, consume the token once and invalidate old sessions.

1. Choose Forgot password.

2. Request a code for the verified email.

3. Enter the code and a conforming new password; then sign in again.

Expected outcome and limits: Reset lifetime 15 minutes. Actual email delivery/provider authorization was not tested. Reset is not an in-profile password-change screen.

UM 2.015 Suspension appeal access

Access and purpose: Suspended traveler. A suspended traveler shall read account notices/history and submit one pending appeal with a bounded message.

1. Sign in to open Account appeal.

2. Read the account notices.

3. Enter a 10-2000-character appeal, submit once and check its status.

Expected outcome and limits: Three attempts/hour; a pending appeal blocks duplicates. Approval/restoration is an administrator decision.

UM 3 Destination selection

After traveler sign-in, use Plan a trip and confirm a locality. Identical names can occur in different countries or administrative areas: read the country, parent and type before selecting. Country/admin selection alone is insufficient for a generated city plan. Missing screenshot: current authenticated destination selector and area comparison.

UM 3.016 Worldwide direct search

Access and purpose: Traveler. Destination search shall return source-backed place identity, country, hierarchy and coordinates without inventing missing source records.

1. In Plan a trip, enter Where do you want to go? Read the parent/country/type of each result.

2. Select the intended locality, or use verified manual resolution.

Expected outcome and limits: Full imported catalogue records 5,721,309 records/250 source country entries; fallback compressed index is bounded. Coverage is not proof of complete geography.

UM 3.017 Country and administrative browsing

Access and purpose: Traveler. The destination browser shall follow recorded parent IDs/types with pagination and preserve an existing confirmation on cancellation.

1. Choose Browse by country.

2. Select a country and available administrative level, use search or Load more, then choose a final city/town.

3. Close or Back to abandon a browsing draft.

Expected outcome and limits: Available levels/terminology vary. Philippine independent cities retain official parents; coordinates absent from official data require separate resolution.

UM 3.018 Manual resolution and clear

Access and purpose: Traveler. Manual destination input shall require verified disambiguation; clearing shall remove confirmation and dependent plan/stay state.

1. Use the manual-resolution action for a missing direct result.

2. Choose among candidates if ambiguous.

3. Clear or delete the input to remove confirmation before selecting another place.

Expected outcome and limits: Country/region/island selections need narrowing before generation. Wrong-country or unresolved coordinates cannot be promoted to confirmed destinations.

UM 3.019 Area suggestions and comparison

Access and purpose: Traveler. Area browsing shall expose verified child localities and a limited comparison without implying researched prices.

1. Choose an area that needs a locality.

2. Search its cities/towns and use Help me choose where available.

3. Review up to six candidates, then select deliberately.

Expected outcome and limits: One to six city identifiers; comparison uses entered spending allowance and mapped interest samples, not live tourism popularity or confirmed costs.

UM 3.020 Starting location and optional geolocation

Access and purpose: Traveler. An origin shall be entered or explicitly obtained through browser location permission, with manual entry after denial/failure.

1. Open the optional starting-location control.

2. Type a place or choose Use my current location.

3. Grant permission only if desired, review the resolved area, or enter it manually.

Expected outcome and limits: Geolocation requires secure context/permission. Origin selection clears when account changes; it is optional for current generation.

UM 3.021 Currency detection and fallback

Access and purpose: Traveler. Destination country codes shall map to supported planning currency or require an explicit supported fallback.

1. Select a destination.

2. Review the detected destination currency.

3. If unsupported, select a supported planning currency; keep that code through the plan.

Expected outcome and limits: 20 supported codes; JPY/KRW/VND use whole units. Fallback is a planning choice, not currency conversion.

UM 3.022 Reference-currency equivalents

Access and purpose: Traveler. Reference conversion shall preserve the trip currency and label source/date/freshness, retry limits and unavailable rates.

1. Choose Show equivalents in.

2. Read the reference rate and its date beside relevant amounts.

3. Keep the original currency as the budget authority.

Expected outcome and limits: Frankfurter mid-market reference, not card/cash conversion. Same-code rate is 1; wrong pair/invalid rate is rejected; Retry-After is honored.

UM 3.023 Destination transport guidance

Access and purpose: Traveler. The planner shall label curated destination transport modes as guidance and allow progress when no guidance is available.

1. After selecting a locality, review Transportation options.

2. Choose a relevant mode if known or continue without one.

3. Refresh options only when needed.

Expected outcome and limits: Guidance exists for selected country/city patterns. It is not a live timetable, capacity or fare source.

UM 3.024 Destination inspiration carousel

Access and purpose: Traveler. An inspiration selection shall populate its source destination/preferred activity for review before generation.

1. Browse destination spot cards.

2. Choose the planning action for a spot.

3. Review the selected destination, dates and budget before generating.

Expected outcome and limits: Inspiration is not a booking or a guarantee the attraction fits the final schedule.

UM 4 Trip planning and enrichment

Start with a confirmed destination, dates, party and total group budget. New plans use Balanced defaults. Expand optional fields only when relevant; choosing a suggestion does not book accommodation or establish price/availability. Generation and saving are separate outcomes. Missing screenshots: current authenticated planner, generation results and cost breakdown.

UM 4.025 Required trip brief and bounds

Access and purpose: Traveler. Generation shall require a confirmed locality, positive supported-currency group budget, valid party and 1-31 inclusive future/current dates.

1. Open Plan a trip.

2. Confirm destination, dates, Traveling as and whole-party budget.

3. Resolve any field-specific generation message before submitting.

Expected outcome and limits: Maximum 20 travelers; solo=1, couple=2, family/friends=2-20. Budget ceiling 1,000,000,000. Server start-day check uses UTC while browser uses local date (D-03).

UM 4.026 Preferences with Balanced generation policy

Access and purpose: Traveler. New/regenerated plans shall use Balanced style/pace while accepting interests, preferred activities, stay preference and transport.

1. Set optional comma-separated interests/activities, stay preference and transportation.

2. Generate after reviewing the brief.

3. Historical style metadata may be displayed on reopened plans.

Expected outcome and limits: No current travel-style or pace selector; old style values remain readable. Arrays max 20; bounded item lengths.

UM 4.027 Optional accommodation suggestions

Access and purpose: Traveler. The current planner shall browse relevant property suggestions without requesting room prices or requiring a stay selection.

1. Browse accommodations after selecting a locality.

2. Inspect name, actual location, photos and amenities where supplied.

3. Select an unpriced stay or choose it later.

Expected outcome and limits: Current UI passes suggestionsOnly. LiteAPI sandbox / Rapid / OSM sources vary; no room reservation. Empty/provider-error stays do not block generation.

UM 4.028 Manual stay and retained property details

Access and purpose: Traveler. A manual/mapped stay shall persist as an unpriced planning preference and remain recognizable after reopen.

1. Choose manual accommodation entry if the property is missing.

2. Enter its name and location.

3. Save the generated trip and inspect Saved accommodation after reopening.

Expected outcome and limits: Provider identity/allowed photos can be retained. Selection does not establish availability, rate, star quality or a completed booking.

UM 4.029 Expense assumptions

Access and purpose: Traveler. Explicit room, food, fare, fee and contingency assumptions shall be validated and calculated without invented prices.

1. Expand Budget details.

2. Enter known estimates and their shared/per-person basis.

3. Review the resulting known/estimated/unpriced amounts after generation.

Expected outcome and limits: Default contingency 10%; permitted 0-50%. Missing costs are excluded from known subtotal, not free. Currency precision applies.

UM 4.030 Optional pre-trip expenses

Access and purpose: Traveler. Pre-trip user estimates shall be included once as whole-party costs with matching starting location and currency.

1. Choose Add pre-trip expenses.

2. Enter applicable transfers, passport, visa/authorization, travel tax, insurance and other amounts.

3. Review the committed-cost breakdown.

Expected outcome and limits: No government fee advice is generated. Amounts are entered estimates; verify actual fees separately.

UM 4.031 Place grounding and geographic validation

Access and purpose: Traveler. The backend shall select physical attractions from supported records and reject invented, wrong-country or unsupported administrative/object entries.

1. Submit a valid brief.

2. Read the returned place sources and geographic context.

3. Confirm any nearby/outside-city transfer and venue details before travel.

Expected outcome and limits: Wikipedia/Wikidata/OSM references can be incomplete. Max 30 km straight-line is only one filter; membership evidence remains required. Coordinates alone do not prove a venue.

UM 4.032 AI structured schedule and repair

Access and purpose: Traveler. AI shall arrange only supplied place IDs/times in batches of at most seven days and pass independent schema, identity, date and transfer checks.

1. Generate the plan and wait for the request outcome.

2. Review source/provider labels, day order and suggested times.

3. Retry an explicitly retryable failure after it settles.

Expected outcome and limits: OpenAI default gpt-4o-mini / Gemini default gemini-3.1-flash-lite are code defaults and can be configured. One invalid-output repair per batch, sequential provider fallback; 45s per adapter call, 540s outer generation signal.

UM 4.033 Deterministic full-day scheduling

Access and purpose: Traveler. The scheduler shall reconcile catalog-backed visits, travel buffers, meal/free-time blocks and honest incomplete days within usable local-day windows.

1. Review every generated day.

2. Expand day notes for suggested, incomplete, rest or travel status.

3. Check actual routes/hours rather than treating local clocks as reservations.

Expected outcome and limits: Normal current Balanced cap is three catalog visits/day; no place repetition to fill a long trip. Arrival/departure/rest constraints exist in API but current planner has no input controls (see the API-only constraint inventory entry).

UM 4.034 Catalog fallback and incomplete results

Access and purpose: Traveler. When AI fails or references are sparse, the system shall retain catalog-backed suggestions/allowances with explicit incomplete or unavailable disclosures.

1. Read a provider/fallback notice after generation.

2. Inspect incomplete days and known costs.

3. Revise the brief or manually refine rather than assuming missing attractions or prices.

Expected outcome and limits: Catalog fallback is not a guest chatbot. Mock label is possible only with explicit non-production flag; production does not invent replacement venues.

UM 4.035 Generation idempotency and fencing

Access and purpose: Traveler. Repeated identical generation keys shall replay completed results, reject conflicting input and fence stale concurrent attempts.

1. Wait for a pending generation.

2. Retry the same unchanged brief only after a visible failure settles.

3. Change the brief deliberately when creating a distinct request.

Expected outcome and limits: 8-100-character keys. A pending record under 10 minutes conflicts; failed/stale attempts use compare-and-set fencing. This does not guarantee paid remote computation cancellation.

UM 4.036 Automatic save after generation

Access and purpose: Traveler. Successful generation shall attempt an owned atomic initial save or regeneration update and preserve a recoverable draft if saving fails.

1. Generate the plan.

2. Read whether it was saved automatically.

3. Open Saved trips to confirm; if auto-save failed, keep the draft and use the available save action.

Expected outcome and limits: Generation success and save success are separate. Save key is generated by saveIdentity; ordinary API calls without one may have different deduplication behavior.

UM 4.037 Activity photographs and provenance

Access and purpose: Traveler. Activity photos shall retain exact place/source identity, reusable rights and credits, with separately labeled illustrative/destination fallbacks.

1. Open a day or activity.

2. Read photo/source credits.

3. If a photo fails, allow recovery or retry without assuming a different image proves the exact venue.

Expected outcome and limits: No photo upload endpoint. Stored selection -> matched Wikimedia/Wikipedia -> Unsplash illustration -> trip destination -> neutral placeholder. Ordinary production flow bypasses Google provider.

UM 4.038 Date-matched weather

Access and purpose: Traveler. Weather shall distinguish current reference conditions from the dates actually covered by a forecast and disclose missing or stale coverage.

1. Read weather beside a trip/day.

2. Check forecast date and freshness.

3. Treat future dates outside provider coverage as unavailable rather than repeated forecasts.

Expected outcome and limits: OpenWeatherMap when configured, then Open-Meteo (14-day request). Thirty-one trip days do not imply 31 forecast days.

UM 4.039 Crowd estimates

Access and purpose: Traveler. Crowd context shall remain an explicitly low-confidence calendar heuristic rather than live congestion data.

1. Read crowd level and its explanatory note.

2. Consider suggested earlier visits, but review the actual venue before changing plans.

Expected outcome and limits: Weekend and broad Philippines/December heuristics only; no foot-traffic feed or automatic live rerouting.

UM 4.040 Advisory refresh without overwriting edits

Access and purpose: Traveler. Refreshing conditions shall update advisory metadata without regenerating or replacing manual activity content.

1. Choose Refresh conditions.

2. Wait for the result.

3. Verify the activities, notes and edits remain while forecast/crowd metadata changes or reports failure.

Expected outcome and limits: A refresh is not confirmation of unchanged future conditions or automatic rescheduling.

UM 5 Saved trips and traveler records

Use Saved trips to reopen the stored plan. A completed trip and a recorded actual visit are different records. Archive/restore and complete/reopen have distinct effects. Review confirmation dialogs before destructive actions. Missing screenshots: current saved-trip list, versions, manual edits, visited history, feedback and notices.

UM 5.041 Saved-trip list and pagination

Access and purpose: Traveler. Saved trips shall show owned summary cards with status/source/cost context and load further pages without exposing other accounts.

1. Open Saved trips.

2. Use status filters and inspect a card.

3. Load more when offered; open only the intended plan.

Expected outcome and limits: Summary responses omit full itinerary/weather. Internal tab key bookings means Saved trips, not an active booking feature.

UM 5.042 Reopen, day navigation and return

Access and purpose: Traveler. Reopening and navigating days shall preserve the selected trip/draft and never silently regenerate missing saved content.

1. Choose View itinerary.

2. Open View day itinerary, then Back to itinerary.

3. Use Back to saved trips from the overview; native Back/Forward should retain the same trip context.

Expected outcome and limits: Opening a sparse day does not repair/populate it. Legacy URL/history normalization is separate from AI generation.

UM 5.043 Activity editing

Access and purpose: Traveler. The editor shall permit bounded title, time, description, location, category and estimated-cost edits, recalculating totals as user estimates.

1. Open an active itinerary’s Edit activity action.

2. Enter the intended fields.

3. Save the dialog, review changed totals and use Update/Save changes to persist the draft.

Expected outcome and limits: Venue identity changes clear stale place/photo evidence. Unsupported changed price claims become estimates; server-owned management history cannot be forged.

UM 5.044 Add, remove and reorder activities

Access and purpose: Traveler. Draft activity changes shall preserve chronological day identity and existing evidence where appropriate, with explicit removal confirmation.

1. Use Add activity or a day’s edit controls.

2. Reorder with the available controls or confirm removal.

3. Inspect all affected days and save the changes deliberately.

Expected outcome and limits: At most eight saved activities/day in relevant validation. Completed/skipped management records are protected against silent removal.

UM 5.045 Save manual changes and recover drafts

Access and purpose: Traveler. Manual edits shall remain a browser draft until saved, with owner-scoped recovery and guards against unintended navigation loss.

1. Use Update/Save changes after reviewing edits.

2. If a leave/recovery dialog appears, choose deliberately to save, keep/discard or restore the draft.

3. Retry a failed save without navigating away.

Expected outcome and limits: Browser storage may be blocked/full. A browser draft is not a server backup; invalid/wrong-owner drafts are rejected.

UM 5.046 Regenerate existing trip

Access and purpose: Traveler. Regeneration shall require deliberate input review and produce a version of the same active trip only after valid generation/save.

1. Open an active trip’s regeneration/planning settings.

2. Review its restored brief; move passed dates forward as prompted.

3. Generate deliberately and confirm the saved result/version.

Expected outcome and limits: Current Balanced defaults apply to regenerated plans. Failed generation/save must leave the saved plan available; no automatic regeneration on open.

UM 5.047 Version history and whole-version restoration

Access and purpose: Traveler. Version restoration shall atomically recover the selected owned snapshot and its matching original trip context as a new version.

1. Expand a saved trip’s History.

2. Load more metadata if necessary and choose a version.

3. Review and confirm Restore only for an active/reopened trip.

Expected outcome and limits: 50 metadata records/page. Legacy versions can lack party/context and fail safely; unrecorded context is not guessed.

UM 5.048 Archive and restore trip

Access and purpose: Traveler. Archiving shall retain the trip and its prior completion state; restoring shall return it to active/completed accordingly.

1. Use a card’s More options -> Archive.

2. Inspect Archived trips.

3. Restore when needed before editing or recording a batch of visits.

Expected outcome and limits: Archived trips cannot be directly completed/reopened; restore first. Archiving is distinct from permanent deletion.

UM 5.049 Complete and reopen trip

Access and purpose: Traveler. Trip completion/reopening shall update lifecycle state only and shall not automatically create/remove visit history.

1. Choose Mark completed for a finished plan.

2. Use Reopen to make changes again.

3. Record actual visited places separately.

Expected outcome and limits: Completion does not prove attendance. Completed itineraries remain readable but editing/management requires reopen.

UM 5.050 Delete owned trip

Access and purpose: Traveler. Deletion shall require deliberate confirmation, delete the owned trip/version records and preserve detached visit/feedback history according to foreign keys.

1. Open More options -> Delete.

2. Read the confirmation.

3. Confirm only when permanent removal is intended; cancel to retain the trip.

Expected outcome and limits: No normal UI undelete is implemented. Documentation task did not execute deletion.

UM 5.051 Saved cover photos and stay details

Access and purpose: Traveler. Saved cards shall preserve relevant destination cover/photo attribution and any selected unpriced stay without altering itinerary content.

1. Inspect the Saved trip card cover and selected accommodation.

2. Open its source credits or details.

3. If media is unavailable, retain the plan and its identity.

Expected outcome and limits: Cover selection is not attraction verification. Unsplash selection tracking is separate and must not run on every render.

UM 5.052 Manual visited-place recording

Access and purpose: Traveler. A traveler shall explicitly record a named place and past/present actual visit date under their own account.

1. Open Visited places.

2. Add a place name, destination and actual visit date.

3. Save and inspect the personal history.

Expected outcome and limits: Future visit dates are rejected using the supplied valid timezone. Manual records are not automatically geographically verified.

UM 5.053 Record places from a saved trip

Access and purpose: Traveler. The batch visit modal shall list actual saved place candidates across days, prevent stale/duplicate selections and save selected actual visits atomically.

1. On an active/completed trip choose Record visited places.

2. Select actual places and an actual visit date, then confirm.

3. Reopen to inspect Already recorded states.

Expected outcome and limits: Allowances/rest/placeholders excluded. Archived trip requires restore. Batch duplicate prevention is stronger than the date-specific database unique key.

UM 5.054 Visited history search and removal

Access and purpose: Traveler. Visit history shall be owner-scoped, searchable, date-ordered and paginated, with explicit individual removal.

1. Open Visited places.

2. Search a place/destination, load more if offered, inspect retained media and remove a record only if intended.

Expected outcome and limits: Deleting a trip leaves its visited records detached. Removing a visit does not remove the saved itinerary.

UM 5.055 Itinerary feedback and updates

Access and purpose: Traveler. Feedback shall bind one 1-5 rating, supported category and 10-2000-character comment to an owned saved trip, updating the same record on resubmission.

1. Choose Rate trip or Feedback and support.

2. Select the intended saved itinerary, rating/category and comment.

3. Submit; reopen to edit existing feedback.

Expected outcome and limits: Category budget_accuracy/weather_info/ai_cohesion/route_efficiency. Sentiment remains not_analyzed; no sentiment AI is implemented.

UM 5.056 Traveler problem reports

Access and purpose: Traveler. Travelers shall submit bounded problem reports for administrator review and receive resolution notices.

1. Open Feedback and support or the account report form.

2. Enter a 5-120-character subject and 10-2000-character details.

3. Submit and monitor notifications.

Expected outcome and limits: Reports are not an itinerary chat command or a guarantee of support response time.

UM 5.057 Traveler notifications

Access and purpose: Traveler. The traveler notification view shall display owned notices and permit marking one/all read without changing another recipient's data.

1. Open the bell or Notifications.

2. Read a notice and follow its available destination.

3. Mark a notice or all notices read as intended.

Expected outcome and limits: Traveler updates use request/polling behavior; administrator live SSE is a distinct feature. Notices are not push/email delivery guarantees.

UM 5.058 Traveler profile and contact settings

Access and purpose: Traveler. Profile editing shall persist only allowed name/contact/avatar/bio fields and preserve account role/email/security state.

1. Open Your account.

2. Edit name, contact number, HTTPS photo URL or Travel profile.

3. Choose Save profile and inspect the result.

Expected outcome and limits: Name 2-80, phone <= 30, bio <= 500, avatar HTTPS <= 2048. Email is read-only; unknown protected keys do not become writable.

UM 6 Saved trip maps

Open a saved trip, select the intended day and open its map/navigation view. Resolve uncertain stops before asking for a route. GPS is optional and requires your explicit browser permission; stop tracking when finished. Missing screenshots: current authenticated map, exact-stop chooser and stop weather.

UM 6.059 Owned saved-trip map and timeline

Access and purpose: Traveler. The saved-trip map shall derive days/stops from the selected owned itinerary and separate daily allowances from visit stops.

1. Choose View map / Navigate on a saved trip.

2. Select the saved day and timeline activity.

3. Read exact/ambiguous/unavailable location states.

Expected outcome and limits: Opening a map does not complete an activity or record a visit. Skipped activities remain visible but are excluded from relevant routing.

UM 6.060 Exact map-place resolution and photos

Access and purpose: Traveler. An uncertain activity shall require an exact source-matched candidate before navigation, without silently promoting a vague name.

1. Select an unresolved stop.

2. Search/choose an exact place candidate and inspect its location and photo/source details.

3. Continue only after confirming the intended venue.

Expected outcome and limits: Vague allowances/activities cannot become precise stops by guess. Location-provider failure leaves unresolved feedback.

UM 6.061 Selected-day ordered route

Access and purpose: Traveler. Day routing shall follow recorded verified-stop order for the selected walking/cycling/driving mode and disclose omitted unresolved stops.

1. Choose a saved day with at least two exact places.

2. Choose the travel mode and request its day route.

3. Review the stop order and any omitted-stop notice.

Expected outcome and limits: OSRM road/path estimates, no live traffic. Public provider quota, no-route, excessive snap distance and unavailable routes are explicit.

UM 6.062 Location sharing and route guidance

Access and purpose: Traveler. Live route guidance shall begin only after an explicit location-sharing action and permit stopping, manual fallback and mode changes.

1. Choose Navigate/Use current location where offered.

2. Review permission and selected destination/mode.

3. Follow displayed estimated guidance; Stop sharing when finished.

Expected outcome and limits: HTTPS/permission required. Denial, inaccurate/stale position and route failure retain recovery paths. No safety-critical navigation guarantee.

UM 6.063 Activity-specific weather

Access and purpose: Traveler. Activity weather shall keep current and forecast requests distinct and use the exact resolved stop/date/time within provider coverage.

1. Select a mapped activity.

2. Open its weather view and review Current versus scheduled-date weather.

3. Read unavailable/hour-reference messages when applicable.

Expected outcome and limits: Exact location needed; scheduled weather outside actual coverage remains unavailable. Global provider request budgets apply.

UM 6.064 Retired GPS URL compatibility

Access and purpose: Traveler / guest. Legacy GPS URLs shall redirect into Saved trips navigation while retaining only valid trip/day identifiers.

1. Open an old GPS bookmark.

2. Read the moved-navigation message.

3. Open the selected saved-trip map, signing in if required.

Expected outcome and limits: The standalone GPS workspace is removed; compatibility redirect does not restore removed navigation tabs.

UM 7 Managed day activities

Open an active saved trip and the intended day. Managed actions change the saved plan and require current revision/confirmation. Completed and skipped stops constrain remaining-day replanning. A preview can be inspected/canceled before accepting. Missing screenshots: current managed activity dialogs and replan preview.

UM 7.065 Confirm completion, skip or restore activity

Access and purpose: Traveler. Managed activity status changes shall require owner, active trip, current revision, saved activity identity and explicit confirmation.

1. In the saved-trip map select an activity.

2. Choose Mark as Completed, Skip Activity or restore/undo.

3. Review and confirm; inspect the retained activity/history.

Expected outcome and limits: Completed details are protected until undo. Skip retains cost/history and excludes remaining-day routing/replanning. Status change does not record a VisitedPlace.

UM 7.066 Confirmed activity edit/reschedule

Access and purpose: Traveler. Managed edit/reschedule shall reject stale revisions, new duplicate places, time overlaps and changes to completed details until undone.

1. Select Edit Activity or Reschedule Activity.

2. Enter valid fields/day/local time and duration.

3. Confirm; reload if a conflict reports a changed saved trip.

Expected outcome and limits: Duration 15-720 minutes. Moving to another day requires Reschedule; target max 8 activities. Changed venue/photo evidence is revalidated.

UM 7.067 AI remaining-day preview

Access and purpose: Traveler. Replan My Day shall propose times for every remaining saved activity exactly once without writing the itinerary or changing costs/completed history.

1. Choose Replan My Day for an active saved day.

2. Enter an available destination-local time window.

3. Preview and compare proposed times, Flexible deferrals and warnings.

Expected outcome and limits: Three previews per owner/5 minutes. Existing Gemini/OpenAI required; no mock fallback. Missing saved durations must be entered first. Preview can consume provider quota.

UM 7.068 Accept or cancel revised day

Access and purpose: Traveler. Acceptance shall validate a signed owner/trip/revision-bound preview and persist one reviewed change; cancellation shall leave itinerary/version data unchanged.

1. Inspect the full preview.

2. Choose Accept revised day to persist, or Cancel replanning to retain the original.

3. Reload and preview again after expiry/conflict.

Expected outcome and limits: Preview token expires after 15 minutes. Browser abort does not prove remote model billing stopped. Completed/skipped records and saved costs remain.

UM 7.069 Activity history and legacy compatibility

Access and purpose: Traveler. Management shall preserve stable activity identity/history and default legacy records to Planned without writing on open.

1. Expand Activity history after a confirmed management change.

2. Review recorded action/date/time.

3. Use version restore when a whole prior plan is needed.

Expected outcome and limits: Legacy IDs are derived read-only until confirmed persistence. Management metadata is server-owned, separate from a free-form manual edit.

UM 8 Administration

Sign in with a provisioned administrator account to open /admin/dashboard. Public registration creates a traveler only. Read record context before suspension, restoration or report/appeal decisions. System health configuration flags are not proof of a healthy external provider. Missing screenshots: all current authenticated administrator tabs.

UM 8.070 Administrator overview

Access and purpose: Administrator. The overview shall show stored aggregate account/generation/feedback counts and seven calendar days without invented activity.

1. Sign in as an authorized administrator.

2. Open Overview.

3. Read all-time totals, recent generation chart and absent/available average rating.

Expected outcome and limits: Calendar uses UTC+8/Asia/Shanghai. Counts describe stored records, not active concurrent users or measured satisfaction.

UM 8.071 User search and access state

Access and purpose: Administrator. User management shall search/filter permitted directory fields and suspend/restore non-admin accounts with confirmation and notifications.

1. Open Users.

2. Search name/email/status and select filters.

3. Choose a traveler’s suspend/restore action, inspect the confirmation and submit deliberately.

Expected outcome and limits: Admin accounts cannot be changed through this action. No public admin creation/role reassignment is exposed. Unknown status currently normalizes to active (D-04).

UM 8.072 Send traveler warning

Access and purpose: Administrator. A warning shall target an existing traveler with a 10-2000-character message and persist a recipient-scoped notice/audit event.

1. Open Users and the selected traveler’s warning action.

2. Enter a factual bounded warning.

3. Confirm and inspect the outcome.

Expected outcome and limits: In-app notice only; not automatic email/SMS. The documentation task sent no warning.

UM 8.073 Review/resolve problem reports

Access and purpose: Administrator. Administrators shall inspect full traveler reports and resolve only pending problem reports with an audited recipient notification.

1. Open Reports.

2. Read the selected report’s full content and reporter context.

3. Choose Mark resolved after review; observe the resulting status.

Expected outcome and limits: Resolution is a recorded decision, not technical proof the underlying defect is fixed. Already resolved/wrong-kind requests are rejected.

UM 8.074 Review suspension appeals

Access and purpose: Administrator. Appeal review shall validate a pending appeal and current suspended traveler state before approved/rejected decision and bounded explanation.

1. Open Reports and the appeal category.

2. Read the appeal.

3. Enter a 10-2000-character decision message, select the decision and confirm.

Expected outcome and limits: Approve restores account access; reject retains suspension. Stale/double review and already-restored account are refused.

UM 8.075 Live reports and reconnect snapshots

Access and purpose: Administrator. The live report stream shall authenticate repeatedly, send scoped snapshots, reconcile cross-instance changes and clean up closed/stalled connections.

1. Open the admin workspace.

2. Read Live reports connected or reconnect feedback.

3. On a connection failure, retain the view and allow reconnection/refresh.

Expected outcome and limits: SSE (not Socket.IO): 5-second reconciliation, 5-minute connection rotation, event-triggered local refresh and bounded write buffer.

UM 8.076 System Health and selected audit activity

Access and purpose: Administrator. System Health shall distinguish service configuration from verified availability and show recent generation failures and selected account/audit activity.

1. Open System Health.

2. Inspect configuration indicators, failed generations and available audit search.

3. Report a provider outage using its actual evidence.

Expected outcome and limits: No live uptime/latency dashboard, AI prompt editor or automated model curation. Configured means key presence, not a successful provider call.

UM 8.077 Feedback filters and exact details

Access and purpose: Administrator. Feedback review shall paginate/search/filter stored feedback and open exact linked feedback/related itinerary without inferring sentiment.

1. Open User Feedback.

2. Search or filter rating/sentiment/date, page through results and select a record.

3. Open its related itinerary if retained.

Expected outcome and limits: Deleted trip/user links can be null. Sentiment defaults not_analyzed; filters do not prove automated analysis.

UM 8.078 Administrator notifications

Access and purpose: Administrator. Admin notifications shall retain recipient isolation, meaningful category/deep link and accurate unread count while allowing one/all read.

1. Open Notifications or a bell notice.

2. Follow its report/feedback/system destination.

3. Mark read and inspect the unread badge.

Expected outcome and limits: Category is encoded in link metadata for current notices; compatibility handles older notices. Only that admin’s read state changes.

UM 8.079 Administrator own profile

Access and purpose: Administrator. Account Settings shall update the administrator’s own permitted contact/profile fields without exposing password/role mutation.

1. Open Account Settings.

2. Edit allowed name/contact/profile fields and save.

3. Verify the returned profile before leaving.

Expected outcome and limits: Same allowed-field and length constraints as the traveler profile; no email/role/security-field editing.

UM 9 Technical services

Operators should use the setup appendix and actual environment examples, keeping all secrets server-side. End users should follow safe error recovery without changing deployment configuration. Missing screenshot: deployed production status or recovery operation; neither was verified.

UM 9.080 API error contracts and safe recovery

Access and purpose: All callers. The API shall return stable safe error/code/retryability metadata, sanitized malformed-JSON feedback and distinct dependency failures.

1. If a request fails, read the visible message.

2. Retry only after a retryable failure/cooldown; retain unsaved work.

3. Provide time and non-sensitive context to support.

Expected outcome and limits: Database outages can return retryable 503 rather than false logout. No exhaustive penetration/security certification is asserted.

UM 9.081 Database retries and pooled access

Access and purpose: API process. The API shall use PostgreSQL/Prisma pooled TCP access with bounded retries only for recognized connection failures.

1. Operators configure the backend database URL securely.

2. On an outage, inspect safe health/error feedback and restore connectivity before retrying affected operations.

Expected outcome and limits: An ambiguous limiter retry may count twice (fail-closed). Not a blanket retry of every business write or a proven backup/restore service.

UM 9.082 Shared rate limits and request-driven cleanup

Access and purpose: API process. Rate limits shall use shared fixed-window PostgreSQL buckets and opportunistically prune old buckets/revocations.

1. Operators monitor 429 outcomes and database health.

2. Respect Retry-After and avoid rapid repeat submissions.

3. Review provider quotas before scaling.

Expected outcome and limits: Cleanup is a 1% request-triggered chance, not a scheduled worker. Authentication itself depends on database availability for rate limiting.

UM 9.083 Provider cache and unavailable disclosures

Access and purpose: API process / traveler. Provider caching shall coalesce identical in-flight requests, bound entries, honor freshness/stale cutoffs and expose unavailable results honestly.

1. Read source/freshness labels.

2. Refresh only when needed.

3. Treat stale data as a fallback and confirm current prices/availability with the original provider.

Expected outcome and limits: General cache max 300 entries. Flights/hotels 5m fresh/15m cutoff; activities 30m/2h; weather 15m/1h; FX 6h/24h.

UM 9.084 Security headers, HTTPS and same-origin writes

Access and purpose: All callers / operator. Production shall enforce HTTPS/security headers and reject untrusted cross-site browser writes with server-side authorization.

1. Operators configure trusted frontend HTTPS origins and independent secrets.

2. Clients access the normal frontend; an origin or access denial is not solved by bypassing security controls.

Expected outcome and limits: No full encryption-at-rest/legal compliance or exhaustive XSS/CSRF proof. Frontend CSP includes unsafe-inline and is an explicit review item.

UM 9.085 Operational health and graceful shutdown

Access and purpose: Operator. Operational endpoints shall expose basic API identity/database readiness and the process shall close HTTP/database resources on termination.

1. Inspect GET / and /health in an authorized environment.

2. Check dependency state as well as HTTP status; use the documented healthcheck on a release candidate.

Expected outcome and limits: Health is a point observation, not measured uptime or a restore drill. The healthcheck script forwards HTTPS scheme for container checks.

UM 9.086 Contracts and paired-revision validation

Access and purpose: Maintainer. API contracts shall remain aligned with generated frontend snapshots and be checked against a pinned compatible backend revision.

1. Use contracts:check in an isolated candidate.

2. Compare schema differences before authorizing contracts:sync or a release.

3. Record both repository revisions.

Expected outcome and limits: contracts:sync writes files; documentation task uses copies/check mode only. CI definitions are configuration evidence, not proof a run passed.

UM 9.087 Container deployment and rollback configuration

Access and purpose: Maintainer. A proposed release shall build a non-root image, apply reviewed migrations explicitly and require health before retaining the new app image.

1. Prepare an isolated release and required configuration.

2. Review backup/migration/rollback plans.

3. Only after separate release authorization run the documented deployment procedure and verify health.

Expected outcome and limits: Configured pipeline, not executed here. Image rollback does not undo applied database migrations. Domain/hosting/backup acceptance needs actual operational evidence.

UM 10 Compatibility and excluded scope

These entries explain compatibility and absent scope. They are not extra user tasks or hidden marketplace features. Use the current navigation rather than old screenshots/forms. Retained APIs require developer-controlled validation and are not supported current UI instructions.

UM 10.088 Flight and activity comparison API

Access and purpose: Traveler API consumer; UI removed. Existing authenticated comparison endpoints shall normalize source-backed flight/activity results and unavailable states, without claiming a current traveler comparison screen.

1. There is no current Travel options tab.

2. Maintainers can inspect the API contract in an isolated integration environment; ordinary travelers use the retained planner.

Expected outcome and limits: Unreferenced comparison component retained in source. Amadeus 1-9 travelers; configured production/test distinction. Mark partial because API exists but UI is retired.

UM 10.089 Hotel quote/revalidation API

Access and purpose: Traveler API consumer; current UI suggestions only. Existing quote APIs shall bind owner/context/occupancy/expiry and reject changed/sold-out/invalid quotes without asserting current UI booking.

1. The main planner shows suggestions only.

2. In an isolated API test, request a supported quote, revalidate it and inspect changed/expired/provider-error outcomes.

Expected outcome and limits: API occupancy 1-8 rooms, adults/child ages 0-17 and total <= 20. LiteAPI remains sandbox; Rapid rights gated. No current price/occupancy controls in main planner.

UM 10.090 Generation arrival/departure/rest API constraints

Access and purpose: Traveler API consumer; no current form controls. The generation API shall validate optional destination-local arrival/departure clocks and in-range rest dates without implying an exposed UI.

1. No current planner controls set these fields.

2. Maintainers may specify them in an isolated API case and inspect rest/travel/incomplete day output.

Expected outcome and limits: HH:MM clocks; <= 31 in-range rest dates. Arrival adds 90 minutes; departure reserves 120. API-only capability is partial for normal users.

UM 10.091 Removed owner/marketplace/payment workflows

Access and purpose: No active user role. Removed owner registration, booking, payment, promotion and trust/profile-moderation actions shall remain unavailable rather than be documented as working product features.

Current availability: unavailable. Do not look for an Owner workspace, checkout or payment action. Use the retained trip planner and obtain actual reservations from external providers. Schema presence is not feature availability. PayMongo is not integrated; owner role no longer exists. Legacy data is not deleted by this documentation work.

UM 10.092 Retired preferences/comparison tabs and duplication

Access and purpose: Traveler. Retired Preferences/Travel options tab URLs shall normalize safely, and the removed duplicate-trip action shall reject requests.

Current availability: unavailable. Use current sidebar entries. An old preferences/compare bookmark returns to the supported workspace; use new trip planning when a distinct plan is needed. Unreferenced TripPlannerForm still contains a travel-style prop; current dashboard does not render it. Do not infer a current style selector from that file.

UM 10.093 Guest chatbot, live crowd feed and automatic backups

Access and purpose: Unavailable / proposed. Unimplemented historical requirements shall be identified explicitly and must not be presented as observed system capabilities.

Current availability: unavailable. Guests can explore the public site and account dialogs. Use authenticated structured planning; do not expect a guest chat, live crowd telemetry or an application-managed daily backup interface. Future work requires separate requirements, provider/security design and execution evidence. Old 99.9% uptime, 500-user capacity and automatic daily backup statements remain unverified targets.

UM 11 Troubleshooting reference

Table 29 Common recovery paths

Symptom	Safe next action
No verification/reset mail	Check email/spam and visible outcome; wait for cooldown; use newest accepted code. Contact the authorized operator for configuration/provider failures.
Destination missing/ambiguous	Choose country/parent/type, use Browse or explicit manual resolution; do not pick an unrelated namesake.
Generation slow/unavailable	Wait for bounded request outcome; read fallback/incomplete labels. Avoid parallel repeated submissions.
Save failed after generation	Retain the owner-scoped local draft, restore connection/session and explicitly retry save; verify Saved trips.

Symptom	Safe next action
Cost looks too low	Inspect unpriced items, assumptions, party basis and omitted unknowns; verify current costs independently.
Wrong or missing photo	Check identity context and attribution; do not substitute a different-country photo. Bir Tawil is an explicit gap.
Map no route/location denied	Resolve stops, select supported mode and use manual context; GPS is optional.
Stale activity/preview	Reload saved plan; discard expired/stale preview; generate a new one if needed.
Old tab or GPS link	Allow normalization/redirect and choose the owned saved trip/day.
Admin stream disconnected	Reconnect via the workspace after checking current session; do not bypass authorization.

Conclusions

The inspected system supports the principal planning-to-saved-trip workflow, with destination identity, constrained generation, explicit money assumptions, versioned edits, maps, managed activities and separate actual-visit recording. The administrator role supplies moderation and record visibility. The current architecture is Next.js/Express/Prisma/PostgreSQL, not the old PHP/MySQL design, and the supported planning span is one to thirty-one days. Marketplace/payment/owner and guest-chat claims are excluded from implemented scope. [1][2][3]

The documentation reconciliation establishes a feature/route/process baseline and repeatable requirements-to-test mapping. The newly executed isolated suites provide evidence for 673 tests across the two suites, but not 673 end-to-end user workflows. No empirical improvement in planning time, accuracy, satisfaction or tourism outcomes was measured. Static concerns and operational acceptance gaps prevent a defensible claim of unconditional production or academic acceptance. [6]

Recommendations

Correct the observed content/contract gaps and reproduce the potential date, status-command, destination-length and currency-normalization defects in disposable tests before changing the application. Preserve the intended raw-input and authorization rules rather than documenting the defect as normal behavior. Run targeted regression and retain the actual correction/retest evidence.

Complete authorized two-owner persistence, role, email/OAuth and provider outage rehearsals, and validate media identity/crops across the full code manifest. Keep Bir Tawil explicit until a verified exact-area image is available. Review provider terms, pacing, attribution and the privacy notice before operational use. Avoid increasing fallback coverage by borrowing unrelated places or photos.

Approve and measure performance, accessibility and recovery targets in a controlled staging environment. Rehearse database restore and migration compatibility rather than relying on image rollback. Confirm team/academic/management details and obtain documented UAT/reviewer decisions. Finally, refresh Word contents/page fields and render every DOCX page in a compatible engine before academic submission. These are proposed next actions, not completed development work.

References

[1] TravelMate frontend working tree. travelmate-frontend-new; HEAD 1d55ba3c2eb2a9387cb34c0cb6ca5edef8bfbc96 with recorded working changes; inspected 15 October 2026. Source snapshot, status and feature references in the evidence package.

[2] TravelMate backend working tree. travelmate-backend-api; HEAD 234e08d2317526f772854296bfaf6a33ca5dd0fa with recorded working changes; schema, 24 migration directories, controllers/services/config/tests inspected 15 October 2026.

[3] TravelMate. Supplied TravelMate-SRS.docx, documentation/mine. Historical requirements and administrative statements, not proof of present implementation. Original SHA256 26828397f5a0647ae2ecc0175f45ace40b19bf8e5d8212d6fc07db04df8ff9f6. Original ZIP CRC fault repaired in a separate reading copy only; all source media decoded.

[4] TravelMate. Supplied TravelMate-SPMP.docx, documentation/mine. Archived planning baseline. Original SHA256 cf5b9f119ba6cf7c8f39167dd6ff6208b1d881e73811a840665d038c15094f13.

[5] Supplied Herbal AI Pre Defense Manuscript v5 2026-2027, revised 15 October 2026, and its standalone engineering references. Used solely for organization/format by explicit user confirmation. No project facts, names, results or contributions transferred. Main reference SHA256 c8fde03149ba05b2f98b8a7e08b131ede9629313d1ae2b4114ee304d005df42c.

[6] TravelMate documentation execution evidence, 15 October 2026: final isolated frontend/backend logs, network guard, copied-source manifest, execution register and anonymous current screenshots. Included in the evidence bundle.

[7] TravelMate historical engineering reports: audit/2026-10-15/globe-photo-curation/REPORT.md; audit/2026-10-15/itinerary-overview/REPORT.md; audit/2026-10-13/remove-workspace-sections/REPORT.md. Each report is

limited to its recorded environment/revision; historical counts are not added to current counts.

[8] TravelMate geographic/media source manifests and read-only catalogue metadata: frontend public discovery assets; backend GeoNames/PSGC/local geography and source metadata. Counts describe these snapshots, not guaranteed worldwide completeness.

[9] NASA Earth Observatory and Natural Earth source/licensing links retained in the current TravelMate image and map credits and asset attribution manifests. External provider/author-specific licenses remain asset-specific; footer attribution is not discarded.

[10] Next.js official App Router documentation, https://nextjs.org/docs/app, accessed 15 October 2026. Used for architectural context; inspected project code and dependency declarations govern this version-specific description.

[11] Republic of the Philippines, Republic Act No. 10173, Data Privacy Act of 2012, Official Gazette, https://officialgazette.gov.ph/2012/08/15/republic-act-no-10173/, accessed 15 October 2026. Legal context only; no compliance certification asserted.

[12] MediaWiki API Etiquette, https://www.mediawiki.org/wiki/API%3AEtiquette, accessed 15 October 2026. Provider identification, pacing and courteous API usage context; actual limits are taken from inspected code/configuration and agreements.

Appendices

Appendix A Requirements and feature traceability

Supporting files: TravelMate-Feature-Inventory.csv; TravelMate-Requirements-Traceability-Matrix.csv; TravelMate-Test-Cases.csv; TravelMate-Quality-Requirements.csv; TravelMate-Database-Field-Register.csv; API, Page and Process Reconciliation CSVs; Verification Execution Register; Change Summary; Unresolved Details; and eleven editable diagram sources.

The accompanying editable CSV files are the complete machine-readable inventory and RTM. The compact register below provides document navigation; the RTM additionally contains full implementation references, status and all linked boundary/quality cases. All ninety-three features appear in the SRS, SDD, STD and manual, with unavailable/partial differences explicitly explained.

Table 30 Compact functional traceability register

Requirement and feature	Design	Case	Manual
TM-FR-001 / TM-F-001 Landing and navigation	SDD 4.1	TM-TC-001	UM 1
TM-FR-002 / TM-F-002 Interactive Earth controls	SDD 4.1	TM-TC-002	UM 1
TM-FR-003 / TM-F-003 Country hit detection and border identity	SDD 4.1	TM-TC-003	UM 1
TM-FR-004 / TM-F-004 Country destination photo and identity card	SDD 4.1	TM-TC-004	UM 1
TM-FR-005 / TM-F-005 Image and map credits	SDD 4.1	TM-TC-005	UM 1
TM-FR-006 / TM-F-006 Privacy and terms pages	SDD 4.1	TM-TC-006	UM 1
TM-FR-007 / TM-F-007 Traveler registration	SDD 4.2	TM-TC-007	UM 2
TM-FR-008 / TM-F-008 Email verification	SDD 4.2	TM-TC-008	UM 2
TM-FR-009 / TM-F-009 Resend verification	SDD 4.2	TM-TC-009	UM 2

Requirement and feature	Design	Case	Manual
TM-FR-010 / TM-F-010 Password sign-in and role redirect	SDD 4.2	TM-TC-010	UM 2
TM-FR-011 / TM-F-011 Google OAuth	SDD 4.2	TM-TC-011	UM 2
TM-FR-012 / TM-F-012 Authoritative sessions and access control	SDD 4.2	TM-TC-012	UM 2
TM-FR-013 / TM-F-013 Logout and local cleanup	SDD 4.2	TM-TC-013	UM 2
TM-FR-014 / TM-F-014 Forgot/reset password	SDD 4.2	TM-TC-014	UM 2
TM-FR-015 / TM-F-015 Suspension appeal access	SDD 4.2	TM-TC-015	UM 2
TM-FR-016 / TM-F-016 Worldwide direct search	SDD 4.3	TM-TC-016	UM 3
TM-FR-017 / TM-F-017 Country and administrative browsing	SDD 4.3	TM-TC-017	UM 3
TM-FR-018 / TM-F-018 Manual resolution and clear	SDD 4.3	TM-TC-018	UM 3
TM-FR-019 / TM-F-019 Area suggestions and comparison	SDD 4.3	TM-TC-019	UM 3
TM-FR-020 / TM-F-020 Starting location and optional geolocation	SDD 4.3	TM-TC-020	UM 3
TM-FR-021 / TM-F-021 Currency detection and fallback	SDD 4.3	TM-TC-021	UM 3
TM-FR-022 / TM-F-022 Reference-currency equivalents	SDD 4.3	TM-TC-022	UM 3
TM-FR-023 / TM-F-023 Destination transport guidance	SDD 4.3	TM-TC-023	UM 3
TM-FR-024 / TM-F-024 Destination inspiration carousel	SDD 4.3	TM-TC-024	UM 3
TM-FR-025 / TM-F-025 Required trip brief and bounds	SDD 4.4	TM-TC-025	UM 4

Requirement and feature	Design	Case	Manual
TM-FR-026 / TM-F-026 Preferences with Balanced generation policy	SDD 4.4	TM-TC-026	UM 4
TM-FR-027 / TM-F-027 Optional accommodation suggestions	SDD 4.4	TM-TC-027	UM 4
TM-FR-028 / TM-F-028 Manual stay and retained property details	SDD 4.4	TM-TC-028	UM 4
TM-FR-029 / TM-F-029 Expense assumptions	SDD 4.4	TM-TC-029	UM 4
TM-FR-030 / TM-F-030 Optional pre-trip expenses	SDD 4.4	TM-TC-030	UM 4
TM-FR-031 / TM-F-031 Place grounding and geographic validation	SDD 4.4	TM-TC-031	UM 4
TM-FR-032 / TM-F-032 AI structured schedule and repair	SDD 4.4	TM-TC-032	UM 4
TM-FR-033 / TM-F-033 Deterministic full-day scheduling	SDD 4.4	TM-TC-033	UM 4
TM-FR-034 / TM-F-034 Catalog fallback and incomplete results	SDD 4.4	TM-TC-034	UM 4
TM-FR-035 / TM-F-035 Generation idempotency and fencing	SDD 4.4	TM-TC-035	UM 4
TM-FR-036 / TM-F-036 Automatic save after generation	SDD 4.4	TM-TC-036	UM 4
TM-FR-037 / TM-F-037 Activity photographs and provenance	SDD 4.4	TM-TC-037	UM 4
TM-FR-038 / TM-F-038 Date-matched weather	SDD 4.4	TM-TC-038	UM 4
TM-FR-039 / TM-F-039 Crowd estimates	SDD 4.4	TM-TC-039	UM 4
TM-FR-040 / TM-F-040 Advisory refresh without overwriting edits	SDD 4.4	TM-TC-040	UM 4
TM-FR-041 / TM-F-041 Saved-trip list and pagination	SDD 4.5	TM-TC-041	UM 5

Requirement and feature	Design	Case	Manual
TM-FR-042 / TM-F-042 Reopen, day navigation and return	SDD 4.5	TM-TC-042	UM 5
TM-FR-043 / TM-F-043 Activity editing	SDD 4.5	TM-TC-043	UM 5
TM-FR-044 / TM-F-044 Add, remove and reorder activities	SDD 4.5	TM-TC-044	UM 5
TM-FR-045 / TM-F-045 Save manual changes and recover drafts	SDD 4.5	TM-TC-045	UM 5
TM-FR-046 / TM-F-046 Regenerate existing trip	SDD 4.5	TM-TC-046	UM 5
TM-FR-047 / TM-F-047 Version history and whole-version restoration	SDD 4.5	TM-TC-047	UM 5
TM-FR-048 / TM-F-048 Archive and restore trip	SDD 4.5	TM-TC-048	UM 5
TM-FR-049 / TM-F-049 Complete and reopen trip	SDD 4.5	TM-TC-049	UM 5
TM-FR-050 / TM-F-050 Delete owned trip	SDD 4.5	TM-TC-050	UM 5
TM-FR-051 / TM-F-051 Saved cover photos and stay details	SDD 4.5	TM-TC-051	UM 5
TM-FR-052 / TM-F-052 Manual visited-place recording	SDD 4.5	TM-TC-052	UM 5
TM-FR-053 / TM-F-053 Record places from a saved trip	SDD 4.5	TM-TC-053	UM 5
TM-FR-054 / TM-F-054 Visited history search and removal	SDD 4.5	TM-TC-054	UM 5
TM-FR-055 / TM-F-055 Itinerary feedback and updates	SDD 4.5	TM-TC-055	UM 5
TM-FR-056 / TM-F-056 Traveler problem reports	SDD 4.5	TM-TC-056	UM 5
TM-FR-057 / TM-F-057 Traveler notifications	SDD 4.5	TM-TC-057	UM 5
TM-FR-058 / TM-F-058 Traveler profile and contact settings	SDD 4.5	TM-TC-058	UM 5

Requirement and feature	Design	Case	Manual
TM-FR-059 / TM-F-059 Owned saved-trip map and timeline	SDD 4.6	TM-TC-059	UM 6
TM-FR-060 / TM-F-060 Exact map-place resolution and photos	SDD 4.6	TM-TC-060	UM 6
TM-FR-061 / TM-F-061 Selected-day ordered route	SDD 4.6	TM-TC-061	UM 6
TM-FR-062 / TM-F-062 Location sharing and route guidance	SDD 4.6	TM-TC-062	UM 6
TM-FR-063 / TM-F-063 Activity-specific weather	SDD 4.6	TM-TC-063	UM 6
TM-FR-064 / TM-F-064 Retired GPS URL compatibility	SDD 4.6	TM-TC-064	UM 6
TM-FR-065 / TM-F-065 Confirm completion, skip or restore activity	SDD 4.7	TM-TC-065	UM 7
TM-FR-066 / TM-F-066 Confirmed activity edit/reschedule	SDD 4.7	TM-TC-066	UM 7
TM-FR-067 / TM-F-067 AI remaining-day preview	SDD 4.7	TM-TC-067	UM 7
TM-FR-068 / TM-F-068 Accept or cancel revised day	SDD 4.7	TM-TC-068	UM 7
TM-FR-069 / TM-F-069 Activity history and legacy compatibility	SDD 4.7	TM-TC-069	UM 7
TM-FR-070 / TM-F-070 Administrator overview	SDD 4.8	TM-TC-070	UM 8
TM-FR-071 / TM-F-071 User search and access state	SDD 4.8	TM-TC-071	UM 8
TM-FR-072 / TM-F-072 Send traveler warning	SDD 4.8	TM-TC-072	UM 8
TM-FR-073 / TM-F-073 Review/resolve problem reports	SDD 4.8	TM-TC-073	UM 8
TM-FR-074 / TM-F-074 Review suspension appeals	SDD 4.8	TM-TC-074	UM 8

Requirement and feature	Design	Case	Manual
TM-FR-075 / TM-F-075 Live reports and reconnect snapshots	SDD 4.8	TM-TC-075	UM 8
TM-FR-076 / TM-F-076 System Health and selected audit activity	SDD 4.8	TM-TC-076	UM 8
TM-FR-077 / TM-F-077 Feedback filters and exact details	SDD 4.8	TM-TC-077	UM 8
TM-FR-078 / TM-F-078 Administrator notifications	SDD 4.8	TM-TC-078	UM 8
TM-FR-079 / TM-F-079 Administrator own profile	SDD 4.8	TM-TC-079	UM 8
TM-FR-080 / TM-F-080 API error contracts and safe recovery	SDD 4.9	TM-TC-080	UM 9
TM-FR-081 / TM-F-081 Database retries and pooled access	SDD 4.9	TM-TC-081	UM 9
TM-FR-082 / TM-F-082 Shared rate limits and request-driven cleanup	SDD 4.9	TM-TC-082	UM 9
TM-FR-083 / TM-F-083 Provider cache and unavailable disclosures	SDD 4.9	TM-TC-083	UM 9
TM-FR-084 / TM-F-084 Security headers, HTTPS and same-origin writes	SDD 4.9	TM-TC-084	UM 9
TM-FR-085 / TM-F-085 Operational health and graceful shutdown	SDD 4.9	TM-TC-085	UM 9
TM-FR-086 / TM-F-086 Contracts and paired-revision validation	SDD 4.9	TM-TC-086	UM 9
TM-FR-087 / TM-F-087 Container deployment and rollback configuration	SDD 4.9	TM-TC-087	UM 9
TM-FR-088 / TM-F-088 Flight and activity comparison API	SDD 4.10	TM-TC-088	UM 10
TM-FR-089 / TM-F-089 Hotel quote/revalidation API	SDD 4.10	TM-TC-089	UM 10
TM-FR-090 / TM-F-090 Generation arrival/departure/rest API constraints	SDD 4.10	TM-TC-090	UM 10

Requirement and feature	Design	Case	Manual
TM-FR-091 / TM-F-091 Removed owner/marketplace/payment workflows	SDD 4.10	TM-TC-091	UM 10 (retired/unavailable capabilities)
TM-FR-092 / TM-F-092 Retired preferences/comparison tabs and duplication	SDD 4.10	TM-TC-092	UM 10 (retired/unavailable capabilities)
TM-FR-093 / TM-F-093 Guest chatbot, live crowd feed and automatic backups	SDD 4.10	TM-TC-093	UM 10 (retired/unavailable capabilities)

Table 31 Quality traceability register

Requirement	Design	Case	Manual
TM-NFR-001 Authoritative access	SDD 7.1	TM-QT-001	UM 2
TM-NFR-002 Credential protection	SDD 7.1	TM-QT-002	UM 2
TM-NFR-003 Request origin and transport	SDD 7.1	TM-QT-003	UM 9
TM-NFR-004 Financial precision	SDD 6.2	TM-QT-004	UM 4
TM-NFR-005 Grounded identities	SDD 6.1	TM-QT-005	UM 4
TM-NFR-006 Honest degraded service	SDD 7.2	TM-QT-006	UM 9
TM-NFR-007 Atomic persistence	SDD 3.3	TM-QT-007	UM 5
TM-NFR-008 Duplicate prevention	SDD 3.3	TM-QT-008	UM 5
TM-NFR-009 Responsive layout	SDD 5	TM-QT-009	UM 1
TM-NFR-010 Keyboard and motion access	SDD 5	TM-QT-010	UM 1
TM-NFR-011 Timeout and cancellation	SDD 7.2	TM-QT-011	UM 4
TM-NFR-012 Data minimization	SDD 7.3	TM-QT-012	UM 6
TM-NFR-013 Attribution and provenance	SDD 5	TM-QT-013	UM 1
TM-NFR-014 Operational recovery	SDD 8	TM-QT-014	UM 9

Requirement	Design	Case	Manual
TM-NFR-015 Contract consistency	SDD 4.9	TM-QT-015	UM 9
TM-NFR-016 Traceable changes	SDD 3.3	TM-QT-016	UM 7
TM-NFR-017 Shared rate limits	SDD 7.2	TM-QT-017	UM 9
TM-NFR-018 Reproducible verification	SDD 8	TM-QT-018	UM 9
TM-NFR-019 Measured performance	SDD 8	TM-QT-019	UM 9
TM-NFR-020 Date and time consistency	SDD 6.3	TM-QT-020	UM 4

Appendix B Entry points and data dictionary

The API table lists actual mounted operations; see the CSV for handler symbols and exact OpenAPI coverage. OAuth forwarding and framework error/loading boundaries are technical entrypoints, not separate product features. Source-derived payload fields should be checked against validators whenever the API changes. [1][2]

Table 32 Actual API operation register

Method and path	Implementation route	OpenAPI exact operation
GET /api/auth	travelmate-backend-api/src/routes/auth-routes.ts:7	Present
POST /api/auth	travelmate-backend-api/src/routes/auth-routes.ts:8	Present
GET /api/auth/oauth/google/status	travelmate-backend-api/src/routes/auth-routes.ts:9	Missing or different path
GET /api/auth/oauth/google	travelmate-backend-api/src/routes/auth-routes.ts:10	Present
GET /api/auth/oauth/google/callback	travelmate-backend-api/src/routes/auth-routes.ts:11	Present
GET /api/account/appeal	travelmate-backend-api/src/routes/appeal-routes.ts:5	Present

Method and path	Implementation route	OpenAPI exact operation
POST /api/account/appeal	travelmate-backend-api/src/routes/appeal-routes.ts:6	Present
GET /api/platform/reports/stream	travelmate-backend-api/src/routes/platform-routes.ts:9	Present
GET /api/platform	travelmate-backend-api/src/routes/platform-routes.ts:10	Present
POST /api/platform	travelmate-backend-api/src/routes/platform-routes.ts:11	Present
GET /api/platform/notifications	travelmate-backend-api/src/routes/platform-routes.ts:12	Present
GET /api/platform/visited-places	travelmate-backend-api/src/routes/platform-routes.ts:13	Present
GET /api/platform/trips/:id/visit-candidates	travelmate-backend-api/src/routes/platform-routes.ts:14	Present
GET /api/platform/trips/:id/activities	travelmate-backend-api/src/routes/platform-routes.ts:15	Present
POST /api/platform/trips/:id/photos	travelmate-backend-api/src/routes/platform-routes.ts:16	Present
PATCH /api/platform/trips/:id/activities/:activityId	travelmate-backend-api/src/routes/platform-routes.ts:17	Present
POST /api/platform/trips/:id/replan	travelmate-backend-api/src/routes/platform-routes.ts:18	Present
POST /api/platform/trips/:id/replan/accept	travelmate-backend-api/src/routes/platform-routes.ts:19	Present
GET /api/platform/trips/:id	travelmate-backend-api/src/routes/platform-routes.ts:20	Present
GET /api/platform/trips/:id/versions	travelmate-backend-api/src/routes/platform-routes.ts:21	Present
GET /api/feedback	travelmate-backend-api/src/routes/feedback-routes.ts:6	Present

Method and path	Implementation route	OpenAPI exact operation
POST /api/feedback	travelmate-backend-api/src/routes/feedback-routes.ts:7	Present
GET /api/feedback/itinerary/:id	travelmate-backend-api/src/routes/feedback-routes.ts:8	Present
GET /api/feedback/:id/itinerary	travelmate-backend-api/src/routes/feedback-routes.ts:9	Present
GET /api/feedback/:id	travelmate-backend-api/src/routes/feedback-routes.ts:10	Present
POST /api/itinerary	travelmate-backend-api/src/routes/itinerary-routes.ts:6	Present
POST /api/itinerary/images	travelmate-backend-api/src/routes/itinerary-routes.ts:7	Present
GET /api/weather	travelmate-backend-api/src/routes/weather-routes.ts:6	Present
GET /api/locations/areas/compare	travelmate-backend-api/src/routes/locations-routes.ts:6	Present
GET /api/locations/areas	travelmate-backend-api/src/routes/locations-routes.ts:7	Present
GET /api/locations/current	travelmate-backend-api/src/routes/locations-routes.ts:8	Missing or different path
GET /api/locations/resolve	travelmate-backend-api/src/routes/locations-routes.ts:9	Present
GET /api/locations	travelmate-backend-api/src/routes/locations-routes.ts:10	Present
POST /api/navigation/search	travelmate-backend-api/src/routes/navigation-routes.ts:7	Present
POST /api/navigation/route	travelmate-backend-api/src/routes/navigation-routes.ts:8	Present
GET /api/navigation/trips/:id/map	travelmate-backend-api/src/routes/navigation-routes.ts:9	Missing or different path

Method and path	Implementation route	OpenAPI exact operation
POST /api/navigation/trips/:id/activity	travelmate-backend-api/src/routes/navigation-routes.ts:10	Missing or different path
POST /api/navigation/trips/:id/day-route	travelmate-backend-api/src/routes/navigation-routes.ts:11	Missing or different path
POST /api/navigation/trips/:id/weather	travelmate-backend-api/src/routes/navigation-routes.ts:12	Missing or different path
GET /api/destination-context	travelmate-backend-api/src/routes/destination-context-routes.ts:6	Present
GET /api/destination-context/exchange-rate	travelmate-backend-api/src/routes/destination-context-routes.ts:7	Missing or different path
GET /api/destination-context/photo	travelmate-backend-api/src/routes/destination-context-routes.ts:8	Missing or different path
POST /api/destination-context/photo-selection	travelmate-backend-api/src/routes/destination-context-routes.ts:9	Missing or different path
POST /api/accommodations/revalidate	travelmate-backend-api/src/routes/accommodations-routes.ts:7	Present
GET /api/accommodations	travelmate-backend-api/src/routes/accommodations-routes.ts:8	Present
GET /api/travel-options	travelmate-backend-api/src/routes/travel-options-routes.ts:6	Present
GET /api/profile	travelmate-backend-api/src/routes/profile-routes.ts:6	Present
PATCH /api/profile	travelmate-backend-api/src/routes/profile-routes.ts:7	Present
GET /	travelmate-backend-api/src/routes/system-routes.ts:5	Missing or different path

Method and path	Implementation route	OpenAPI exact operation
GET /health	travelmate-backend-api/src/routes/system-routes.ts:6	Present

The accompanying TravelMate-Database-Field-Register.csv contains every field, declared type, default, key/relation and model constraint from all twenty Prisma models. The next table summarizes SQL/source history. Raw migration presence is not a live applied-state report.

Table 33 Migration history present in source

Migration directory	Evidence
20260902000000_init	Source SQL; applied state not queried.
20260907000000_add_trip_end_date	Source SQL; applied state not queried.
20260914090000_add_account_tokens	Source SQL; applied state not queried.
20260914120000_add_trip_currency	Source SQL; applied state not queried.
20260916090000_add_trip_destination_context	Source SQL; applied state not queried.
20260918090000_add_trip_lifecycle	Source SQL; applied state not queried.
20260925090000_trip_party_and_integrity_constraints	Source SQL; applied state not queried.
20260926183000_add_google_oauth	Source SQL; applied state not queried.

Migration directory	Evidence
20260929130000_owner_marketplace_operations	Source SQL; applied state not queried.
20260930000000_traveler_admin_roles	Source SQL; applied state not queried.
20261001000000_traveler_reports	Source SQL; applied state not queried.
20261002000000_suspension_appeals	Source SQL; applied state not queried.
20261002000100_pending_appeal_index	Source SQL; applied state not queried.
20261003000000_auth_and_generation_safety	Source SQL; applied state not queried.
20261005000000_user_profile_metadata	Source SQL; applied state not queried.
20261005000100_itinerary_feedback	Source SQL; applied state not queried.
20261006000000_editable_trip_feedback	Source SQL; applied state not queried.
20261006000100_unanalyzed_feedback	Source SQL; applied state not queried.
20261006000200_trip_duration_31_days	Source SQL; applied state not queried.
20261006063703_dropping_table_rate_limit_buckets	Source SQL; applied state not queried.

Migration directory	Evidence
20261006063800_restore_rate_limit_buckets	Source SQL; applied state not queried.
20261008000000_completed_trips	Source SQL; applied state not queried.
20261008120000_visited_places_save_keys	Source SQL; applied state not queried.
20261015000000_visited_place_media	Source SQL; applied state not queried.

Appendix C Installation and operations guide

These are instructions for an authorized operator on a fresh/disposable environment; none was executed against existing data in this documentation task. Inspect scripts and provider agreements first. Do not point setup, integration tests, seeders or migrations at the existing application database.

Table 34 Setup prerequisites and commands

Order	Operator action	Side effects and verification
1	Install compatible Node >=22.13 and PostgreSQL; clone the paired revisions and retain lockfiles.	Fresh workspace only; confirm intended branches/working state. No claimed minimum machine capacity.
2	Backend: copy .env.example to a private .env and set DATABASE_URL, FRONTEND_URL and distinct strong SESSION_SECRET/ACCOUNT_TOKEN_SECRET.	Use a new disposable database; keep credentials outside source/evidence. DIRECT_URL is optional; Prisma Migrate uses DATABASE_URL.
3	Run npm ci in the fresh backend checkout.	Installs dependencies and postinstall runs prisma generate; writes generated files.
4	Review migrations; then npm run db:setup only on the disposable database.	Runs client generation and prisma migrate deploy. Do not run db:seed unless its data effects are separately approved.

Order	Operator action	Side effects and verification
5	Configure mail and optional providers only after operator approval.	Production Resend needs a verified sending domain; Gmail is development-only. Never run email:smoke without an approved recipient.
6	Backend: npm run dev for local work or npm run build then npm start for a controlled production-like run.	Build generates Prisma and dist. Development watcher scripts manage processes; review before use.
7	Frontend: set server-only BACKEND_URL=http://localhost:5000 in private local environment, then npm ci.	Never expose provider secrets with NEXT_PUBLIC_. Frontend predev can clean its image cache.
8	Frontend: npm run contracts:check, then npm run dev:frontend or reviewed npm run dev.	contracts:sync writes generated contracts; --check detects drift. Dev defaults may coordinate servers.
9	Open localhost:3000; use readiness and authorized test fixtures.	Registration/mail/admin provisioning are not part of read-only startup verification.
10	Before deployment, approve secrets, HTTPS/proxy origin, provider rights, migrations, backup and recovery.	Use production Compose/deploy only with an authorized target. Health success is not completed functional acceptance.

Table 35 Configuration groups without secret values

Group	Names and purpose
Core	DATABASE_URL, PORT, FRONTEND_URL, SESSION_SECRET, ACCOUNT_TOKEN_SECRET; server runtime, storage and signatures.
Frontend proxy	BACKEND_URL; server-only Express origin.
Google identity	GOOGLE_CLIENT_ID, GOOGLE_CLIENT_SECRET, GOOGLE_OAUTH_REDIRECT_URI, OAUTH_STATE_SECRET.
Mail	EMAIL_PROVIDER, RESEND_API_KEY, EMAIL_FROM; SMTP_USER/SMTP_PASS for explicit development Gmail; EMAIL_SMOKE_TO only for authorized smoke tests.
AI	AI_PROVIDER, OPENAI_API_KEY/OPENAI_MODEL, GEMINI_API_KEY/GEMINI_MODEL; AI_MOCK_FALLBACK non-production only.

Group	Names and purpose
Geography/reference	DESTINATION_GEOCODER_URL, DESTINATION_AREAS_PLACES_URL, ACCOMMODATION_PLACES_URL/FALLBACK_URL, WIKIMEDIA_USER_AGENT, WIKIMEDIA_PHOTO_INTERVAL_MS.
Weather and optional media	OPENWEATHER_API_KEY, OPEN_METEO_API_KEY; GOOGLE_PLACES_API_KEY and GOOGLE_PLACES_PHOTOS_ENABLED explicit paid opt-in.
Accommodation/retained offers	LITEAPI_API_KEY/LITEAPI_ENV; RAPID_* credentials/environment/content-rights/attribution; ACCOMMODATION_QUOTE_SECRET; AMADEUS_* optional retained comparison.
Legacy/unavailable	PAYMONGO_SECRET_KEY in example config does not establish a current checkout/payment feature.

Production monitoring should distinguish API/DB readiness, provider configuration, actual provider delivery and end-user workflow outcomes. Keep backups outside the running container and verify restoration into a disposable database. The inspected deployment script can roll back the application image after failed health but does not reverse migrated schema. Retention, RPO/RTO, monitoring recipients and maintenance ownership remain unconfirmed. [2]

Appendix D Sanitized source excerpts and diagram sources

The following extracts are selected implementation fragments, not executable setup instructions. Their complete source locations and hashes remain in the snapshot. They contain no user data or secrets. The editable diagrams are supplied as Mermaid text and SVG, with PNG illustrations embedded in the documents.

Registration name and phone validation; travelmate-backend-api/src/schemas/auth/registration-schema.ts:15 [1][2]

```
15: if (input.phone !== undefined && typeof input.phone !== 'string') throw new RegistrationValidationError('Contact number must be text.');
16: const phone = normalizePhilippineMobile(typeof input.phone === 'string' ? input.phone : '');
17: if (phone === null) throw new RegistrationValidationError('Enter a 10-digit Philippine mobile number (e.g. 917 123 4567).');
18: return { name, ...(phone ? { phone } : {}) };
19: }
```

```
20: import { normalizePhilippineMobile } from './philippine-mobile.ts';
21:
```

Exact country code flag selection; travelmate-frontend-new/lib/discovery-country-flags.ts:4 [1][2]

```
4: export function discoveryCountryFlag(code: string): string | null {
5: if (code === 'WD-Q34754') return '/discovery/flags/somaliland.svg';
6: if (code === 'WD-Q23681') return '/discovery/flags/northern-cyprus.svg';
7: // Antarctica and other geographic features have no national flag. Do not invent
one.
8: if (code === 'AQ' || !countryFlagCodes.has(code)) return null;
9: return '/flags/' + code.toLowerCase() + '.svg';
10: }
11:
```

Appendix E Review and acceptance forms

These blank forms are for future authorized review. They do not represent completed tests or surveys. Participant counts and any usability scale must be approved before collecting data.

Table 36 User acceptance session record

Field	To complete
Session/participant code	[TO BE COMPLETED] Use a pseudonymous code, not a full name in public evidence.
Consent and authorized environment	[TO BE COMPLETED]
Role and assigned requirements	[TO BE COMPLETED]
Tasks	Select destination; enter brief; inspect sources/costs; reopen/edit saved trip; map/visit task; role-specific task as applicable.
Observed completion/errors/help	[TO BE COMPLETED AFTER EXECUTION]
User comments and accessibility barriers	[TO BE COMPLETED AFTER EXECUTION]
Evidence, defects and reviewer decision	[TO BE COMPLETED AFTER EXECUTION]

Table 37 Change request and release review form

Field	To complete
Request/requirement/defect IDs	[TO BE COMPLETED]
Proposed correction and affected layers	[TO BE COMPLETED]
Privacy/provider/schema impact	[TO BE COMPLETED]
Test and rollback evidence	[TO BE COMPLETED AFTER EXECUTION]
Reviewer decision, name and date	[TO BE CONFIRMED]

Appendix F Evidence and unresolved details

Use the standalone change summary, unresolved-details list and verification execution register for the complete review queue. Unconfirmed people/year/adviser/panel, actual schedule/effort/budget, UAT, deployment/provider/persistence evidence, privacy/retention/recovery targets and final Word rendering remain open. The companion PDF is supported and visually reviewed separately; it is not a Word-engine export. Original source documents and application files were retained untouched.

Development Team Profile

Final team roster and individual profiles are [TO BE CONFIRMED]. The supplied old SRS lists “CORTES, ROVIC” and separately contains the combined string “GULBE, JOHNMARKJUMAO-AS JOHNLEE”. This revision preserves that provenance here without splitting the string, combining people, assigning roles or claiming a final roster. No credentials, biographies, contribution percentages or attendance have been invented. [3]

Table 38 Team profile completion fields

Field	Record
Final full names and preferred form	[TO BE CONFIRMED]
Academic year and student details	[TO BE CONFIRMED]
Approved role allocation	[TO BE CONFIRMED]
Supported contributions and evidence	[TO BE CONFIRMED]
Credentials or biography authorized for inclusion	[TO BE CONFIRMED]
Adviser/panel names and approval dates	[TO BE CONFIRMED]